Neural Retrieval and Re-ranking

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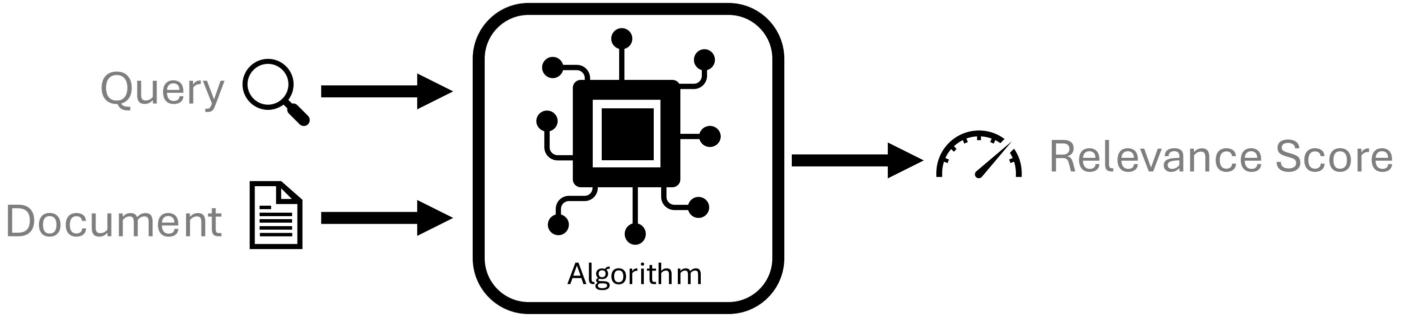
(European Summer School on Information Retrieval)

Neural Retrieval and Re-ranking

Relevance Estimation Models

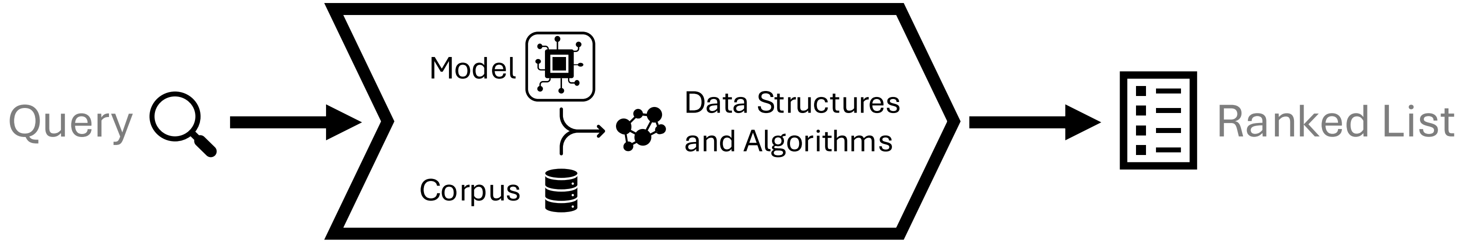
How to estimate relevance?

Lots of research here!
What architecture? How to train?
What inputs/outputs?



Retrieval Engines

How to find the top results for a relevance model?



Also lots of research here!
What data structures? How to scale?
Hybrid systems (multiple models)?

Neural Retrieval and Re-ranking

By the end of this lecture you'll…

be familiar with the main types of **relevance models**.

describe the main strategies for **training** them.

understand how they are used within an **engine**.

recognize the **trade-offs** of various systems.

Let’s dive in!

Neural Retrieval and Re-ranking

Part 1: Relevance Est. Models

Part 2: Training

Part 3: Engines

Wrap-up

A running example...

I want to know Lewis Hamilton's performance in the British Formula One Grand Prix

hamilton british gp

Teams summoned after Lewis Hamilton and Max Verstappen British GP collision

Red Bull have requested a review of the penalty given to Mercedes' Lewis Hamilton after his first-lap collision with Max Verstappen at the British Grand Prix.

Hamilton was ruled to have been "predominantly to blame" for the crash, which resulted in Red Bull driver Verstappen being taken to hospital.

The Briton was given a 10-second penalty but recovered to win the race.

Both teams have been summoned to appear via video link on Thursday.

...

Source: https://www.bbc.co.uk/sport/formula1/57988419

Turkish Grand Prix to replace Canadian Grand Prix on 2021 F1 calendar amid rise in coronavirus cases

The Canadian Grand Prix has been cancelled and will be replaced on its 11-13 June date by a race at Turkey's Istanbul Park track.

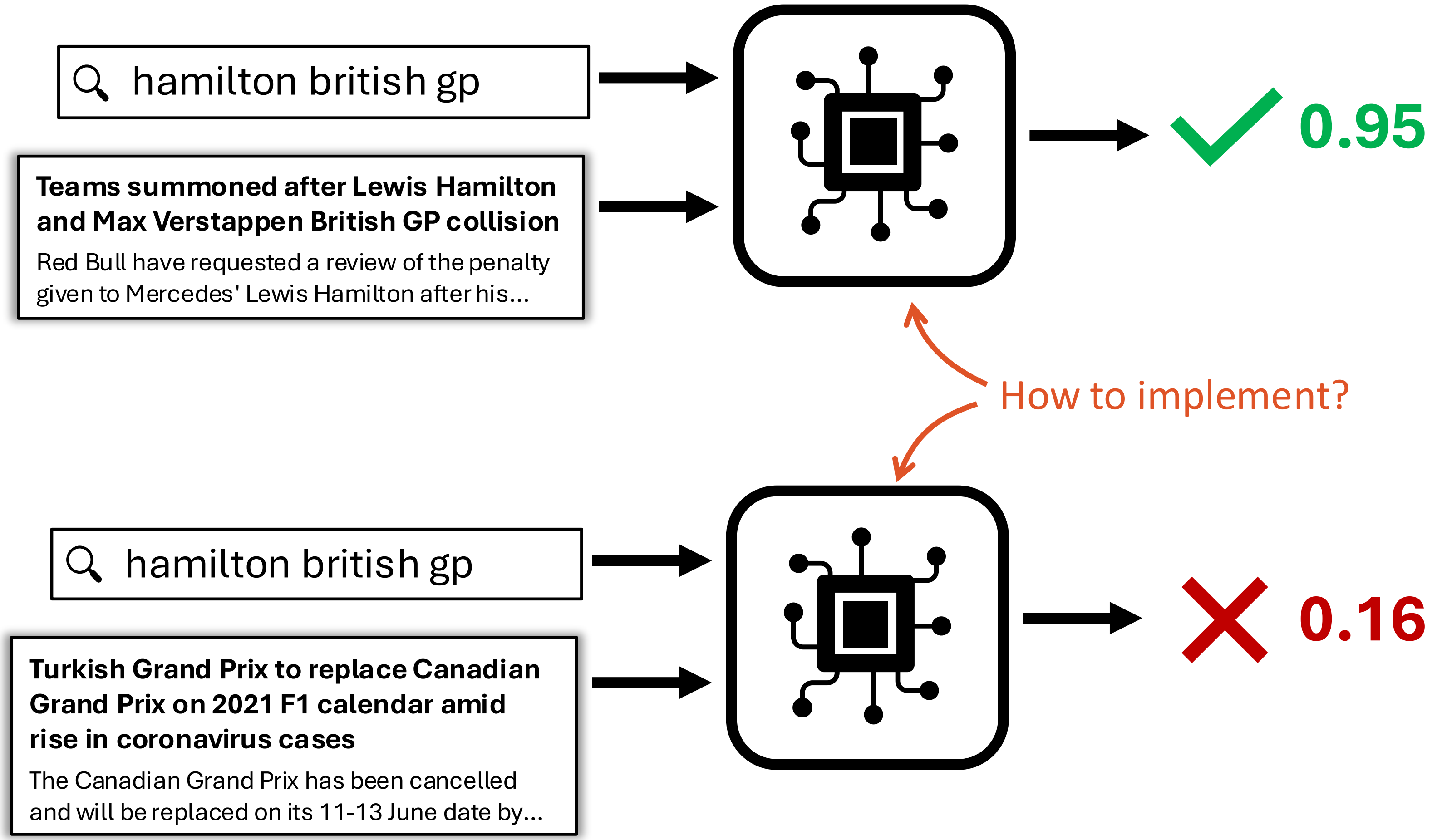
It is the second successive year the Montreal event has been cancelled because of the coronavirus pandemic.

Travel restrictions made it impossible to enter Canada without a mandatory 14-day quarantine, said an F1 statement.

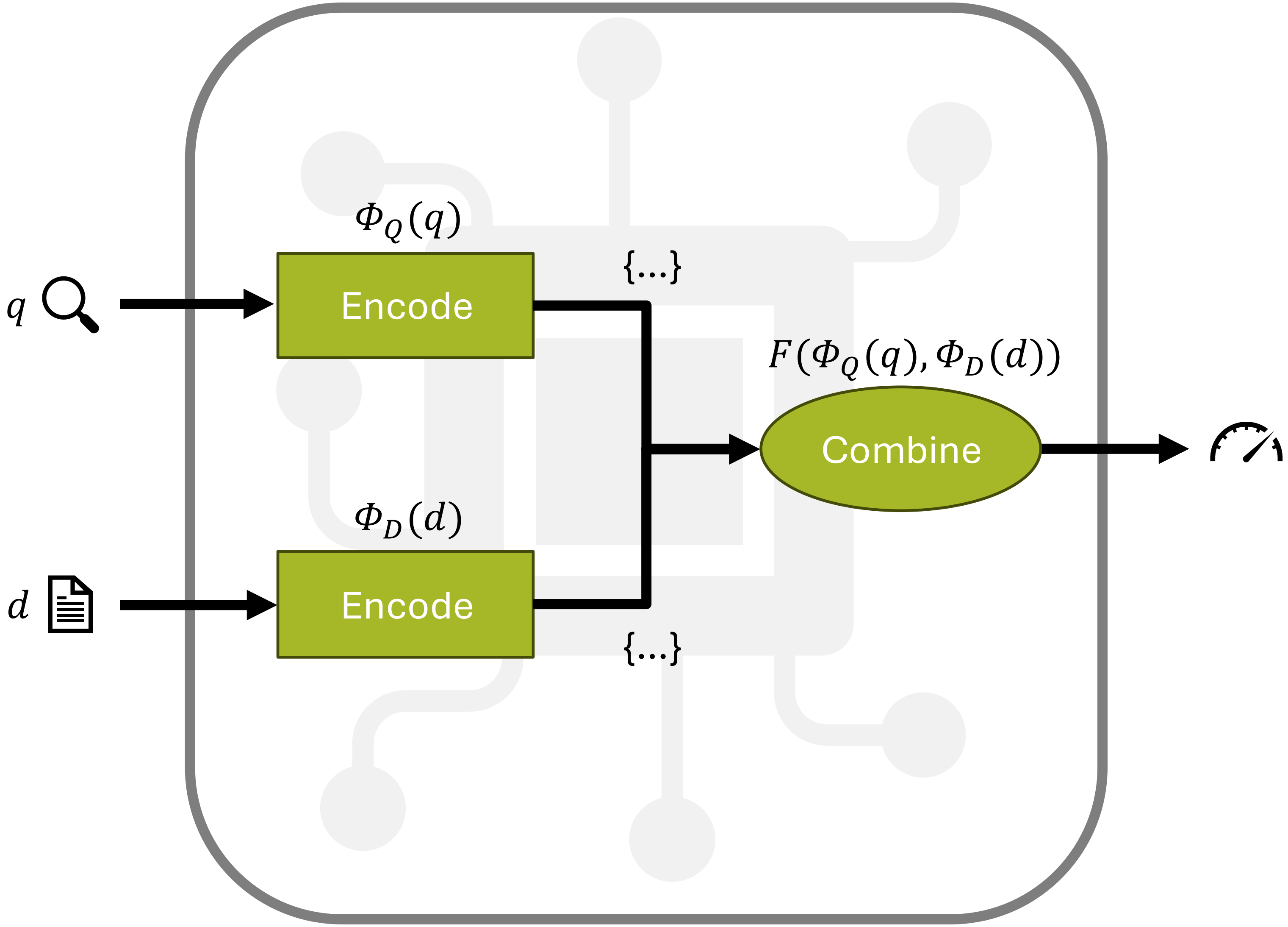
Meanwhile the contract for a race on Montreal's Circuit Gilles Villeneuve has been extended by two years.

...

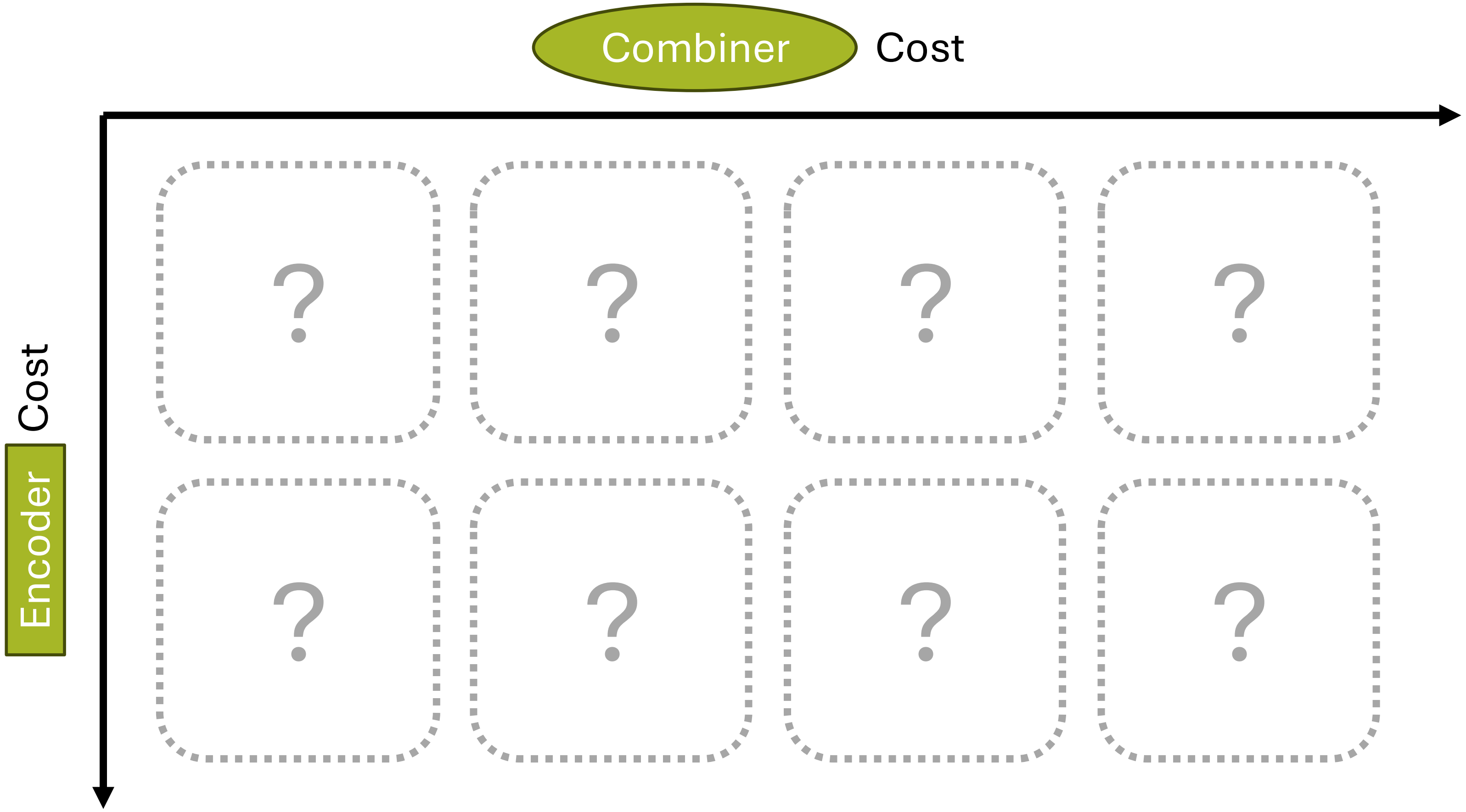
https://www.bbc.co.uk/sport/formula1/56915817


hamilton british gp
Teams summoned after Lewis Hamilton and Max Verstappen British GP collision
Red Bull have requested a review of the penalty given to Mercedes' Lewis Hamilton after his...
0.95
How to implement?
hamilton british gp
Turkish Grand Prix to replace Canadian Grand Prix on 2021 F1 calendar amid rise in coronavirus cases
The Canadian Grand Prix has been cancelled and will be replaced on its 11-13 June date by...
0.16

Generic Relevance Est. Model Structure



The Table of Relevance Est. Models



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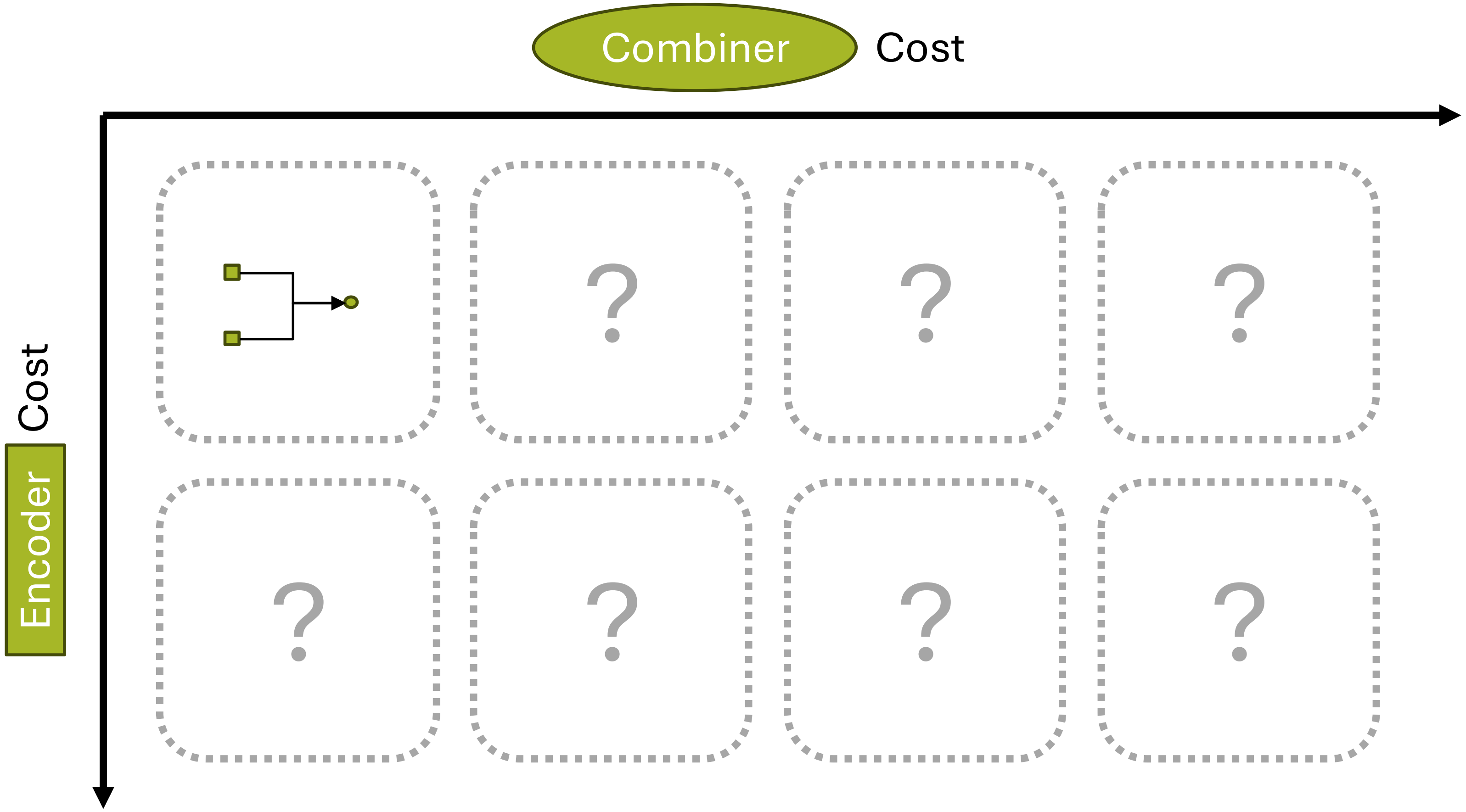

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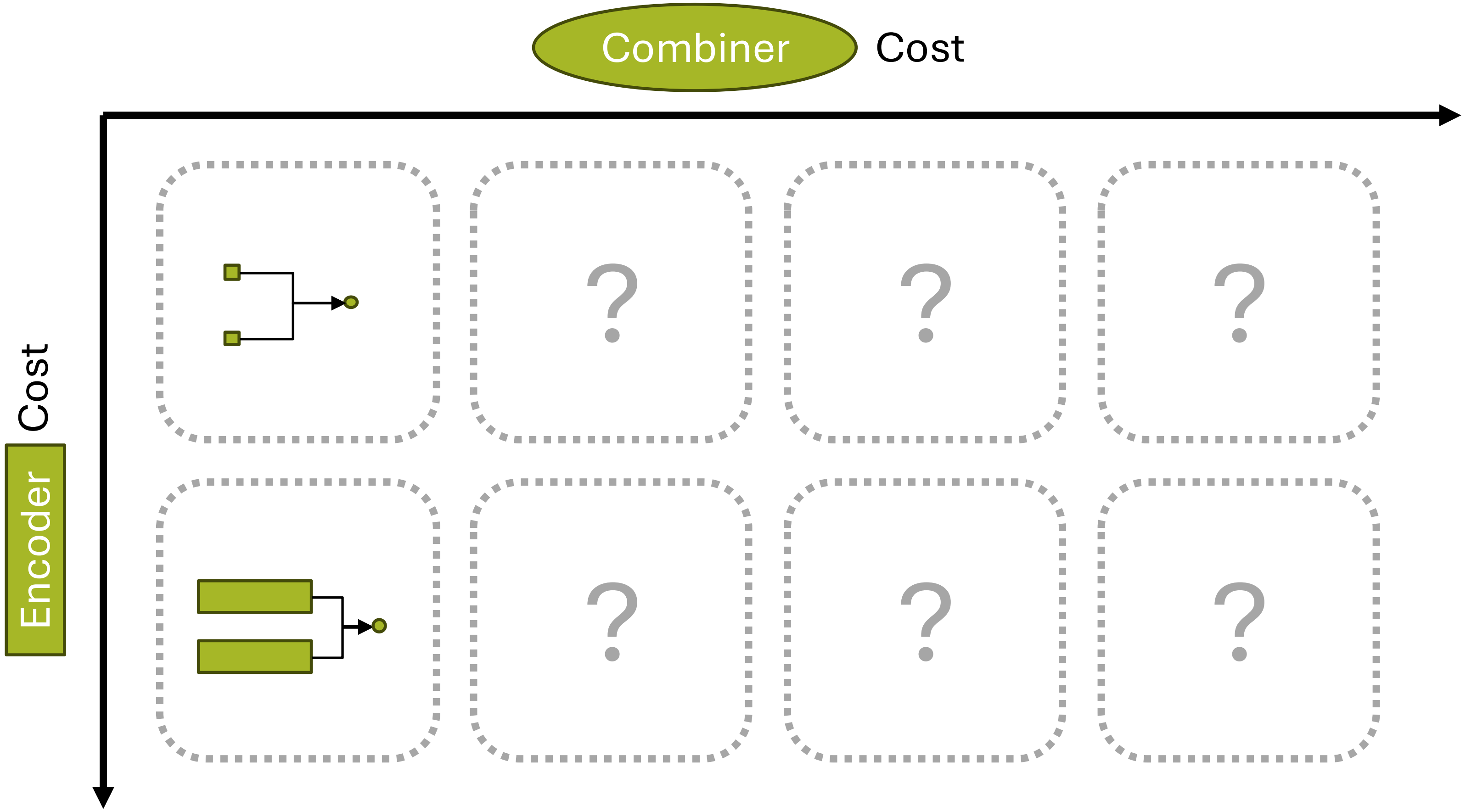

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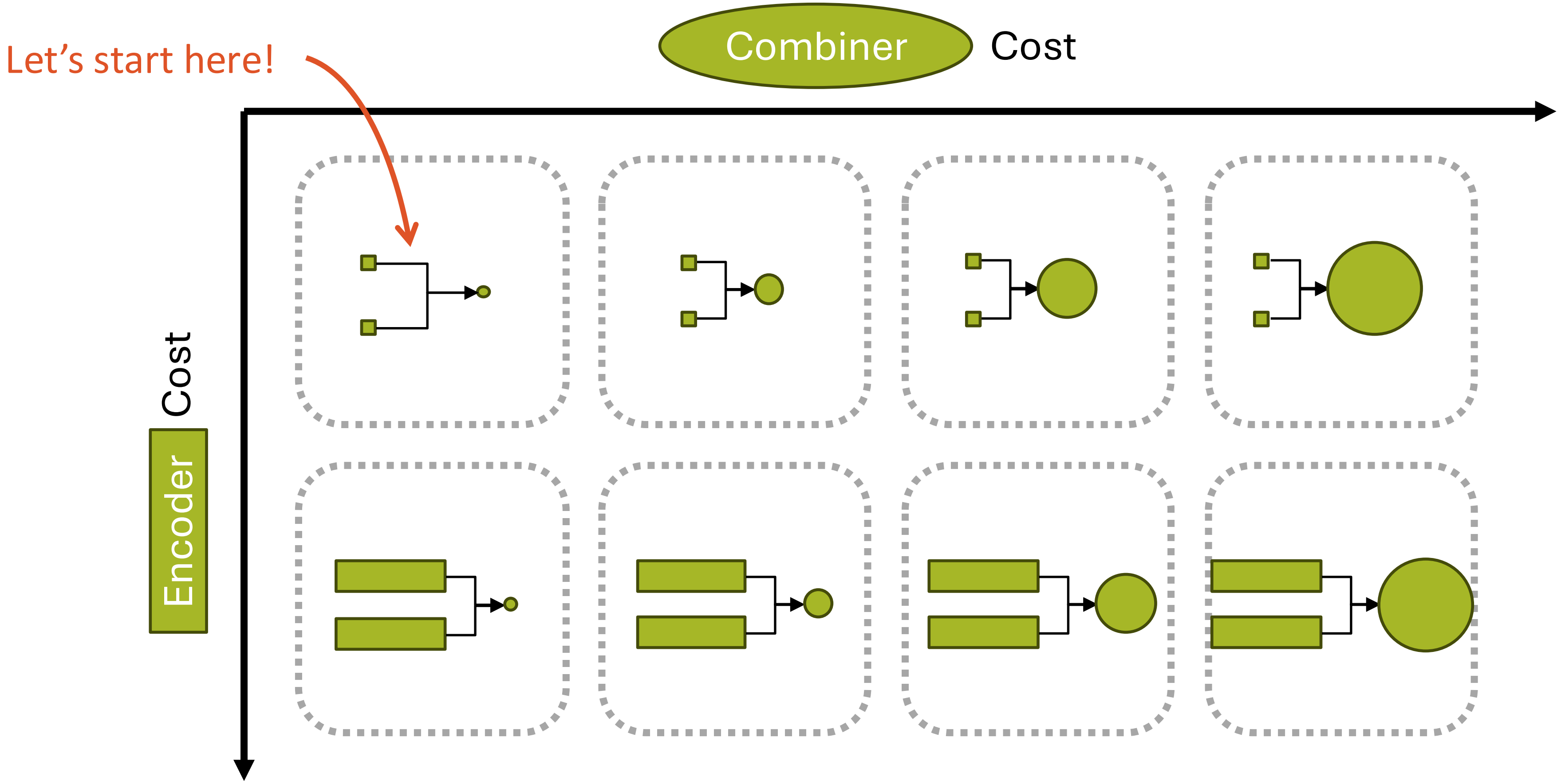

Encoding Strategy: Term Frequency

A sparse representation containing of the number of times each word appears.

hamilton british gp

Encode

{hamilton: 1, british: 1, gp: 1}

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…

Source: https://www.bbc.co.uk/sport/formula1/57988419

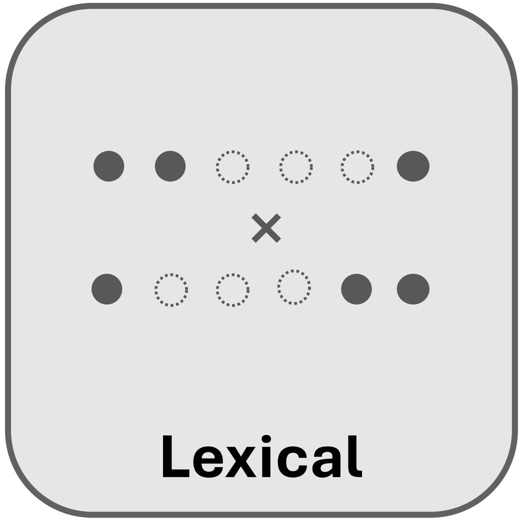
Encode

{to: 6, the: 5, hamilton: 3, verstappen: 3, have: 3, teams: 2, summoned: 2, after: 2, lewis: 2, max: 2, british: 2, collision: 2, red: 2, bull: 2, a: 2, penalty: 2, given: 2, was: 2, been: 2, and: 1, gp: 1, …}

Combination Strategy: Dot Product

Sum of the product of each dimension

Lexical Relevance Models



Representations based on lexical form

Combination based on overlapping terms

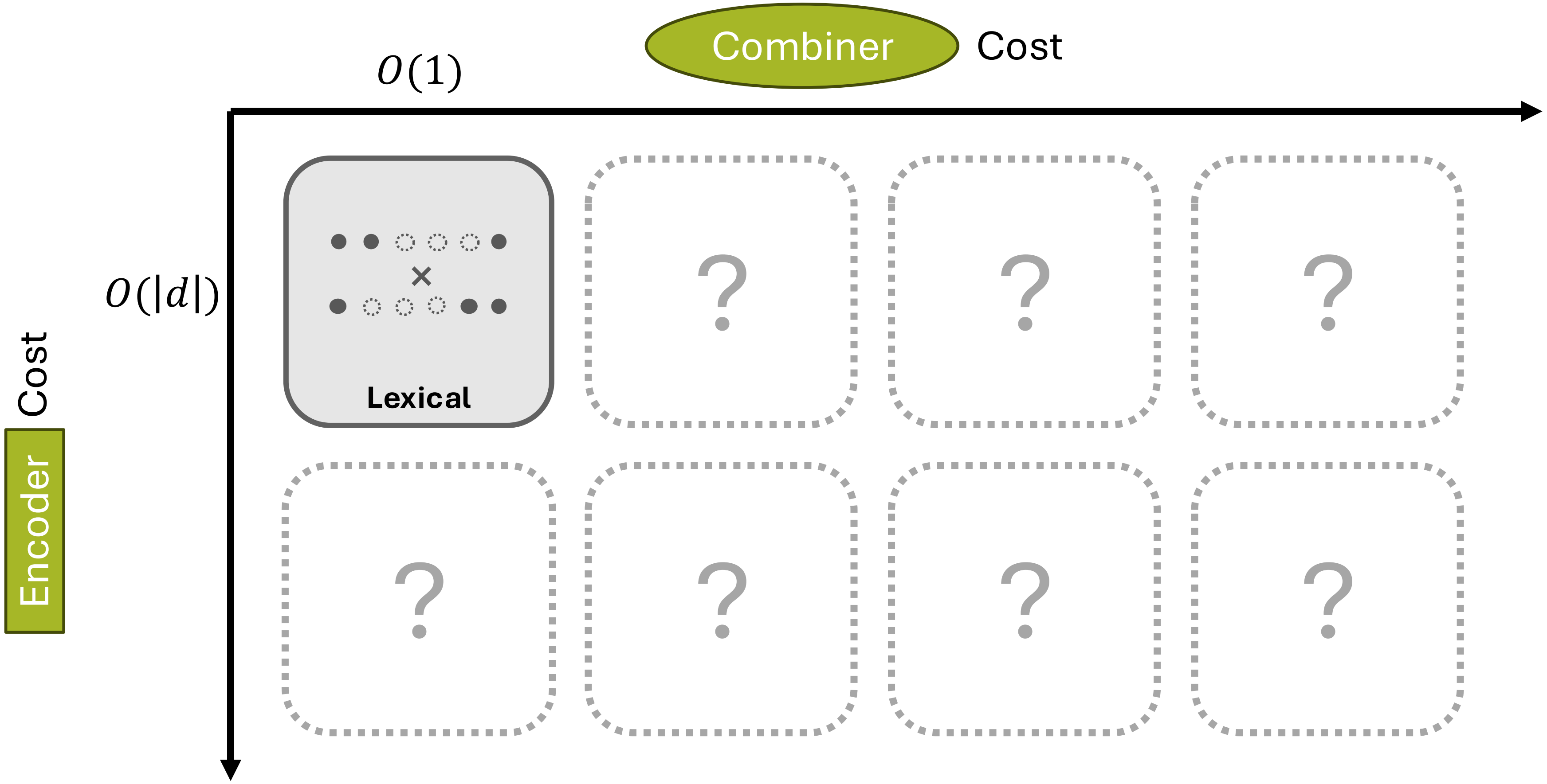
Variations:

- Stopword Removal
- Stemming
- Inverse Document Frequency (IDF)
- Sublinear Term Frequencies
- Document length normalization
- … Others too!

Common Lexical Relevance Models:

- TF-IDF Cosine
- BM25
- Query Likelihood (QL)
- Divergence from Random
- … Others too!

The Table of Relevance Est. Models



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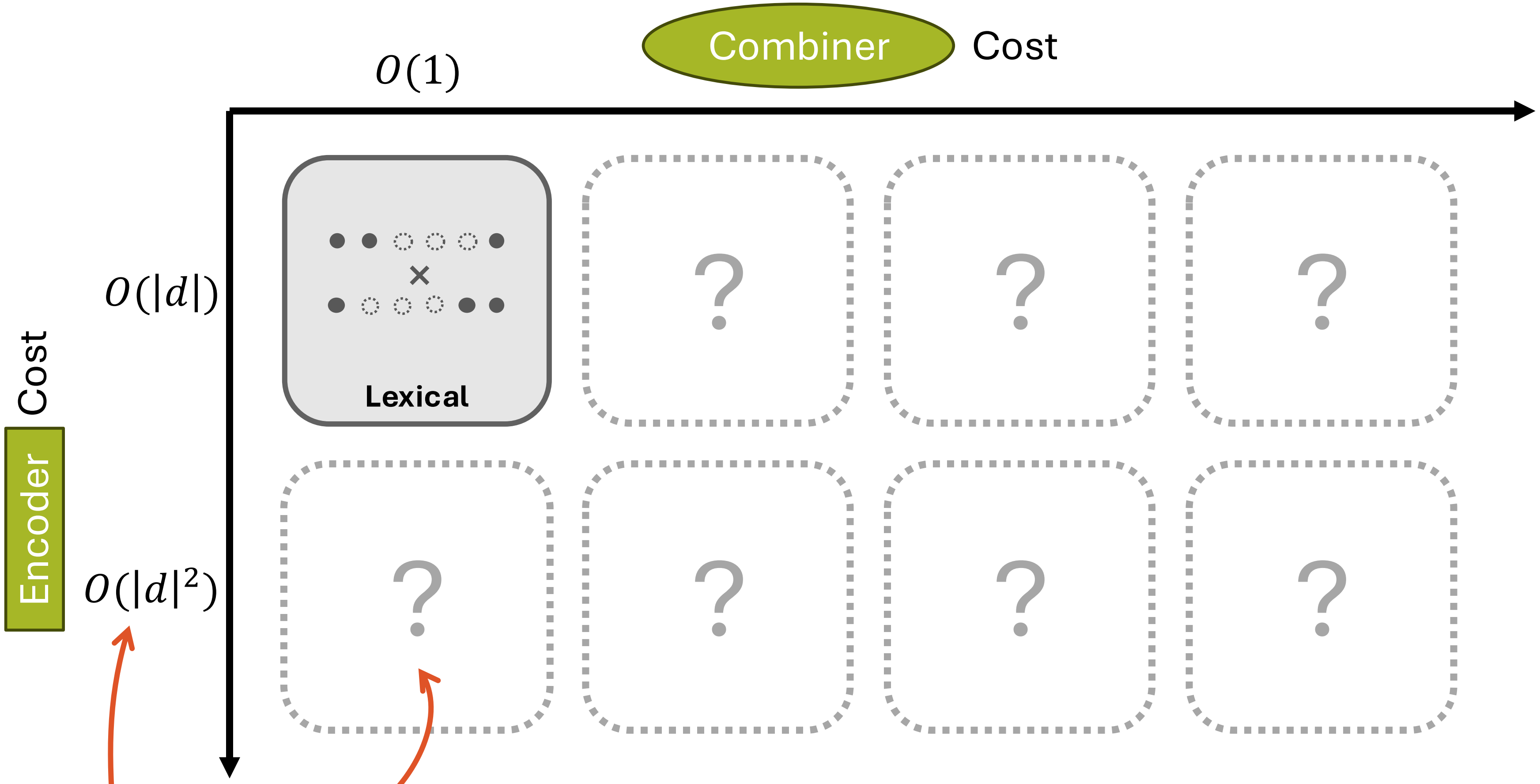

Spend more on the encoder to
improve the quality
(but keep a cheap combiner)

Problem with Lexical Signals: Importance

Teams summoned after Lewis Hamilton and Max Verstappen British GP collision

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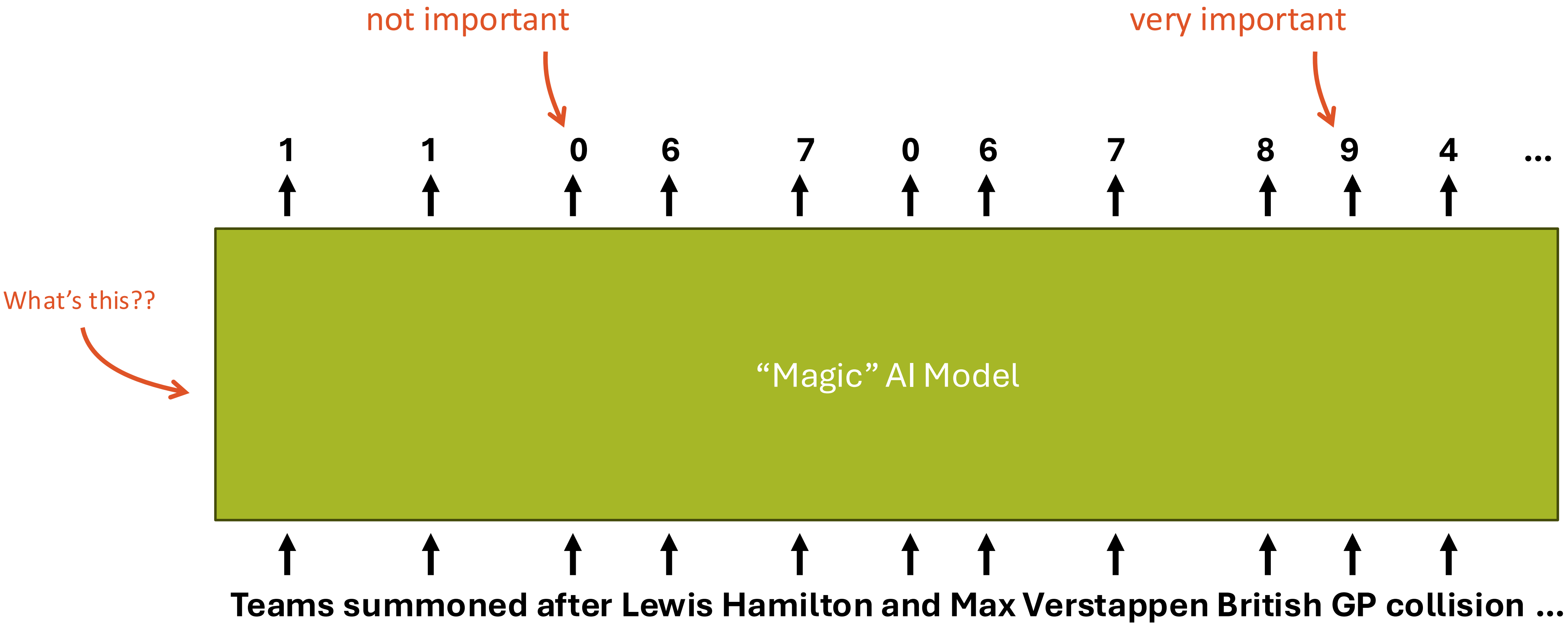
→ Encode →

Terms that are not really important can get higher scores

{to: 6, the: 5, hamilton: 3, verstappen: 3, have: 3, **teams: 2**, summoned: 2, after: 2, lewis: 2, max: 2, **british: 2**, collision: 2, red: 2, bull: 2, a: 2, penalty: 2, given: 2, was: 2, been: 2, and: 1, **gp: 1**, …}

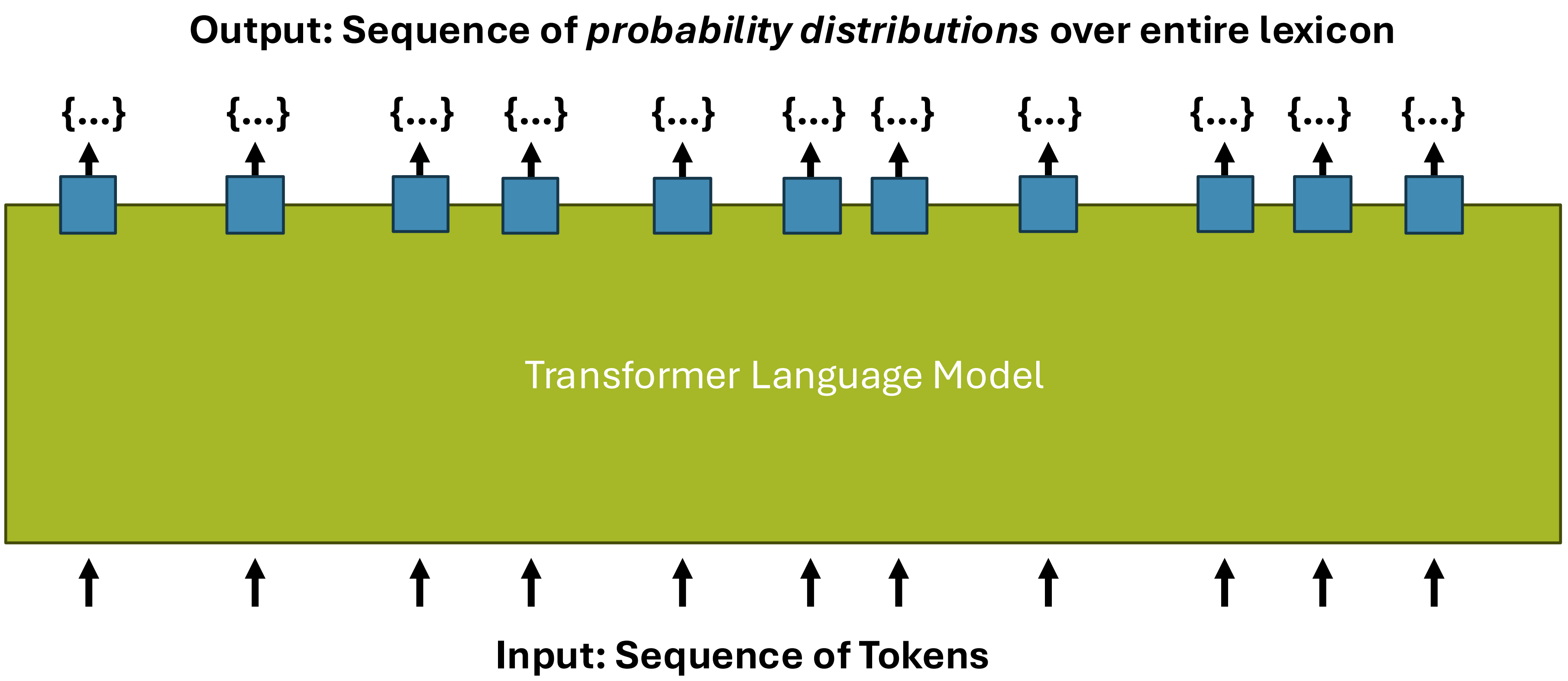
The documents is about the British GP, but not necessarily high scores

Alternative: Use Neural LM to Predict Term Weight



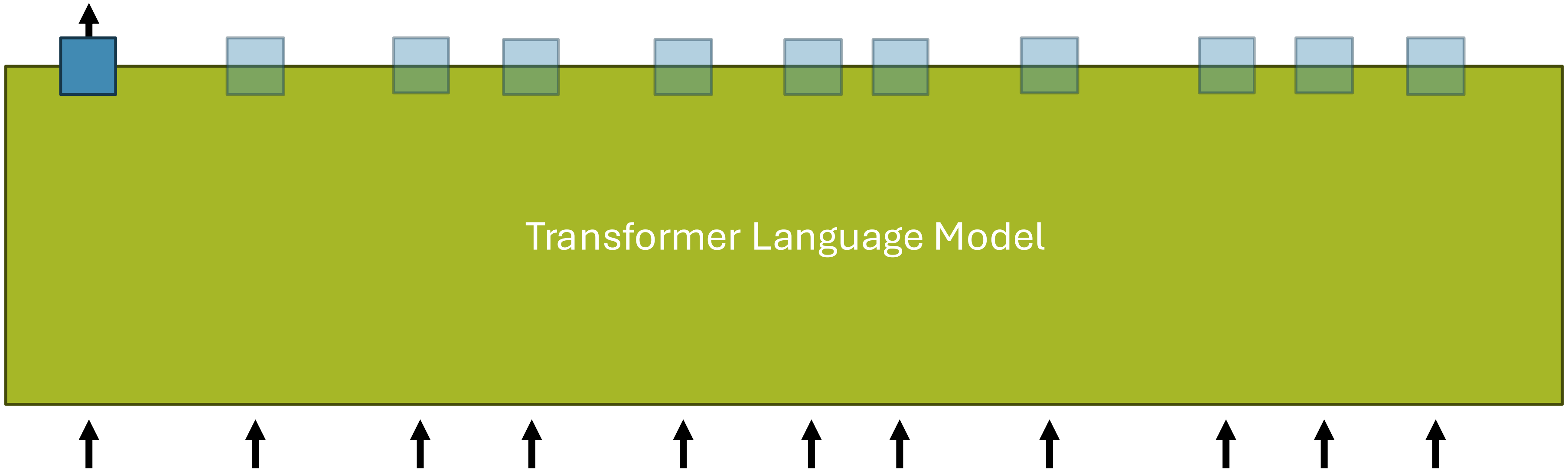
[DeepCT]

Crash Course: Transformer Language Models



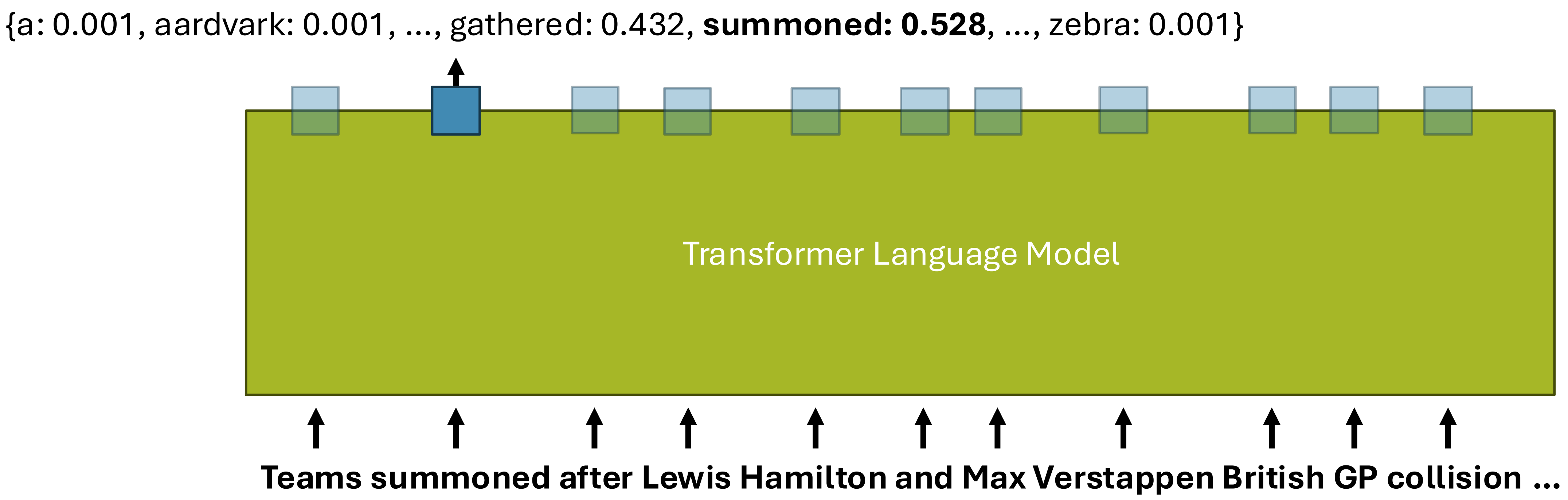
Crash Course: Transformer Language Models

{a: 0.001, aardvark: 0.001, ..., managers: 0.102, **teams: 0.812**, ..., zebra: 0.001}

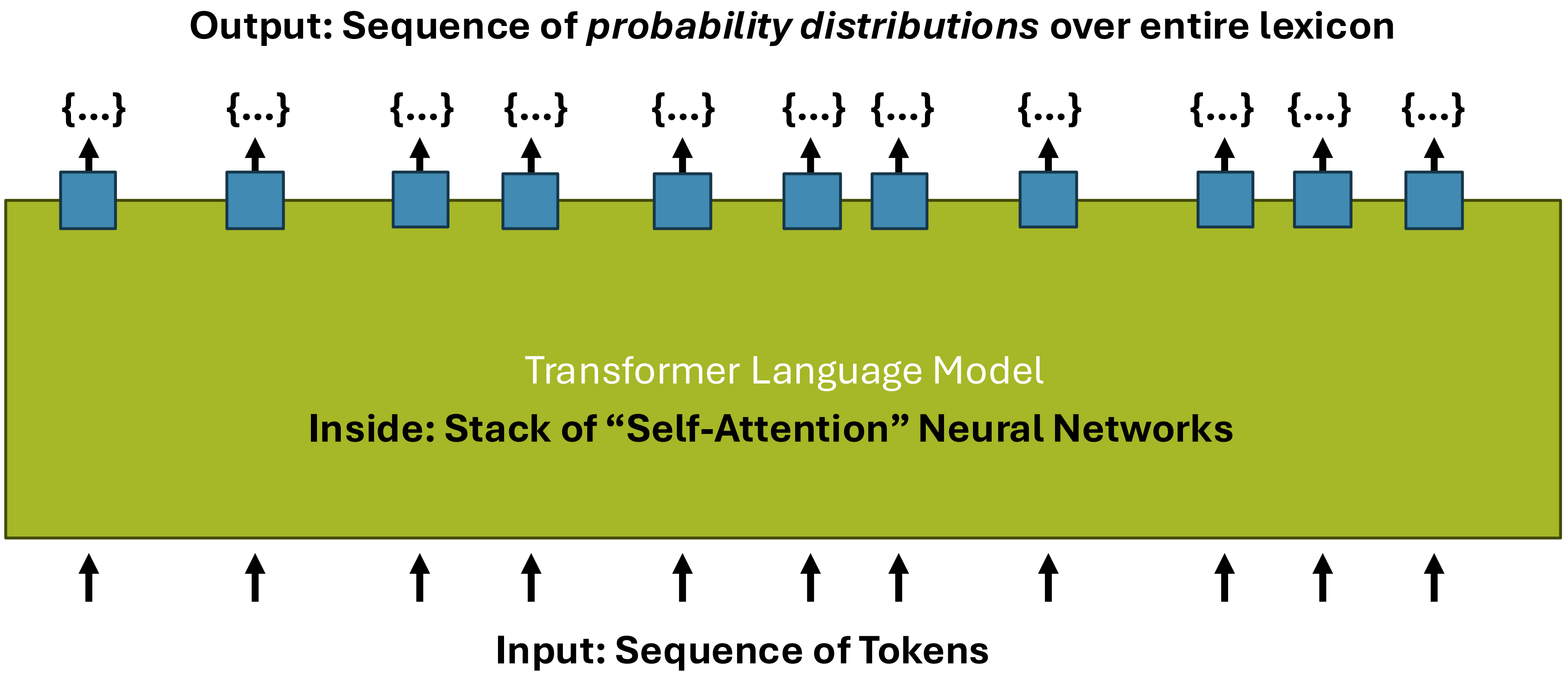


Teams summoned after Lewis Hamilton and Max Verstappen British GP collision ...

Crash Course: Transformer Language Models



Crash Course: Transformer Language Models



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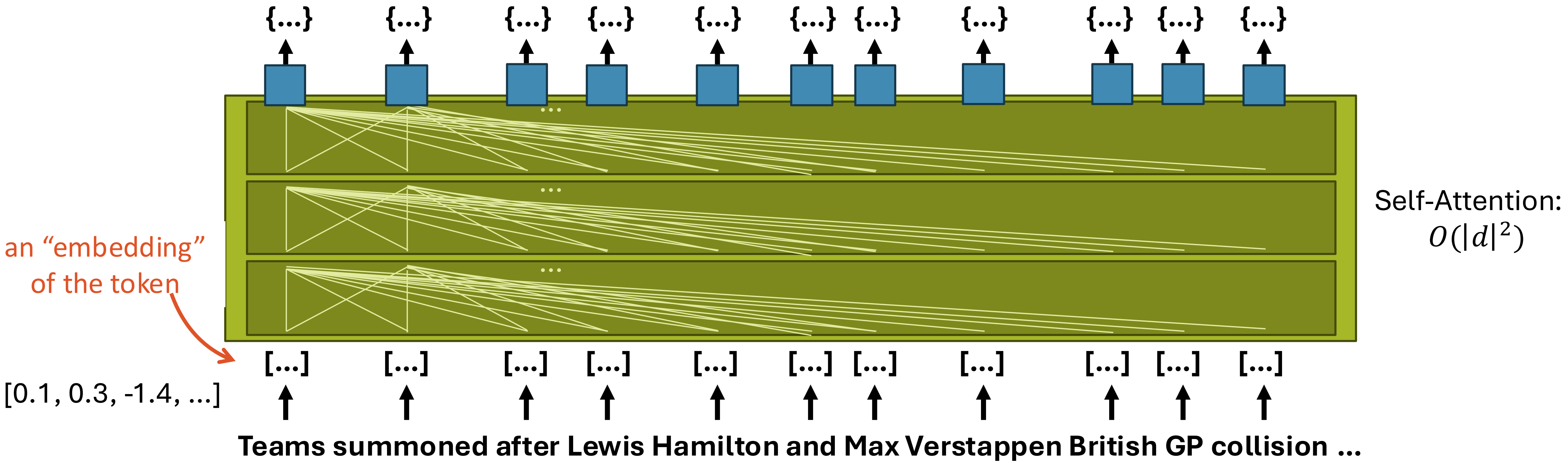
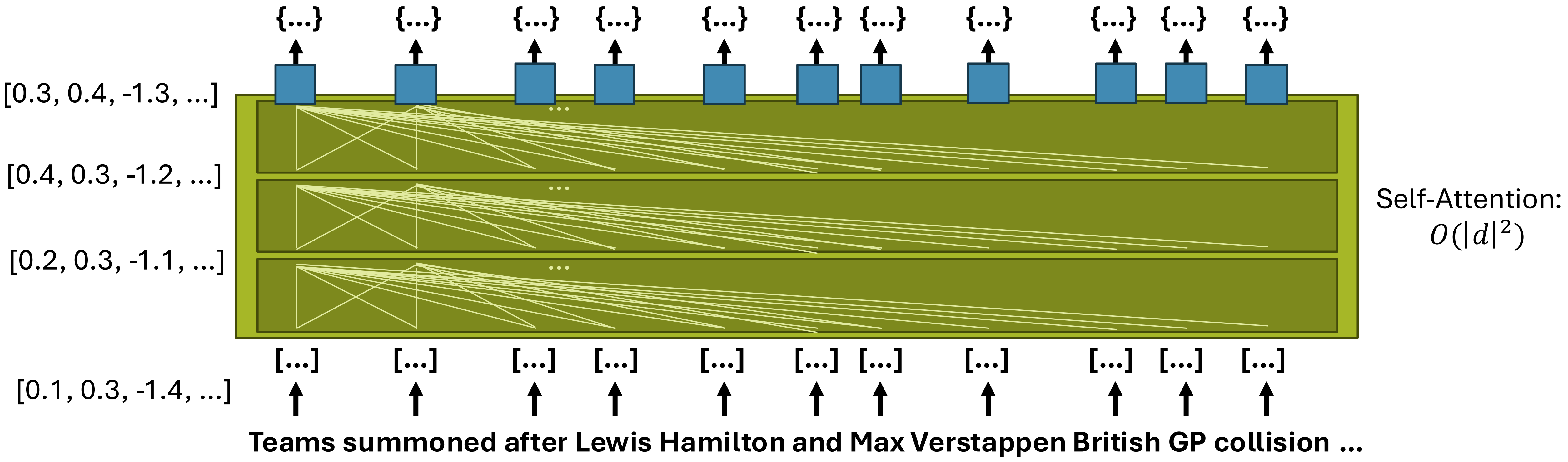


Crash Course: Transformer Language Models



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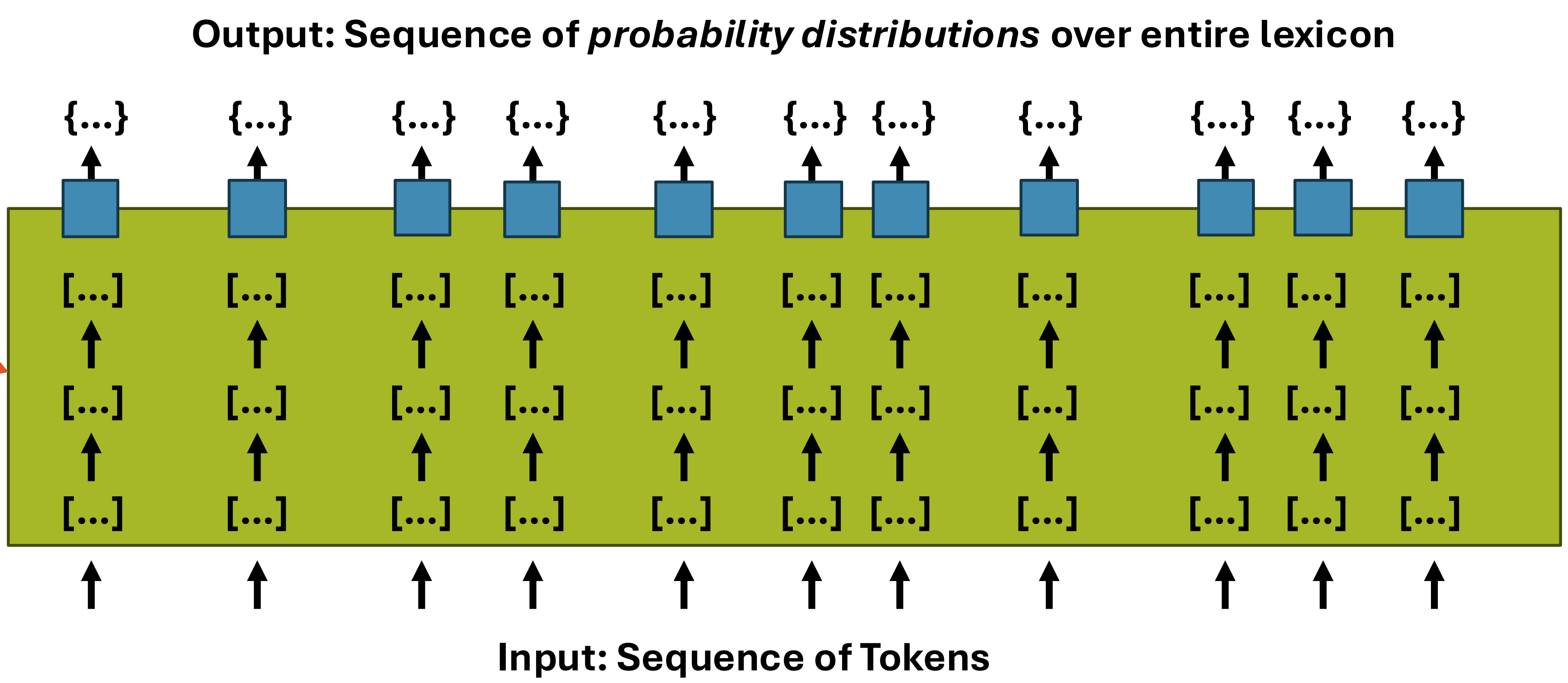
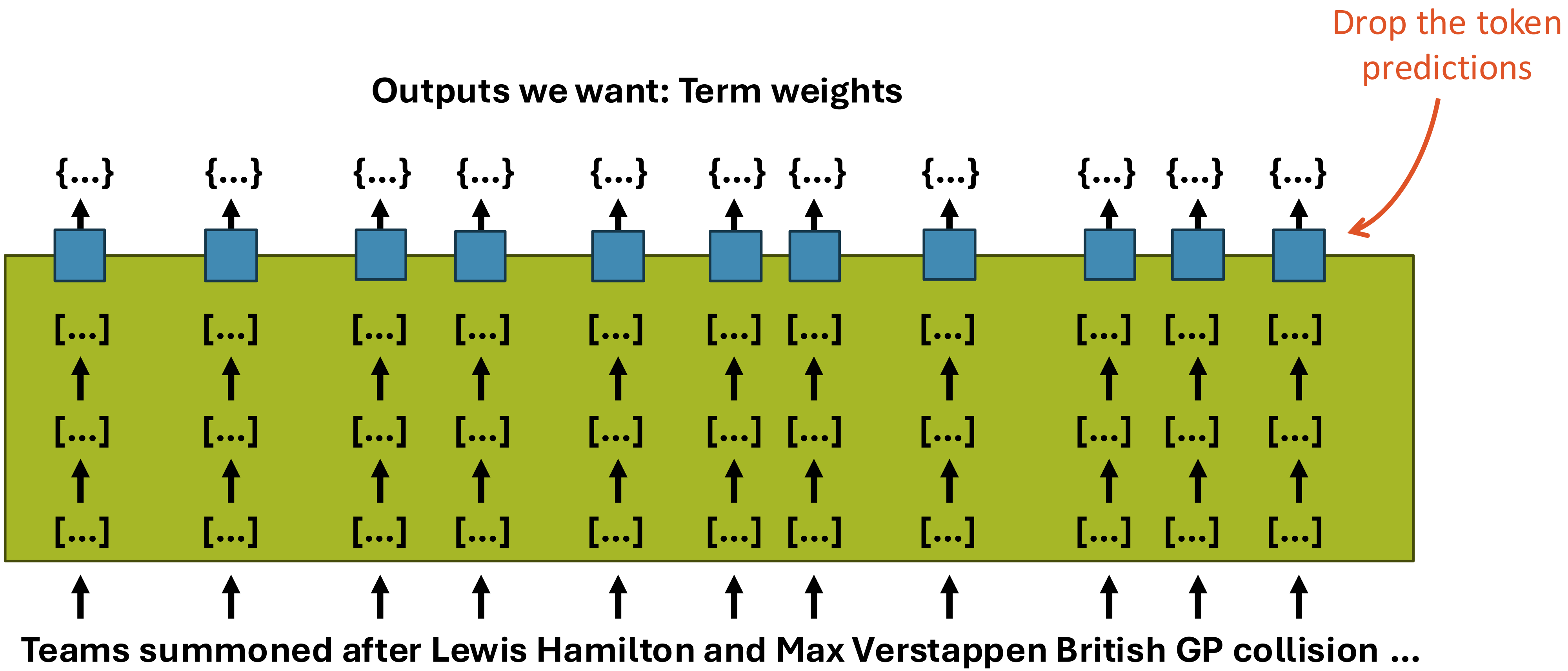


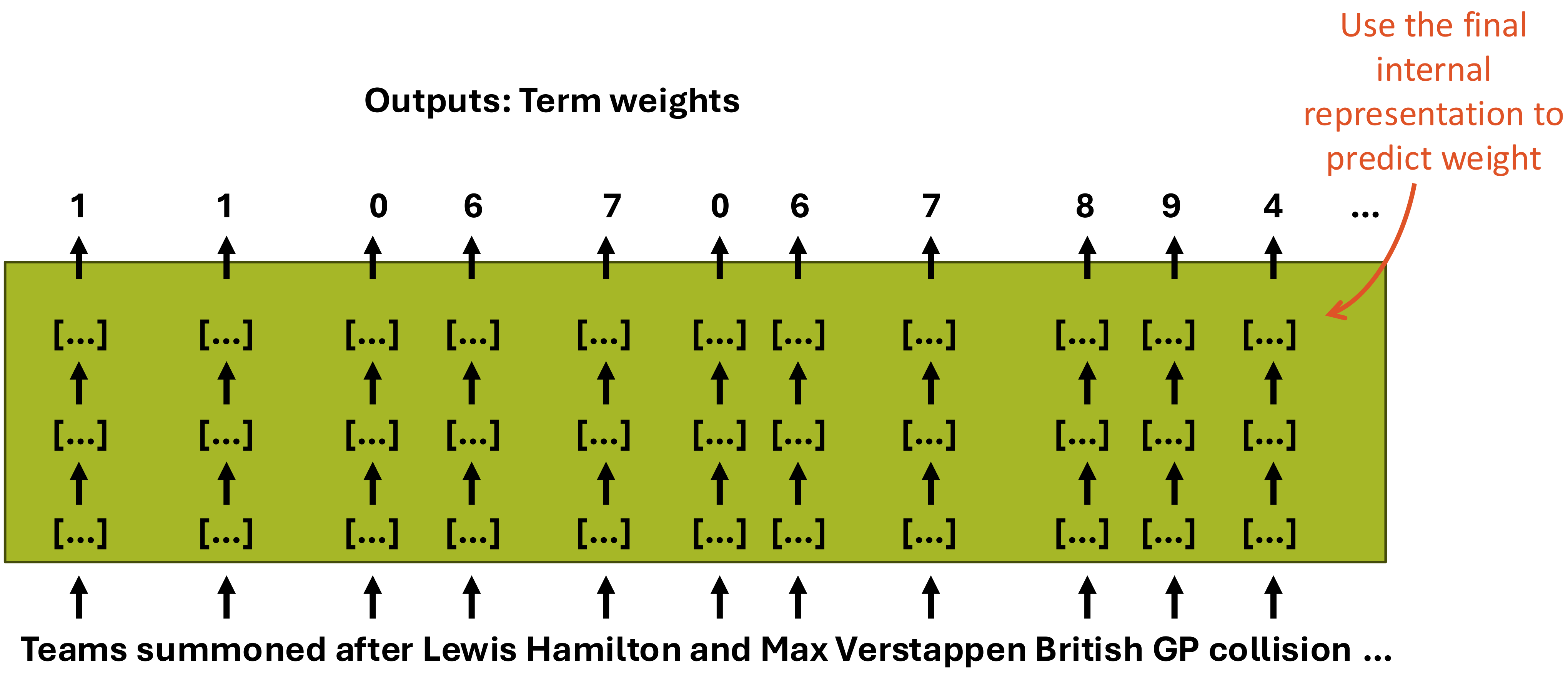
Many models like this: BERT, GPT, etc.

Alternative: Use Neural LM to Predict Term Weight



[DeepCT]

Alternative: Use Neural LM to Predict Term Weight



[DeepCT]

Problem with Lexical Signal: Mismatch

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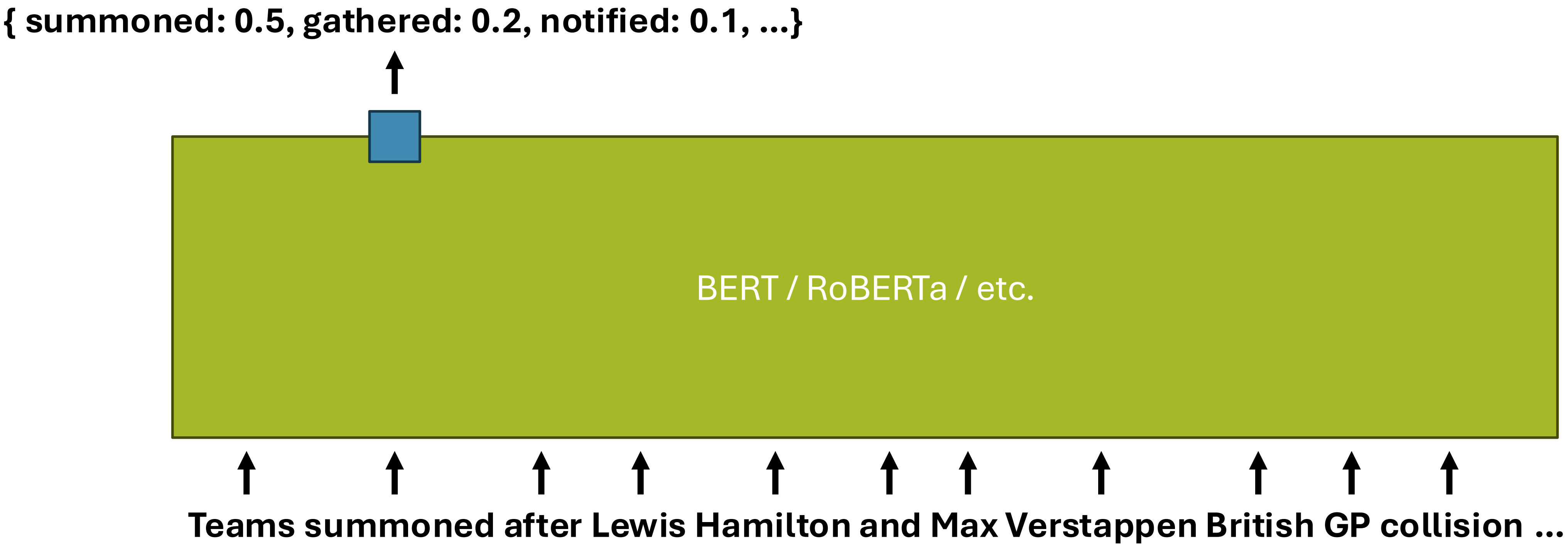
Source: https://www.bbc.co.uk/sport/formula1/57988419

LM Encoder

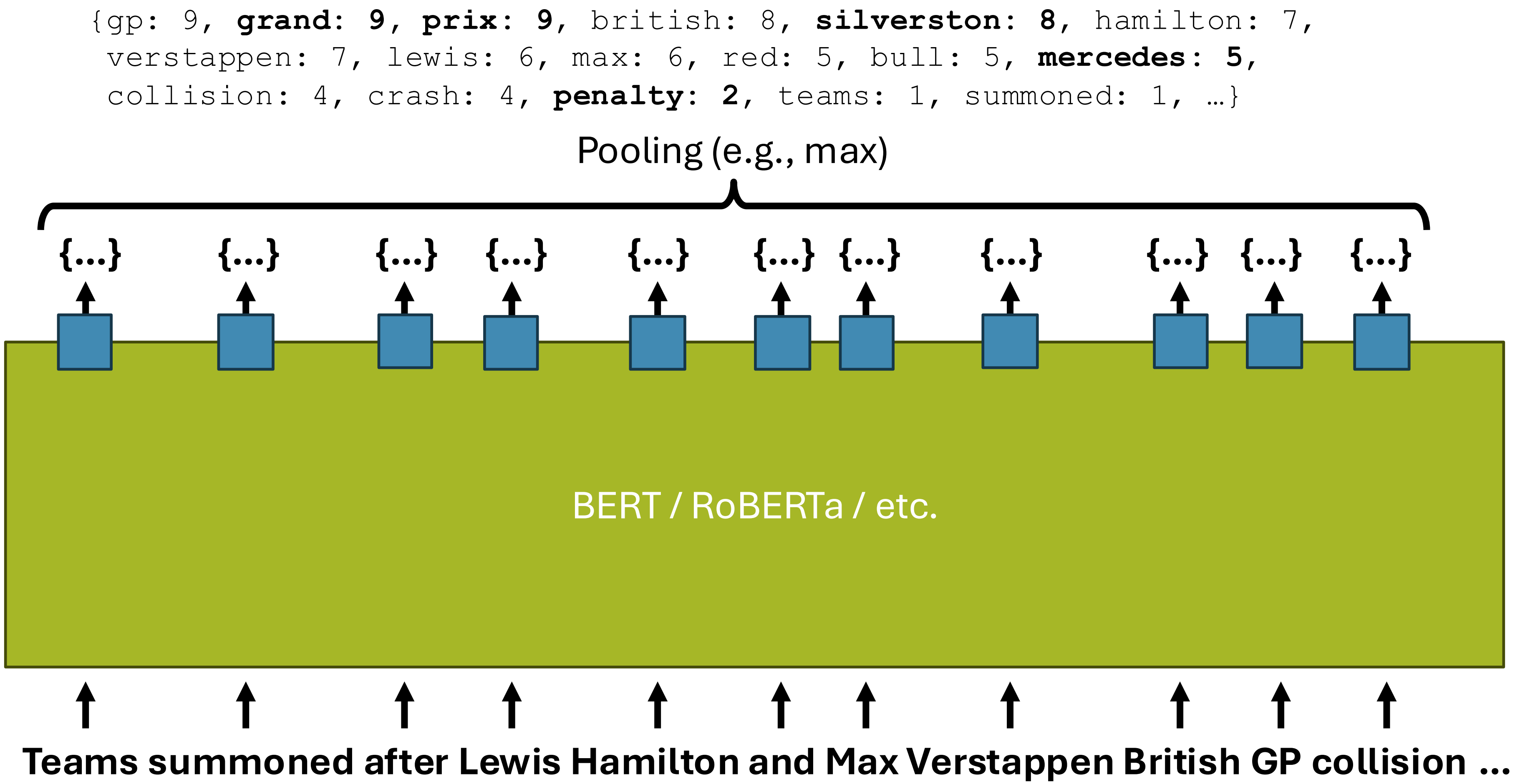
Lexical mismatch:
grand prix != gp

{**gp**: **9**, british: 8, hamilton: 7, verstappen: 7, lewis: 6, max: 6, red: 5, bull: 5, collision: 4, penalty: 2, teams: 1, summoned: 1, …}

We can use the token distributions to “expand”

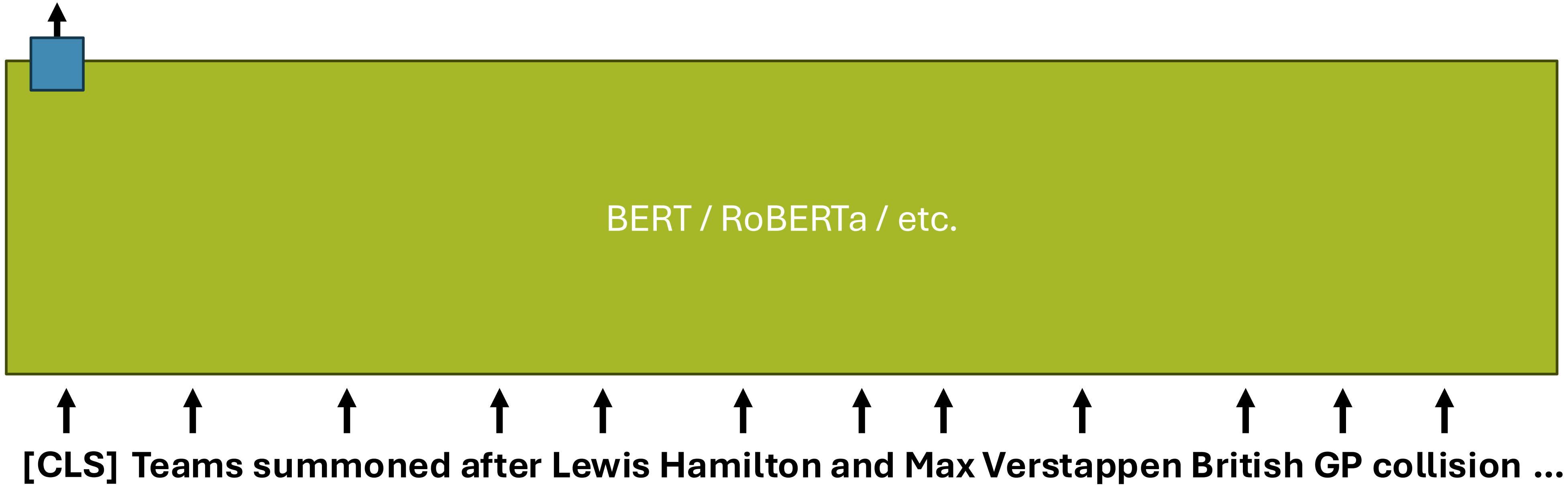


Per-token Language Modelling Expansion

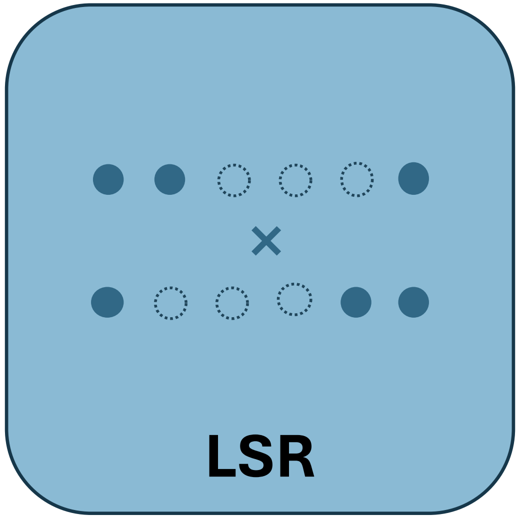


Single-token Language Modelling Expansion

{gp: 9, **grand: 9**, **prix: 9**, british: 8, **silverston: 8**, hamilton: 7,
verstappen: 7, lewis: 6, max: 6, red: 5, bull: 5, **mercedes: 5**,
collision: 4, crash: 4, **penalty: 2**, teams: 1, summoned: 1, …}



Learned Sparse Retrieval (LSR) Models



Representations learned and aligned to lexicon

Combination based on overlapping terms (potentially expanded)

Variations:

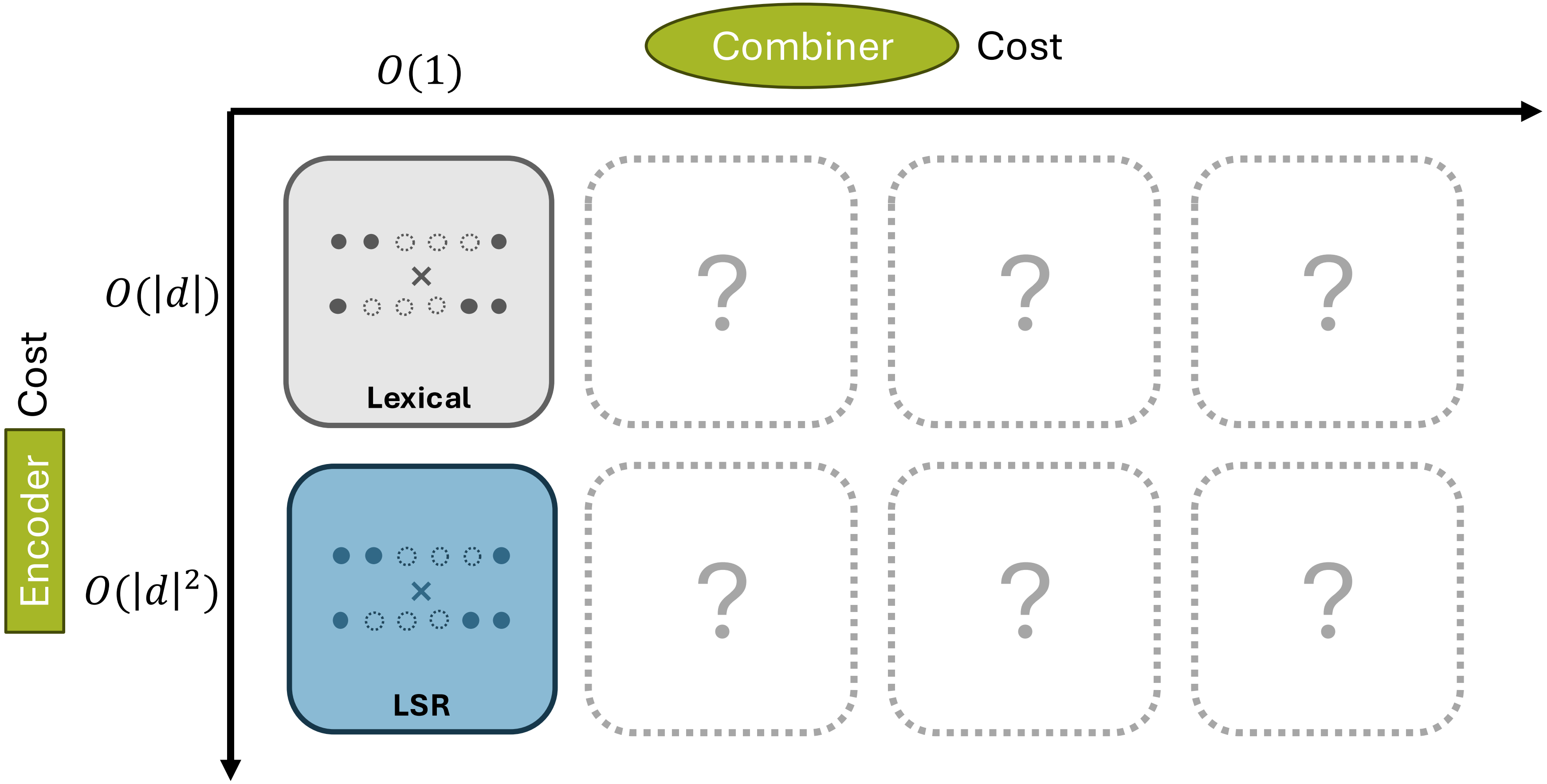
- Token weighting
- Expansion (per-token, single-token, external)
- Sparsification (top-k, regularization)
- ... Others too!

Common LSR models:

- DeepCT [DeepCT]
- EPIC [EPIC]
- SPLADE [SPLADE]
- TILDE [TILDE]
- ... Others too!

[LSR]

The Table of Relevance Est. Models



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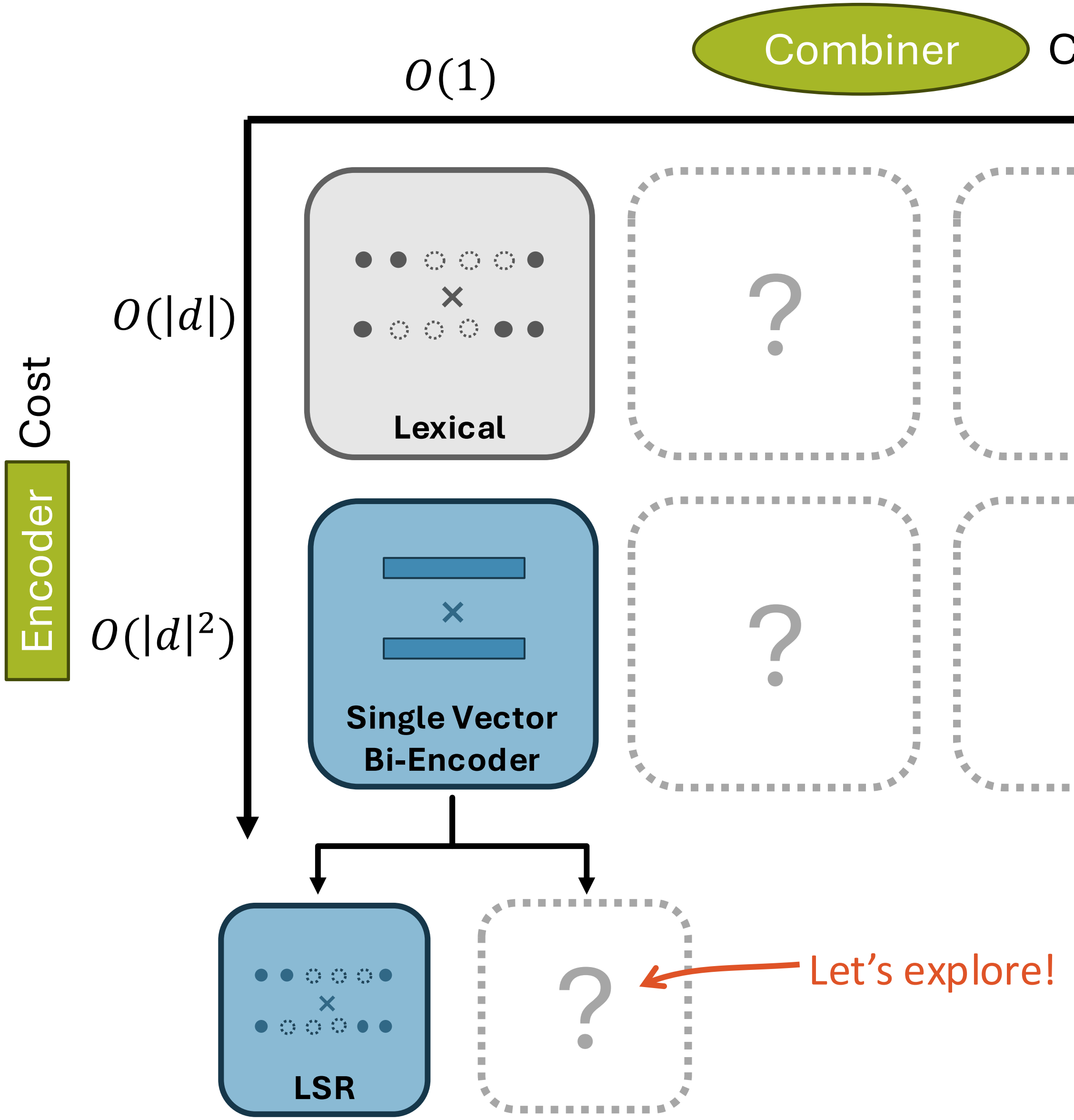

{gp: 9, grand: 9, prix: 9, british: 8, silverston: 8,
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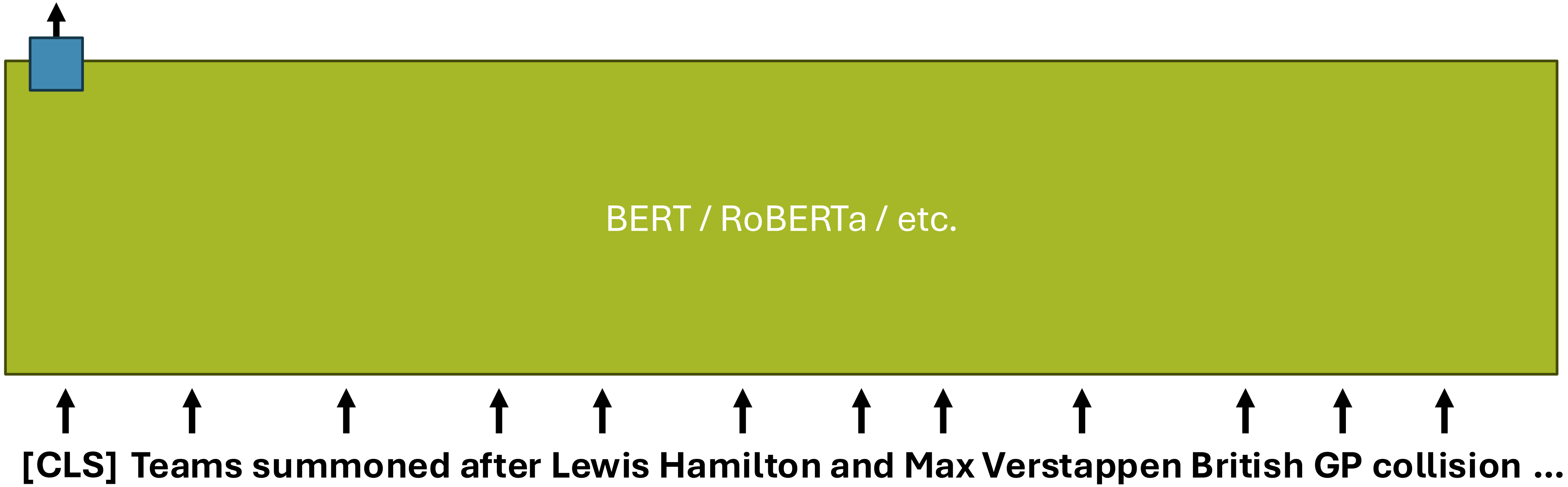

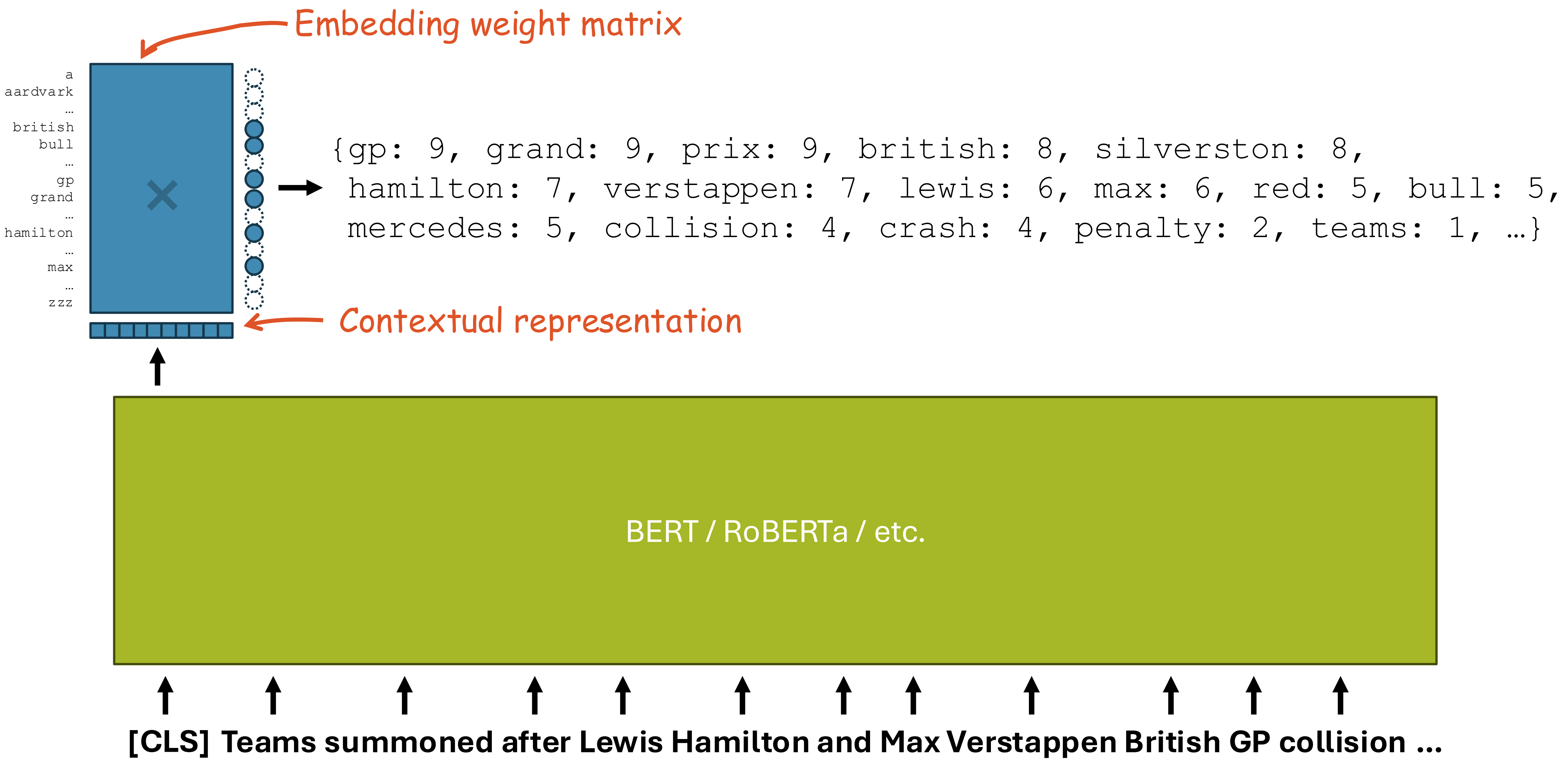
Embedding weight matrix
a
aardvark
…
british
bull
…
gp
grand
…
hamilton
…
max
…
zzz
×
{gp: 9, grand: 9, prix: 9, british: 8, silverston: 8,
hamilton: 7, verstappen: 7, lewis: 6, max: 6, red: 5, bull: 5,
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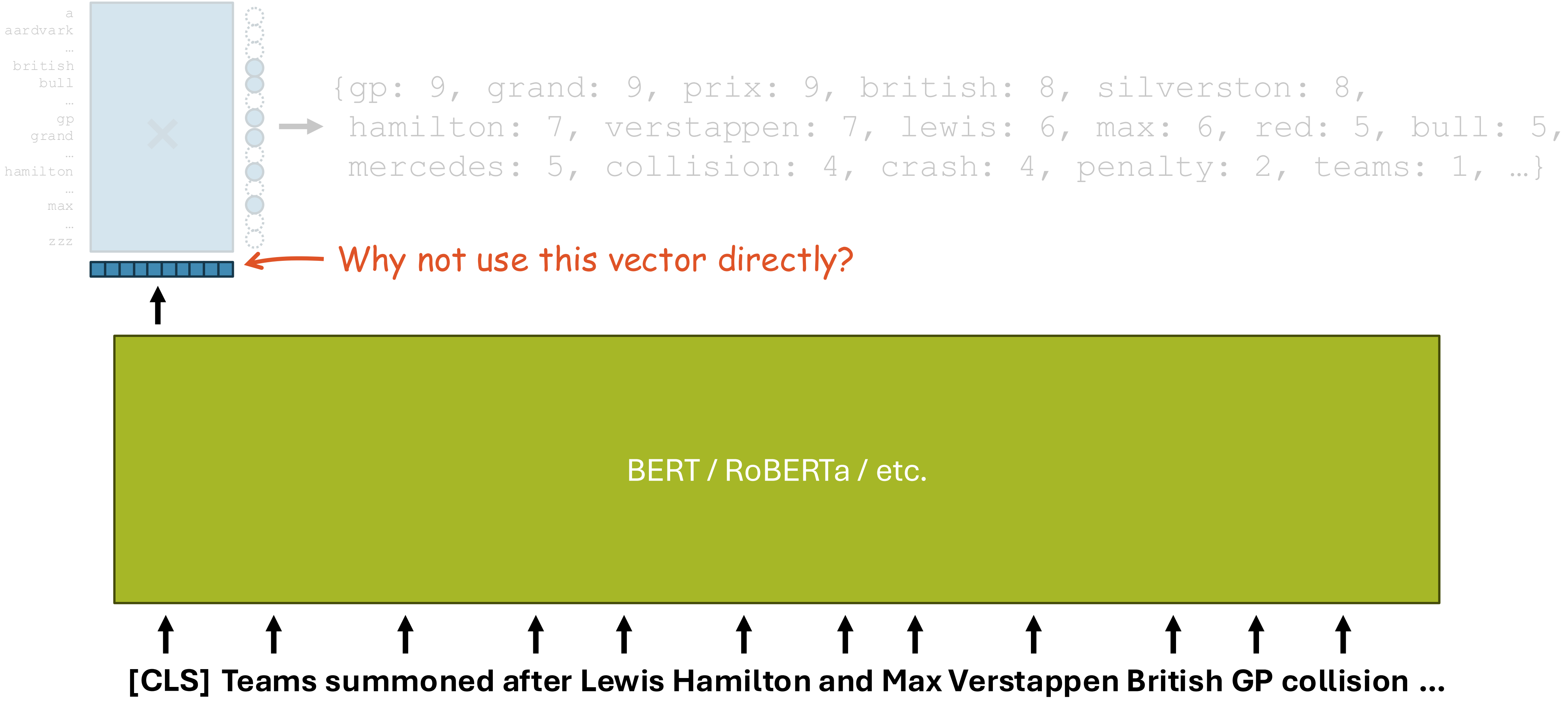
Contextual representation
BERT / RoBERTa / etc.
[CLS] Teams summoned after Lewis Hamilton and Max Verstappen British GP collision ...


a
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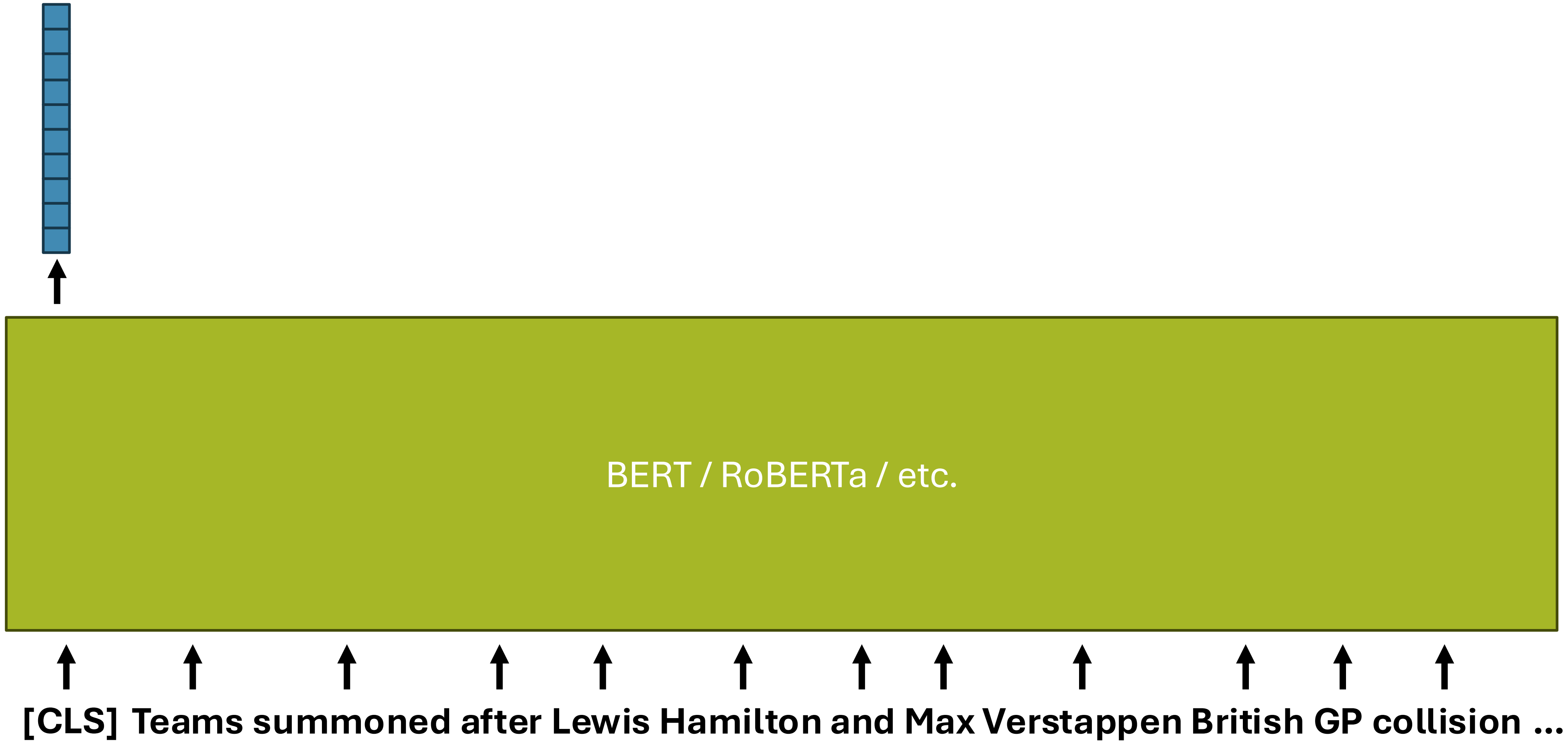
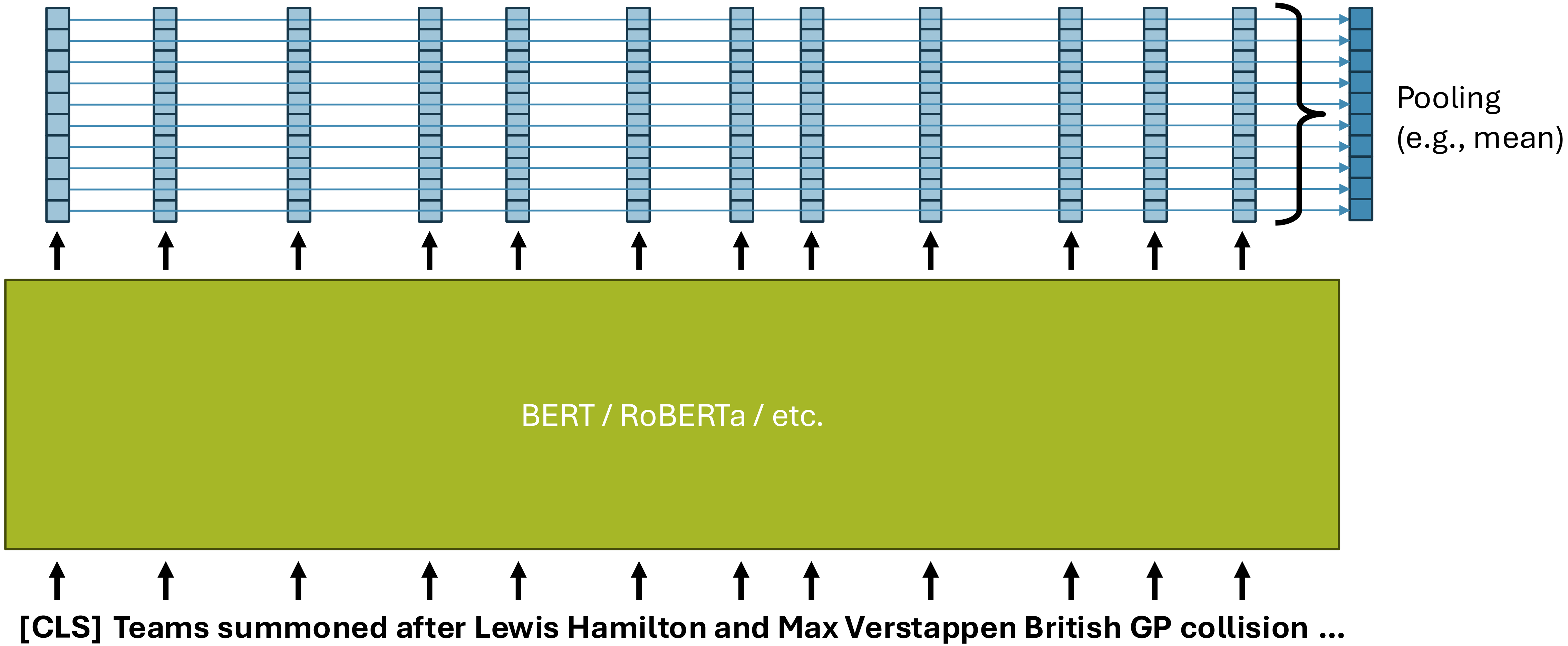
Why not use this vector directly?
BERT / RoBERTa / etc.
[CLS] Teams summoned after Lewis Hamilton and Max Verstappen British GP collision ...

Single-Vector Dense Model: [CLS]



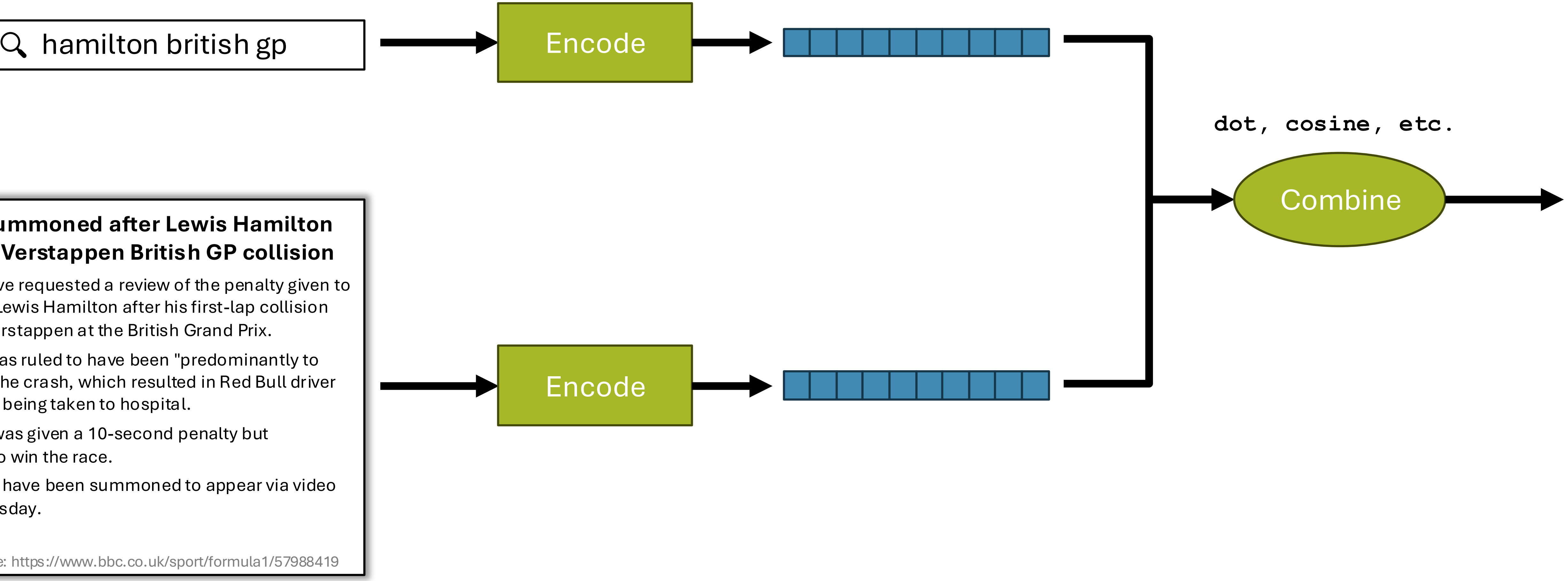
[DPR]

Single-Vector Dense Model: Pooling

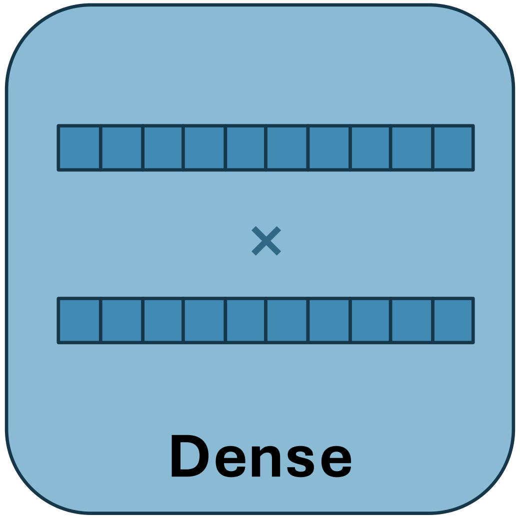


[TCT]

Single-Vector Dense Model



Single-Vector Dense Model



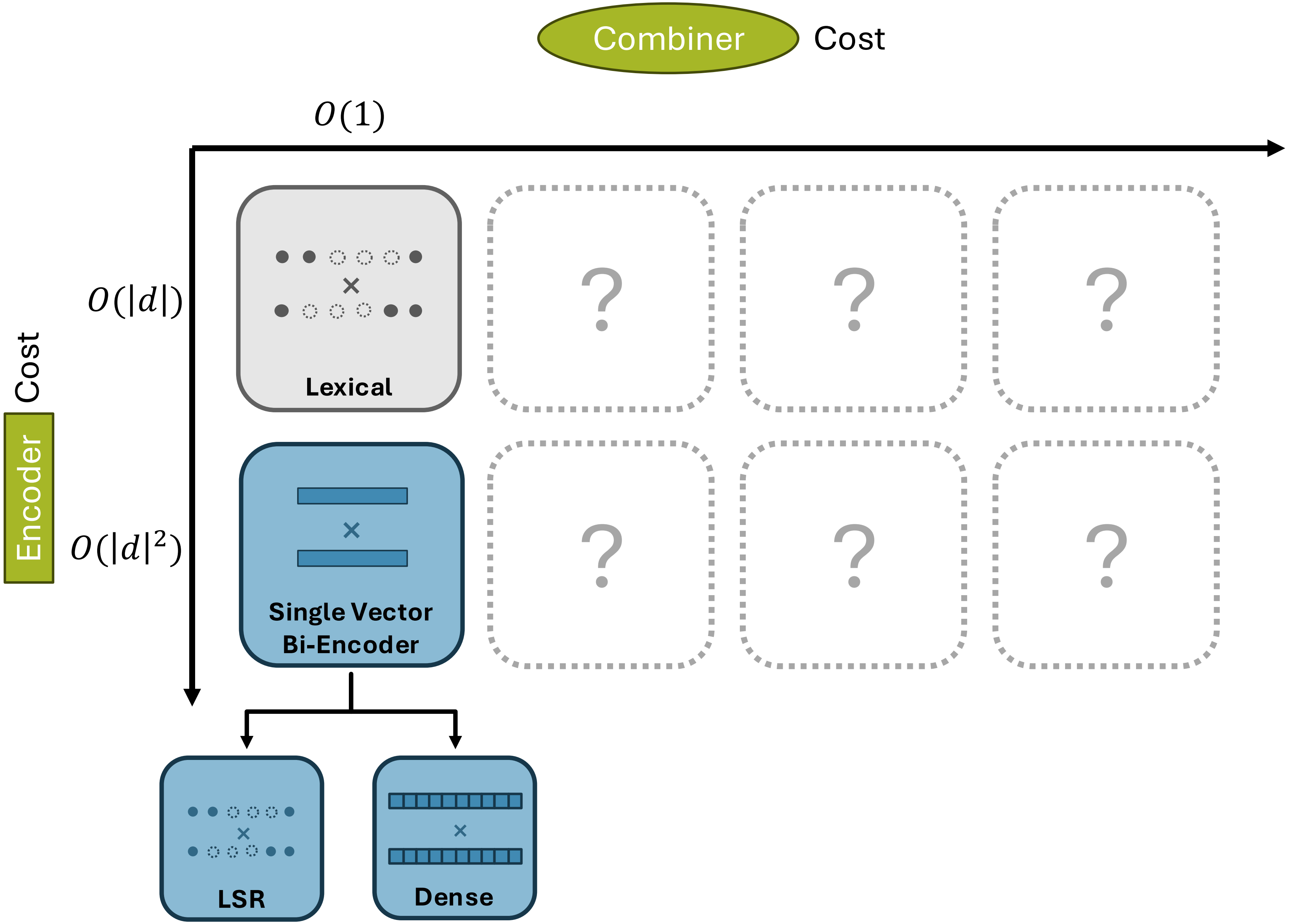
Representations learned as fixed-size dense vectors
(usually ~500-2000 dimensions)

Combination as similarity or distance function between vectors

Variations:
- Pooling: CLS vs mean token vector
- Combination: dot product (common) vs cosine vs Euclidean
- Symmetric vs Asymmetric Encoders

Common Single-Vector Dense Models:
- DPR [DPR]
- Matryoshka [Matryoshka]
- (Lots more, but they differ in training procedures, not architectures)

The Table of Relevance Est. Models



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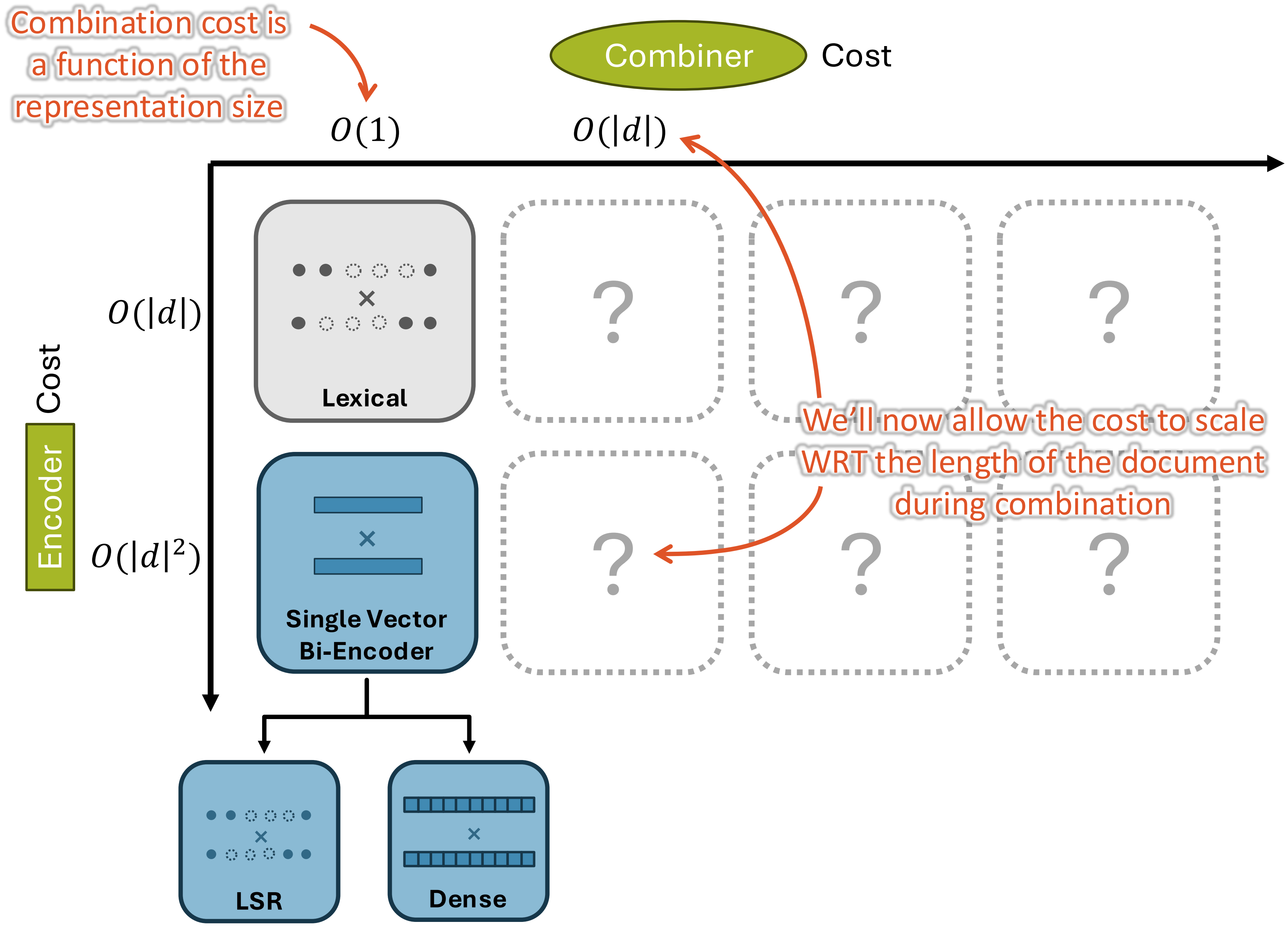
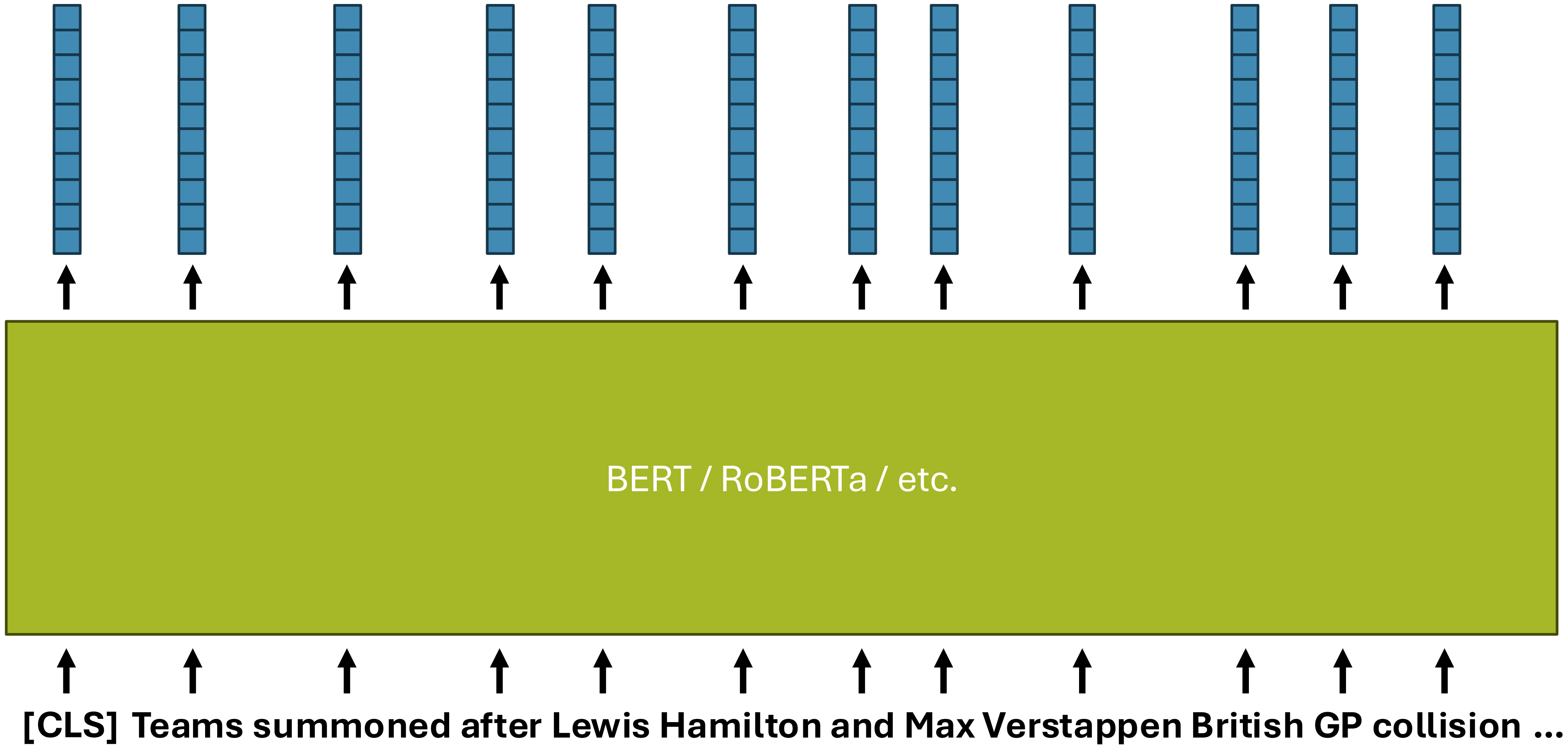


Multi-Vector Encoders



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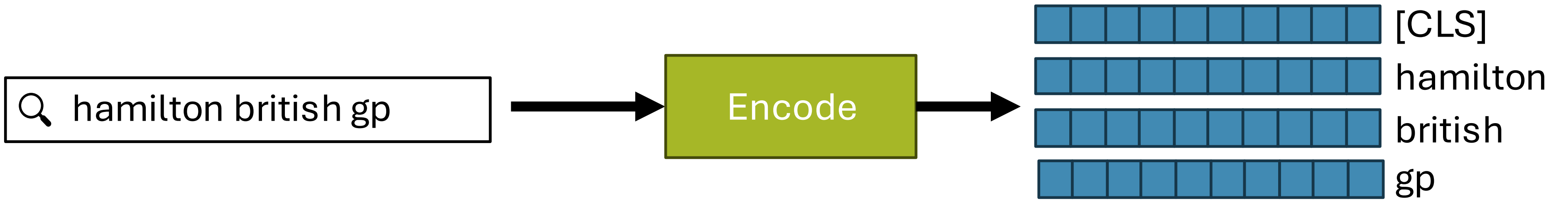

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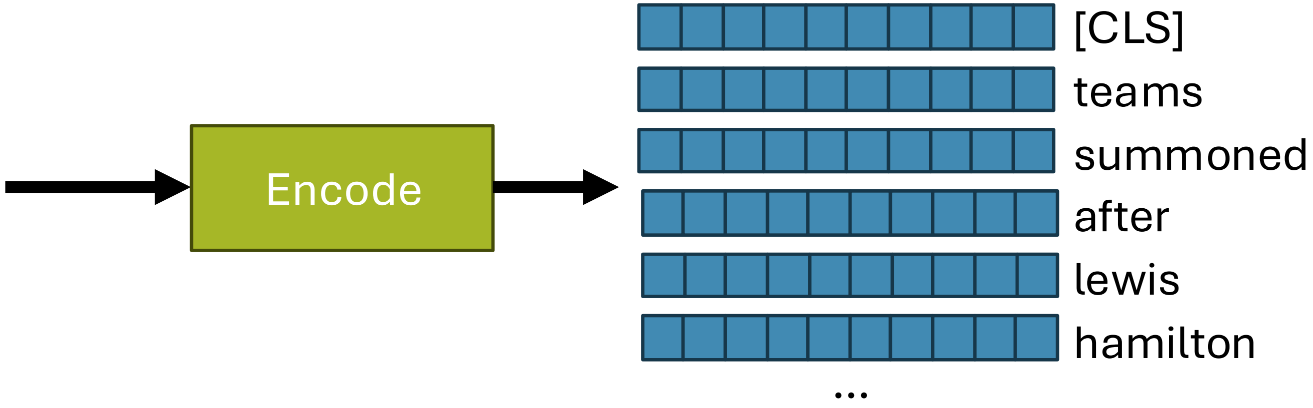
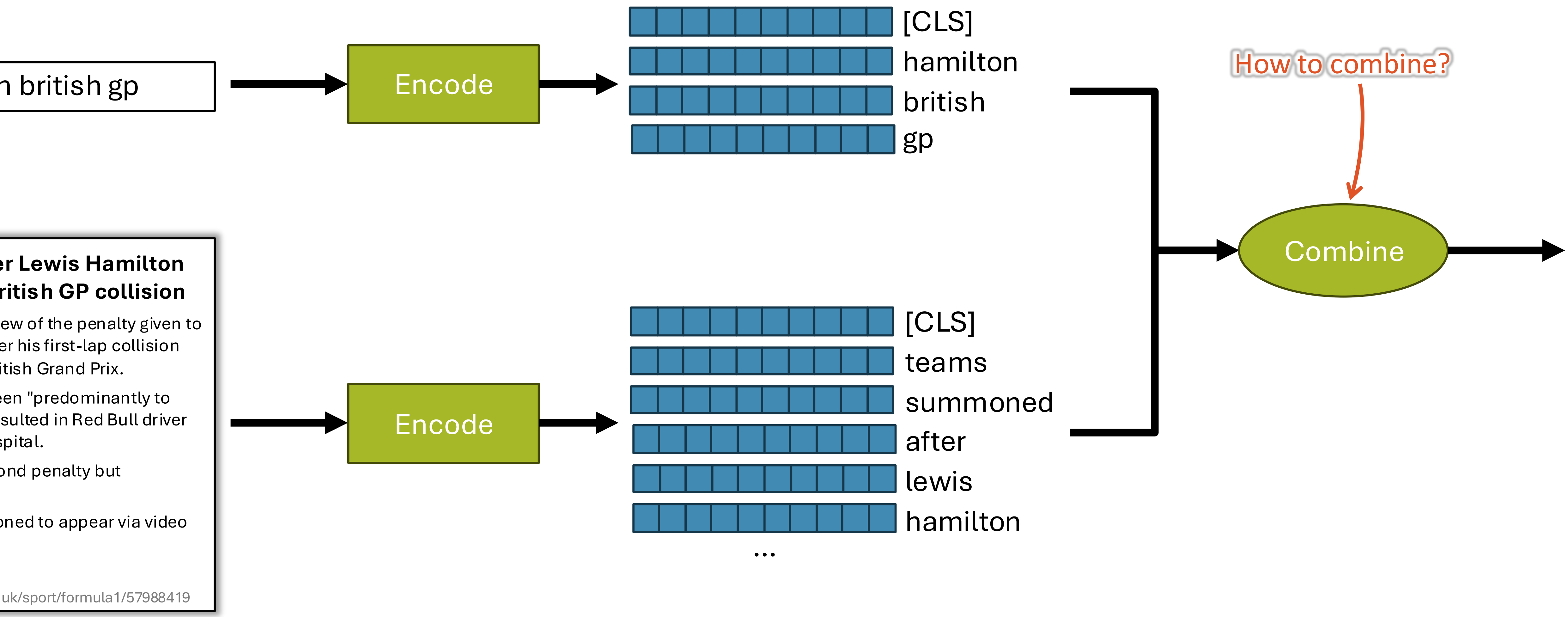
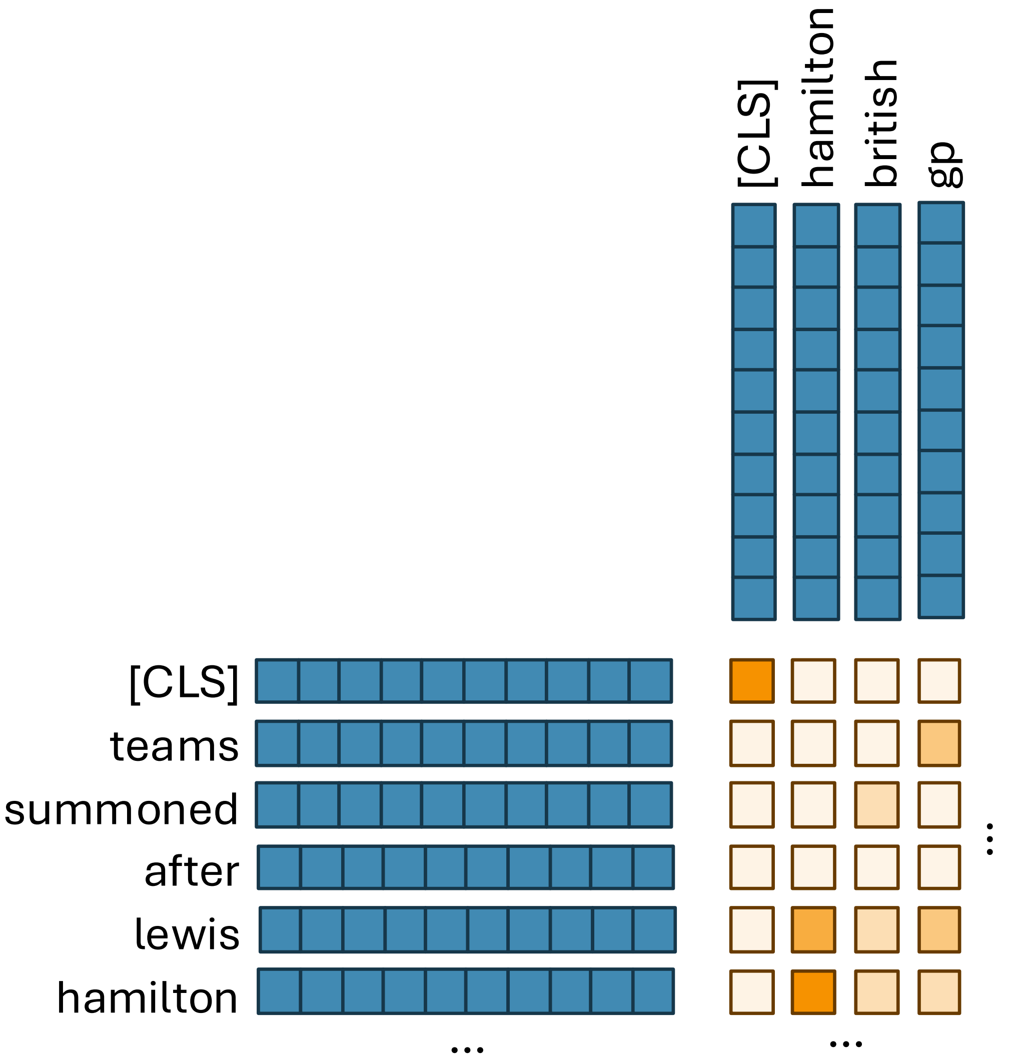


Multi-Vector Combination



Interaction: Similarity between each vector



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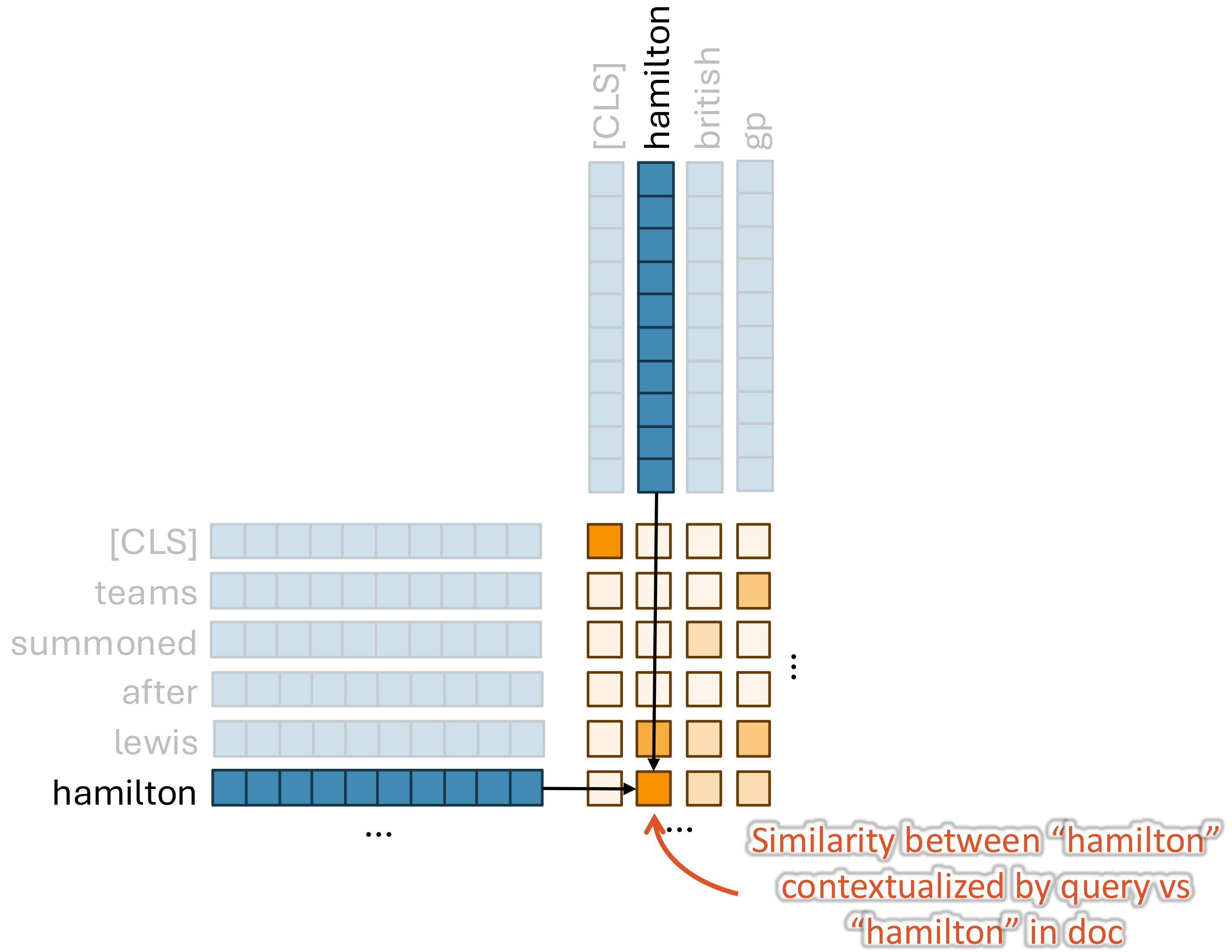

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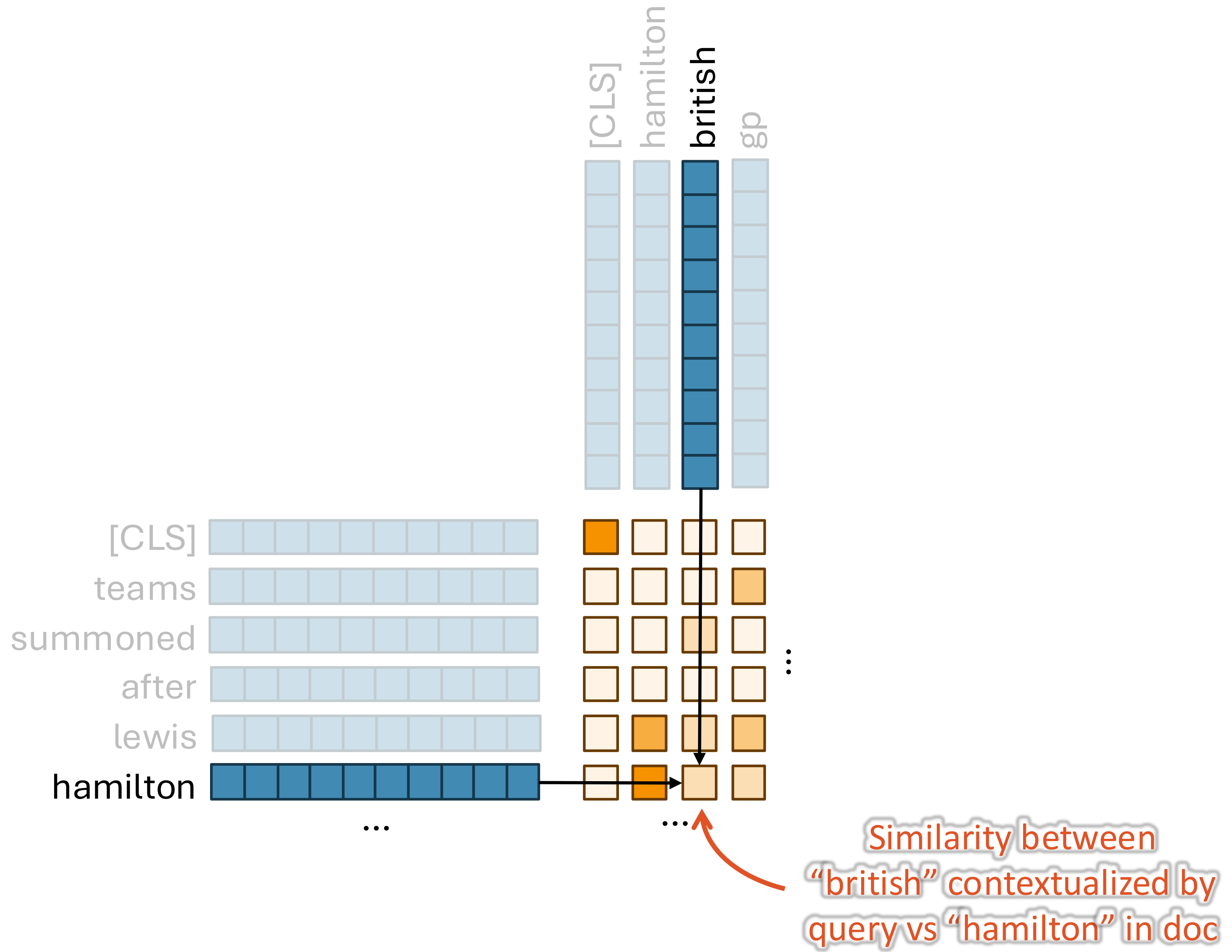
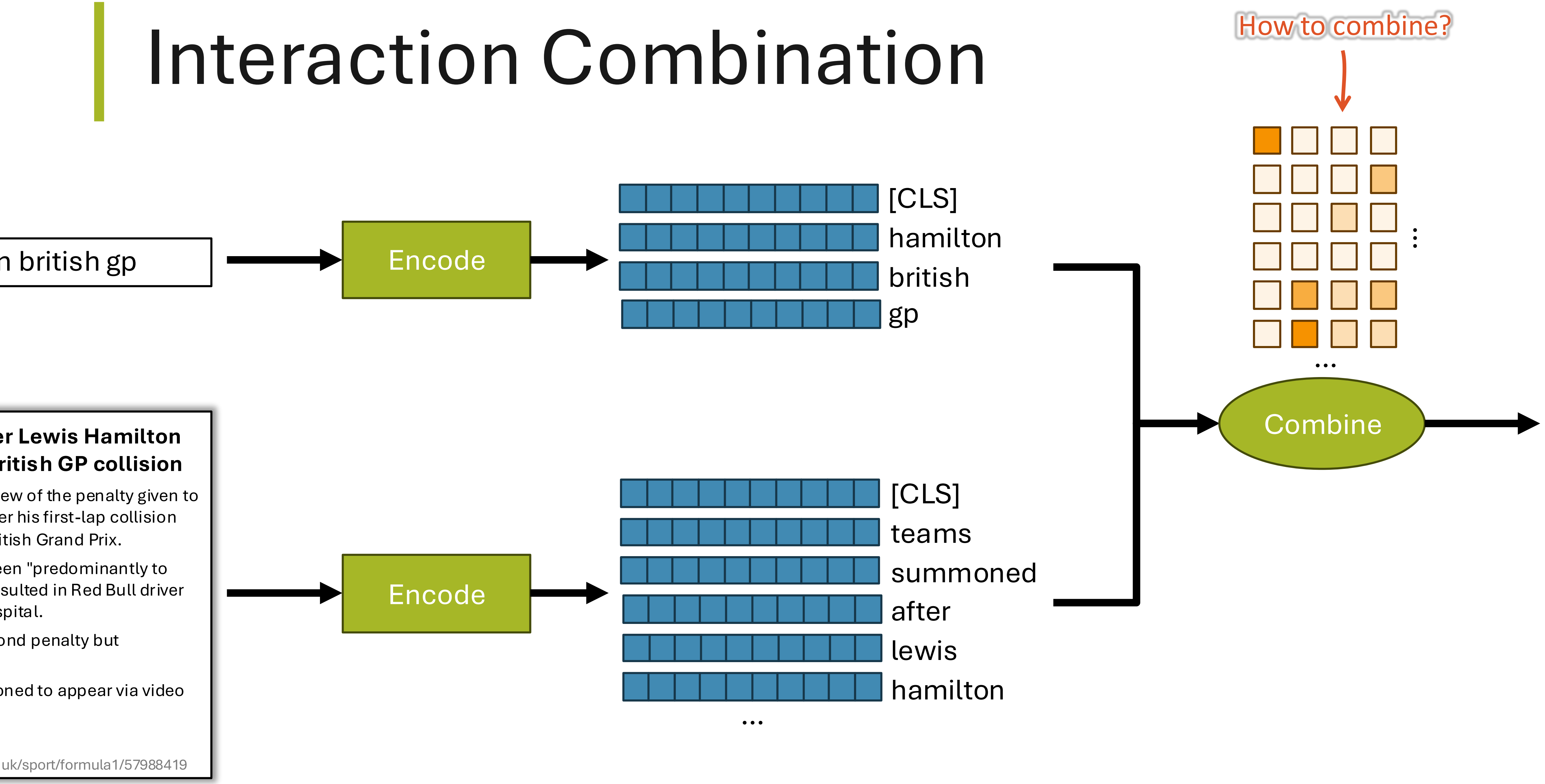
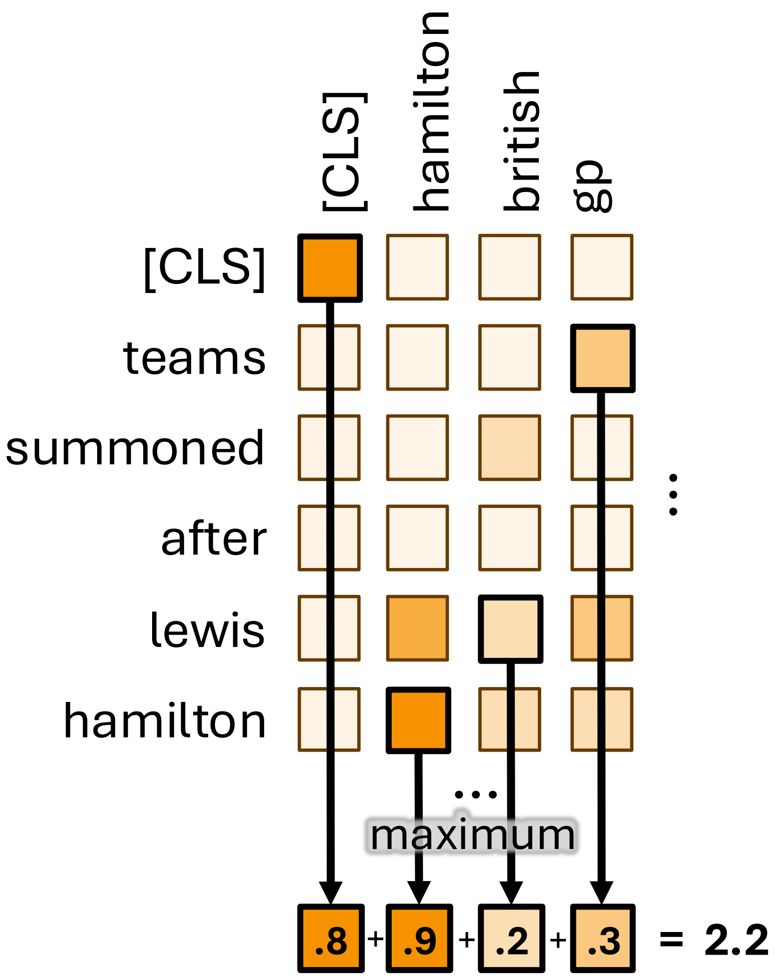


Interaction Combination



Simple but effective: Sum the Maximum by query term
(MaxSim)



$$\text{MaxSim} = \sum_{q_i \in Q} \max_{d_i \in D} sim(q_i, d_i)$$

[ColBERT]

Simple but effective: Sum the Maximum by query term
(MaxSim)

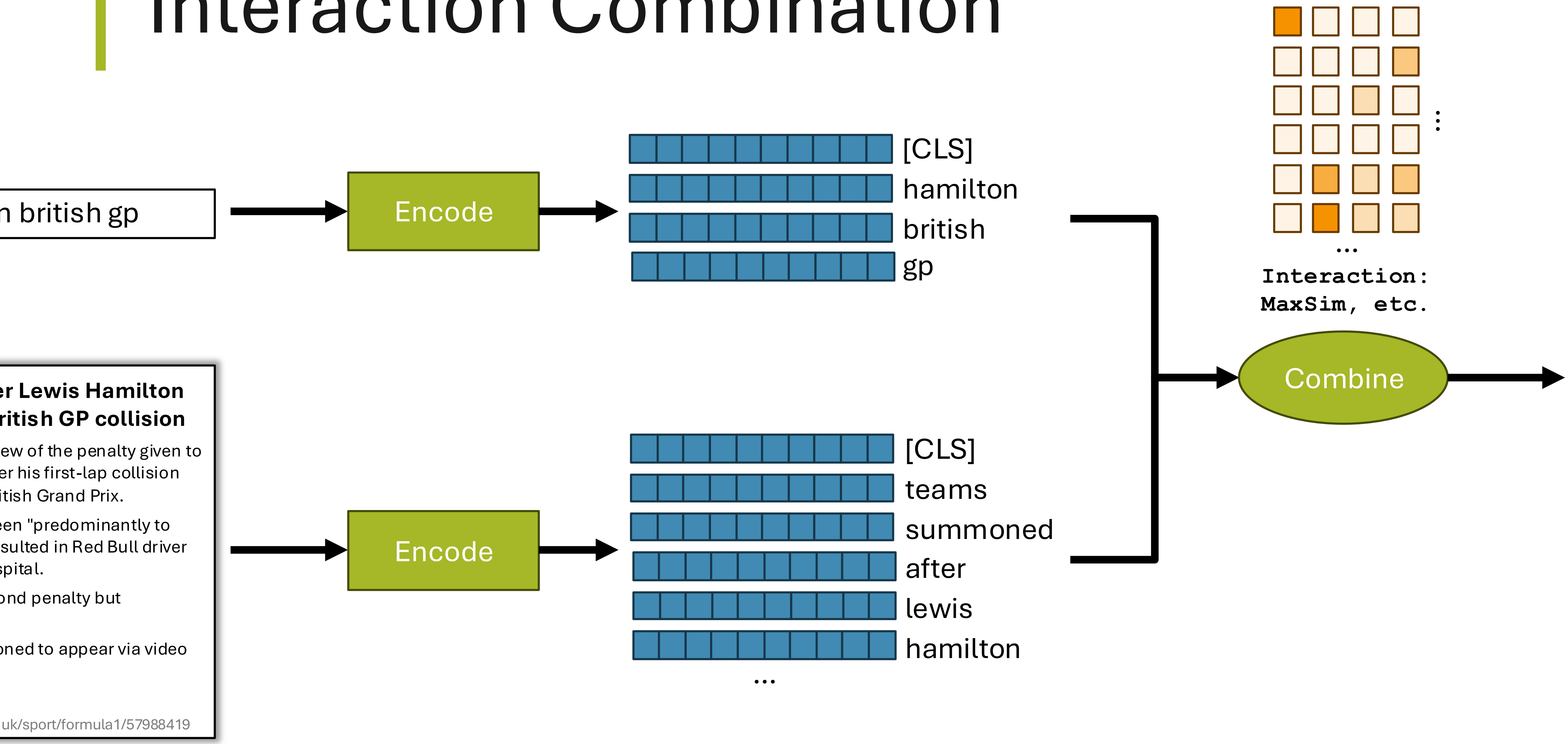
	q_1	q_2	q_3	q_4
d_1	0.2	0.5	0.1	0.3
d_2	0.4	0.1	0.8	0.2
d_3	0.3	0.2	0.4	0.5
d_4	0.1	0.6	0.2	0.4
d_5	0.5	0.3	0.3	0.1
d_6	0.2	0.4	0.5	0.7

$$\text{MaxSim} = \sum_{q_i \in Q} \max_{d_i \in D} sim(q_i, d_i)$$

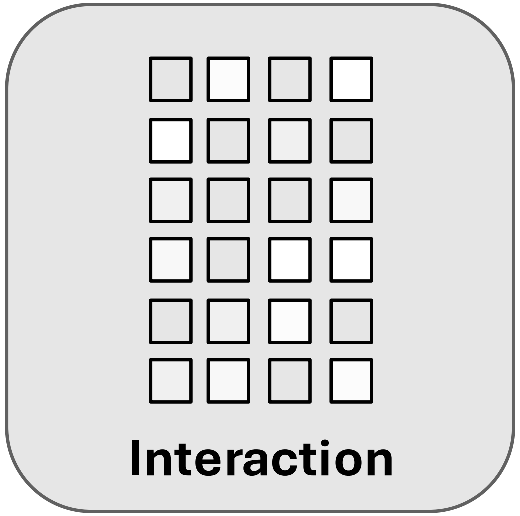
MaxSim = max(0.2, 0.4, 0.3, 0.1, 0.5, 0.2) +
max(0.5, 0.1, 0.2, 0.6, 0.3, 0.4) +
max(0.1, 0.8, 0.4, 0.2, 0.3, 0.5) +
max(0.3, 0.2, 0.5, 0.4, 0.1, 0.7)
= 0.5 + 0.6 + 0.8 + 0.7
= 2.6

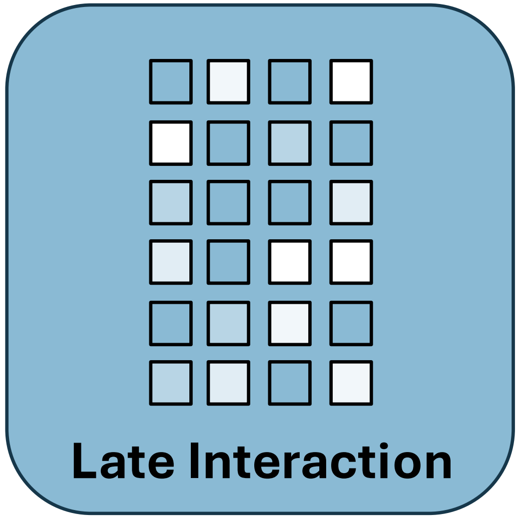
[ColBERT]

Interaction Combination



(Late) Interaction Models





Lightweight "interaction" combination over per-token query/doc vectors

Encoders either static or contextualized ("late" interaction)

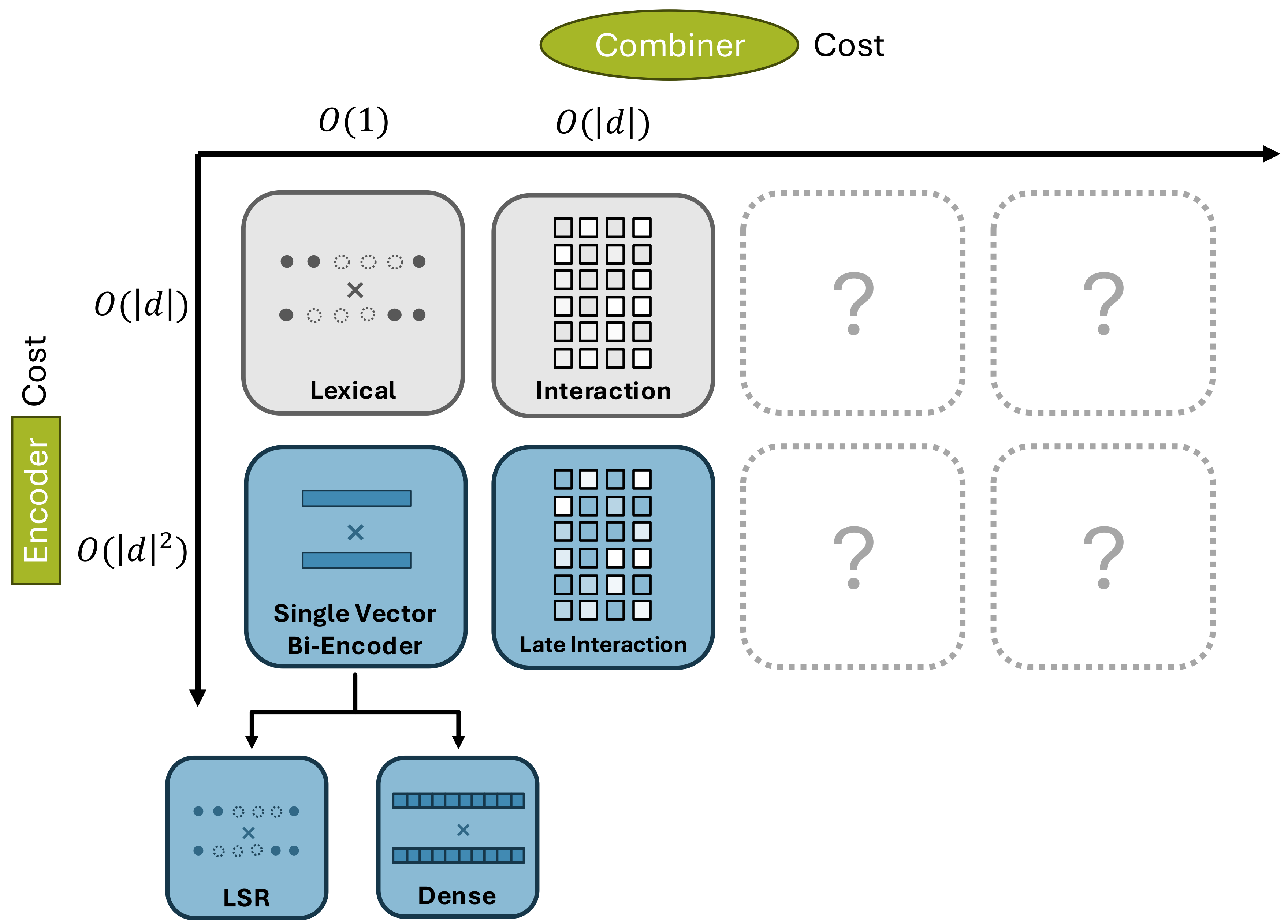
Variations:

Combination function: MaxSim (common) vs kernel pooling vs ...
Encoding strategy: LM (Late Interaction), Static (Interaction)

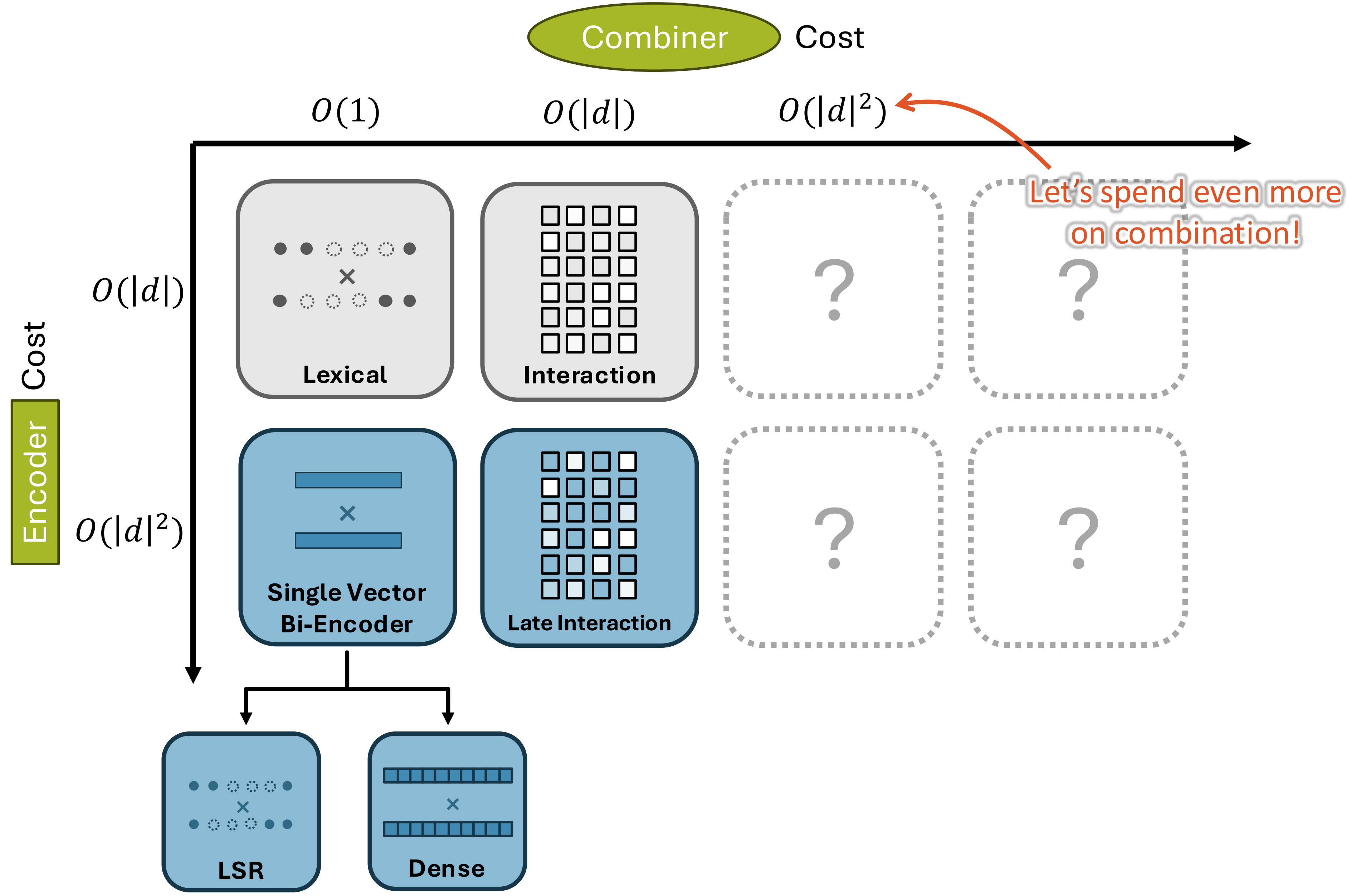
Common Interaction and Late Interaction models:

ColBERT [ColBERT]
DRMM [DRMM]
KNRM [KNRM]
PACRR [PACRR]
CEDR [CEDR]
Others...

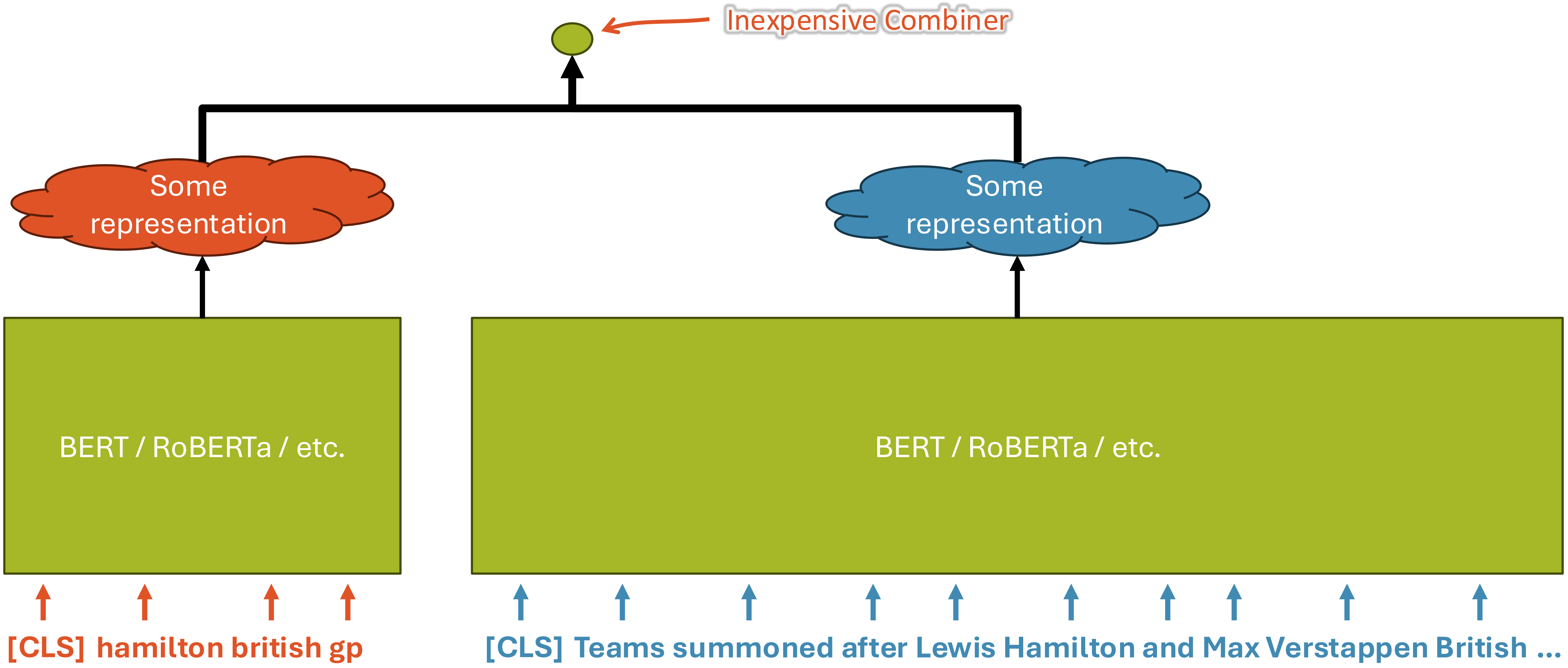
The Table of Relevance Est. Models



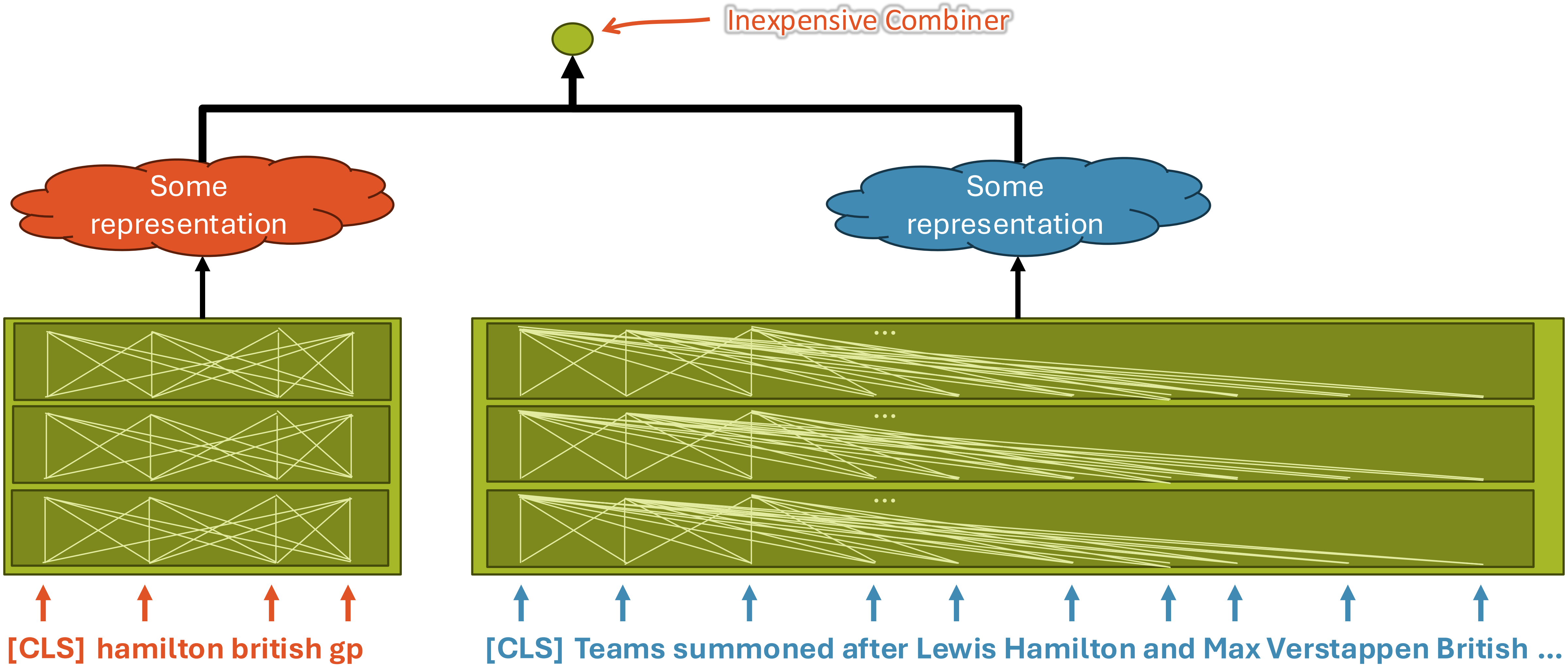
The Table of Relevance Est. Models



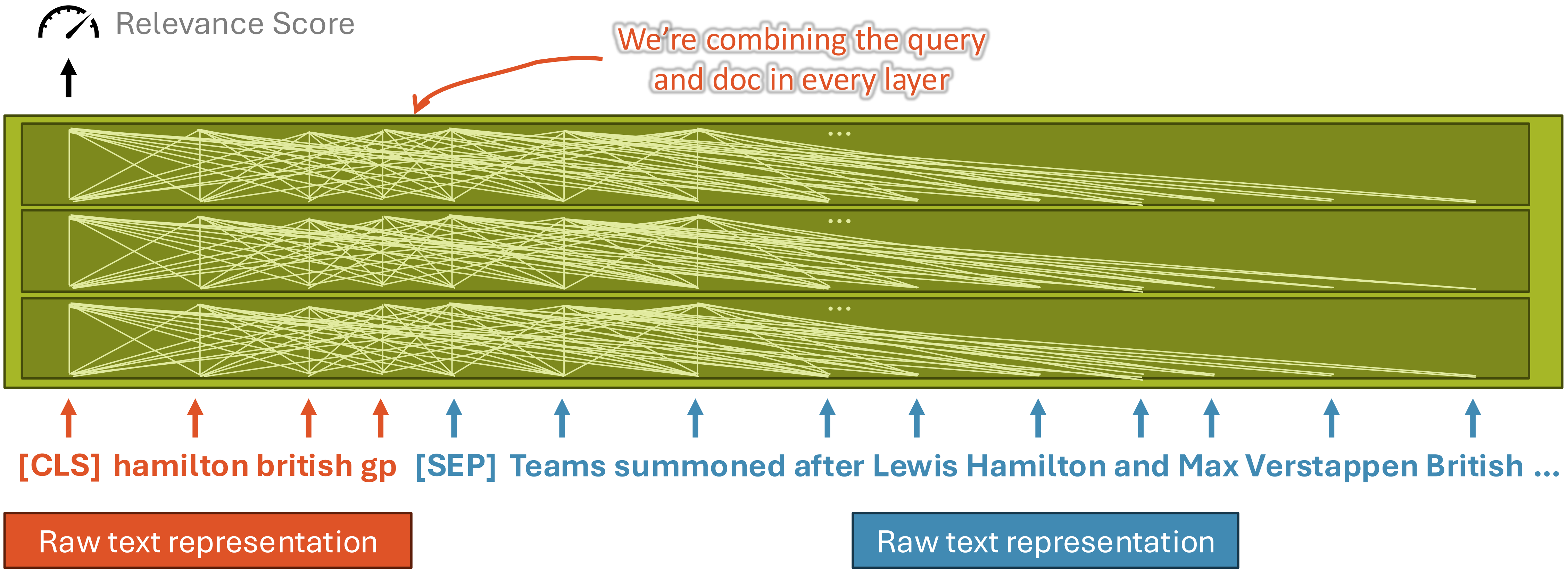
Up to now...


Inexpensive Combiner
Some representation
Some representation
BERT / RoBERTa / etc.
BERT / RoBERTa / etc.
[CLS] hamilton british gp
[CLS] Teams summoned after Lewis Hamilton and Max Verstappen British ...

Up to now...

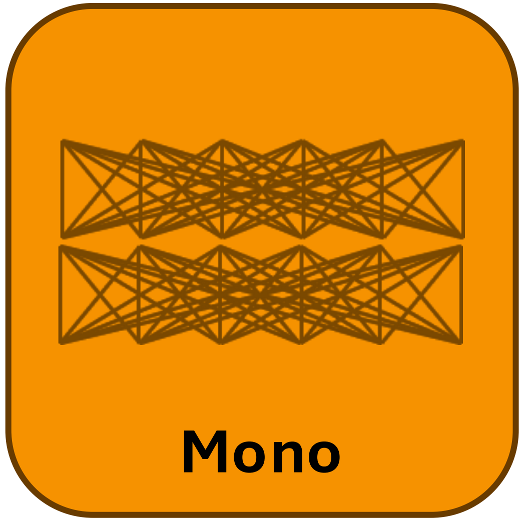

Inexpensive Combiner
Some representation
Some representation
...
...
...
[CLS] hamilton british gp
[CLS] Teams summoned after Lewis Hamilton and Max Verstappen British ...

Cross Encoder Combination



[BertRR]

Mono Relevance Est. Models



Uses cross attention between query and document for combination

Demo! https://huggingface.co/spaces/terrierteam/monot5

Minimal separate representations (i.e. just tokenization)

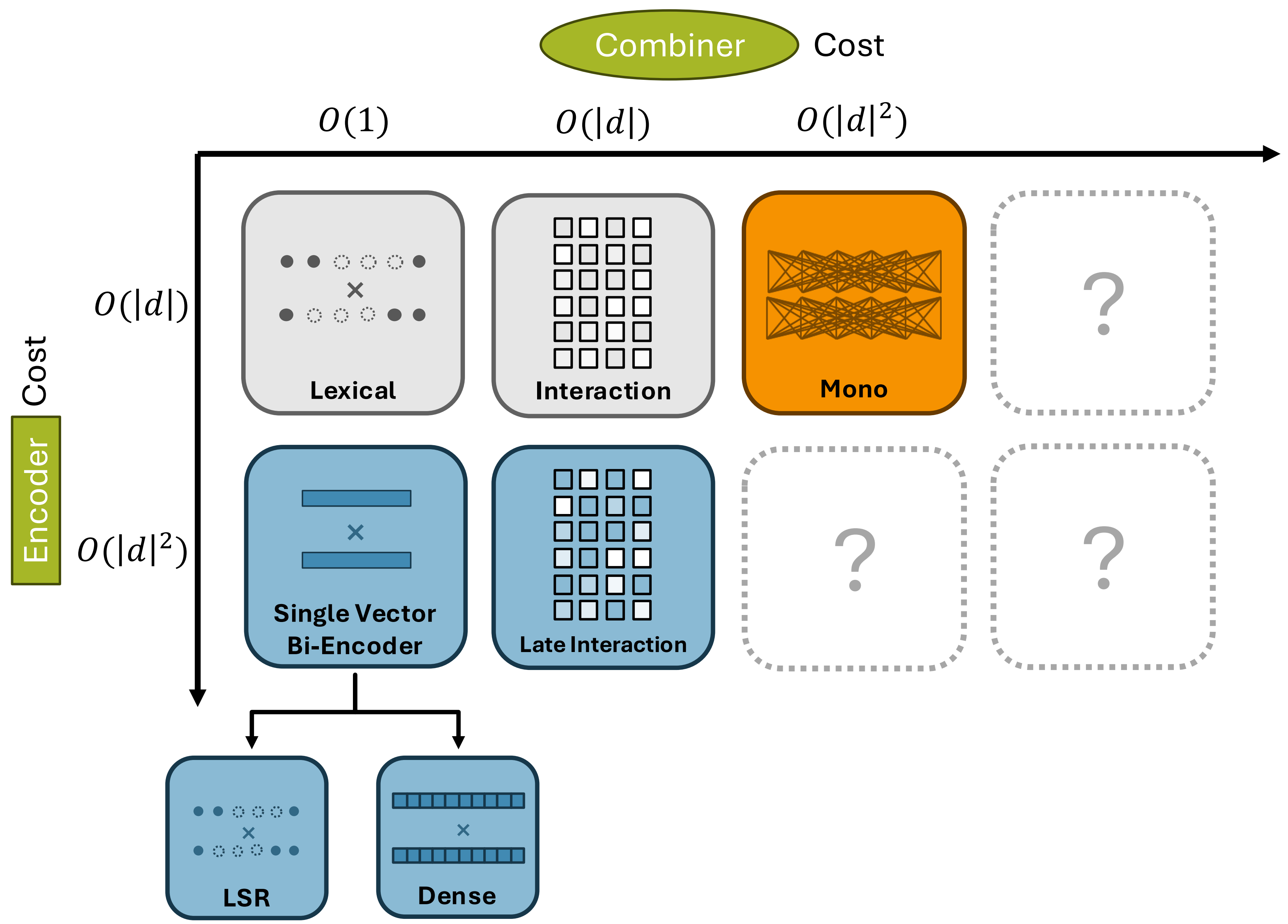
Variations:

- Architecture: Encoder Only, Encoder-Decoder, Decoder Only
- Attention Pruning: Skip some relations `[SparseAttn]`
- Model Size: Shallow networks `[ShallowCE]`

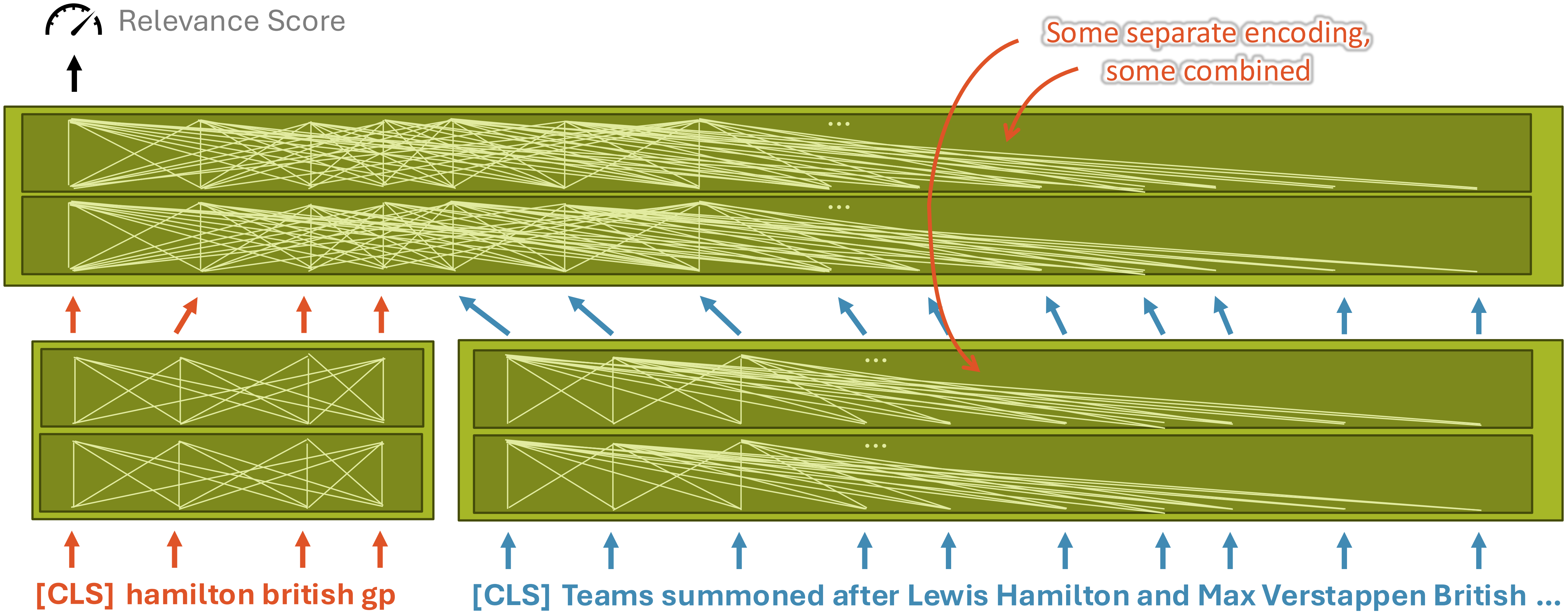
Common models:

- MonoBERT / BERT-cat / Vanilla BERT `[BertRR]`
- MonoT5 `[MonoT5]`
- RankT5 `[RankT5]`

The Table of Relevance Est. Models

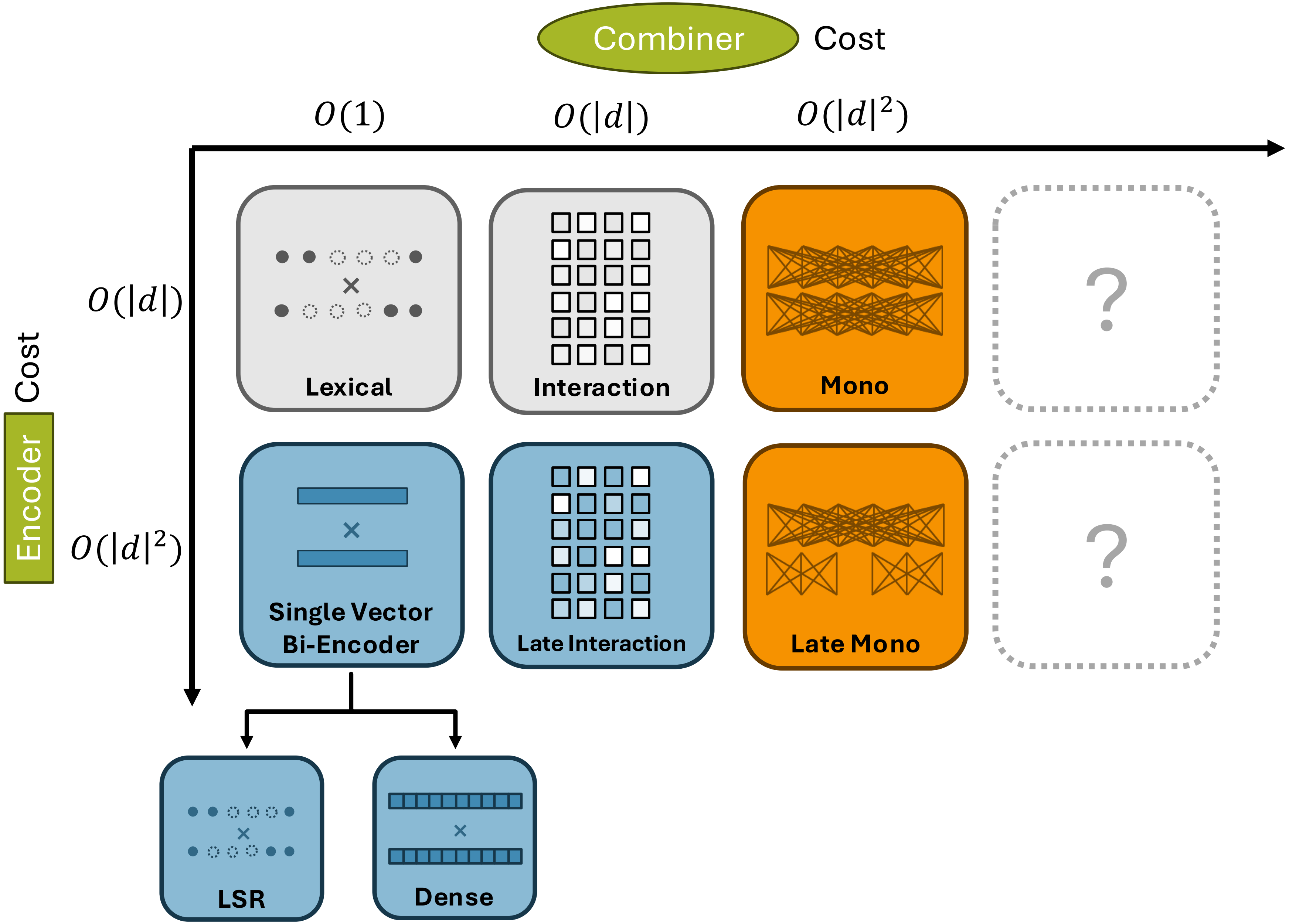


“Late” Mono Relevance Est. Models

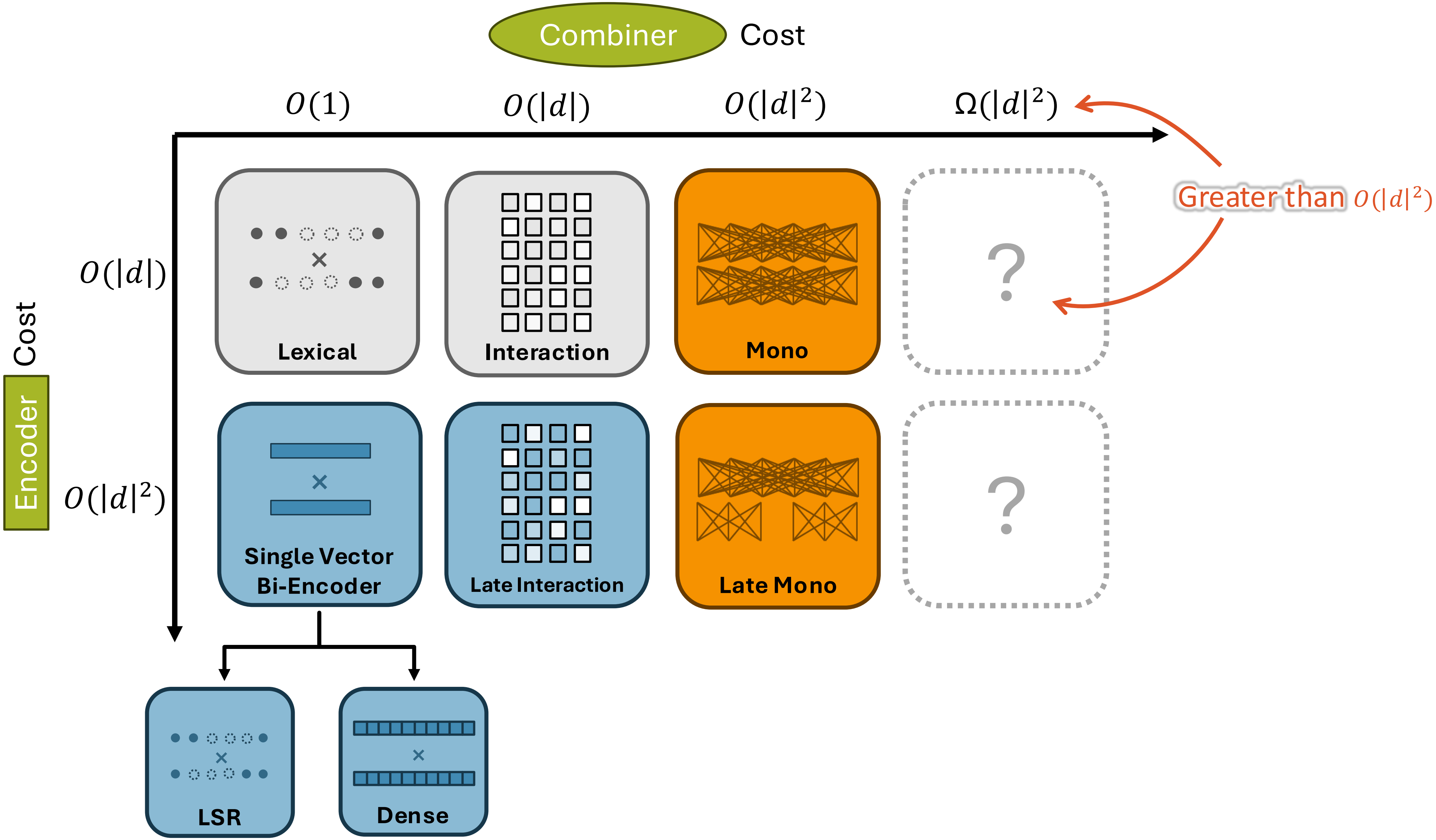


[PreTTR]

The Table of Relevance Est. Models

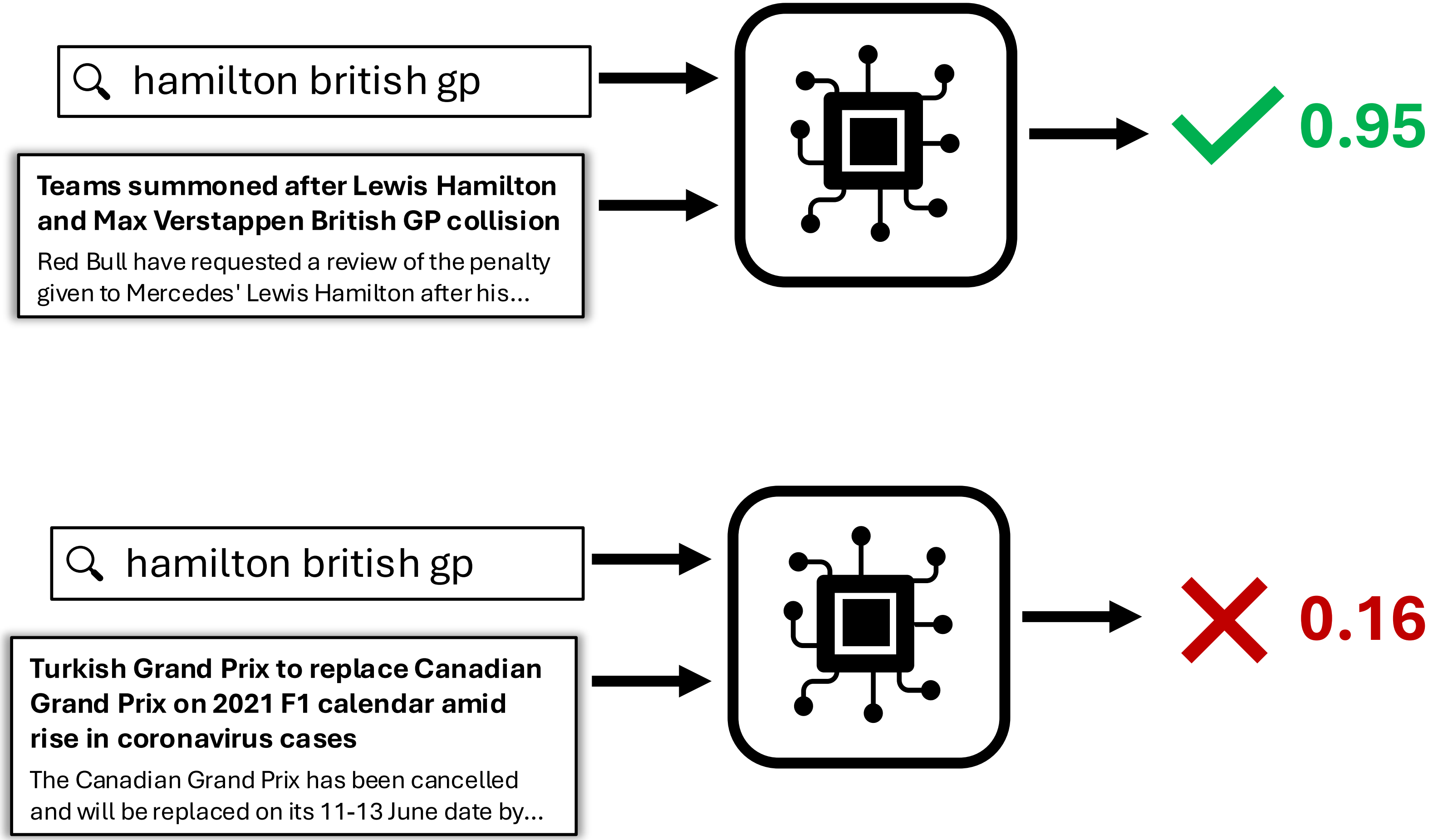


The Table of Relevance Est. Models



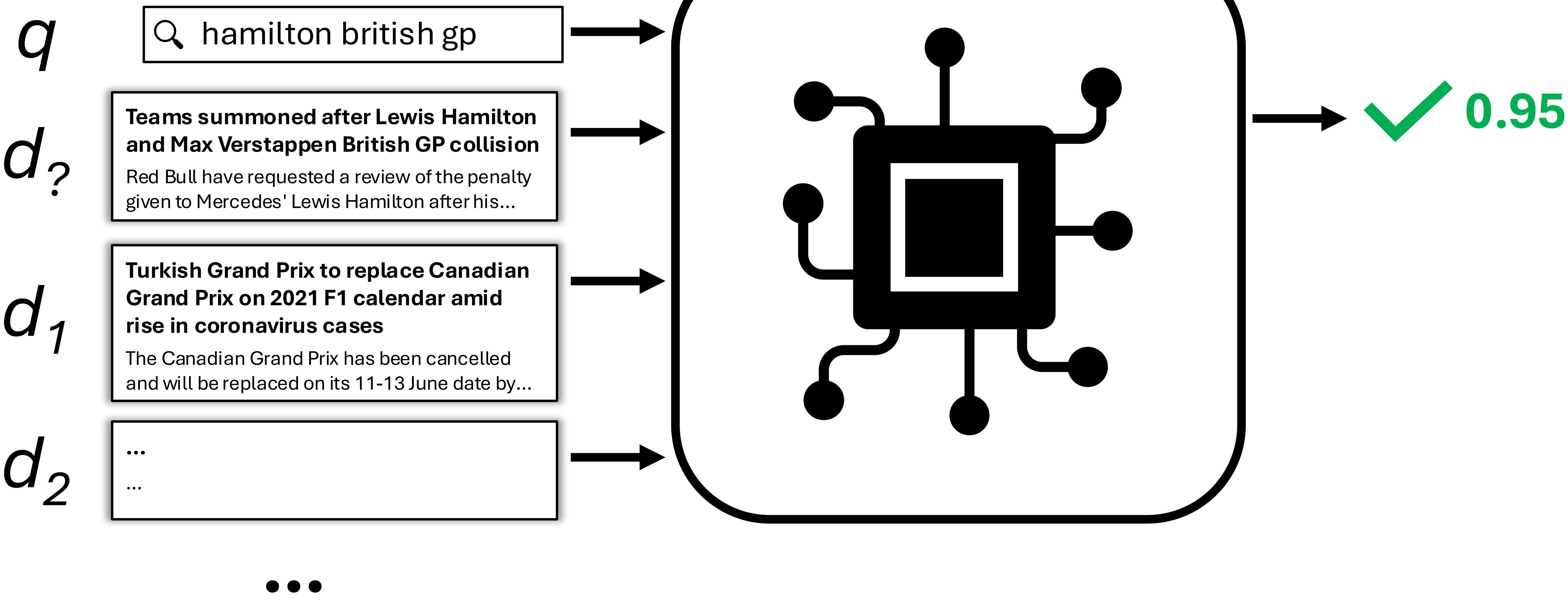
So far we've been asking: Rel($d_?|q$)

Is $d_?$ relevant to q?



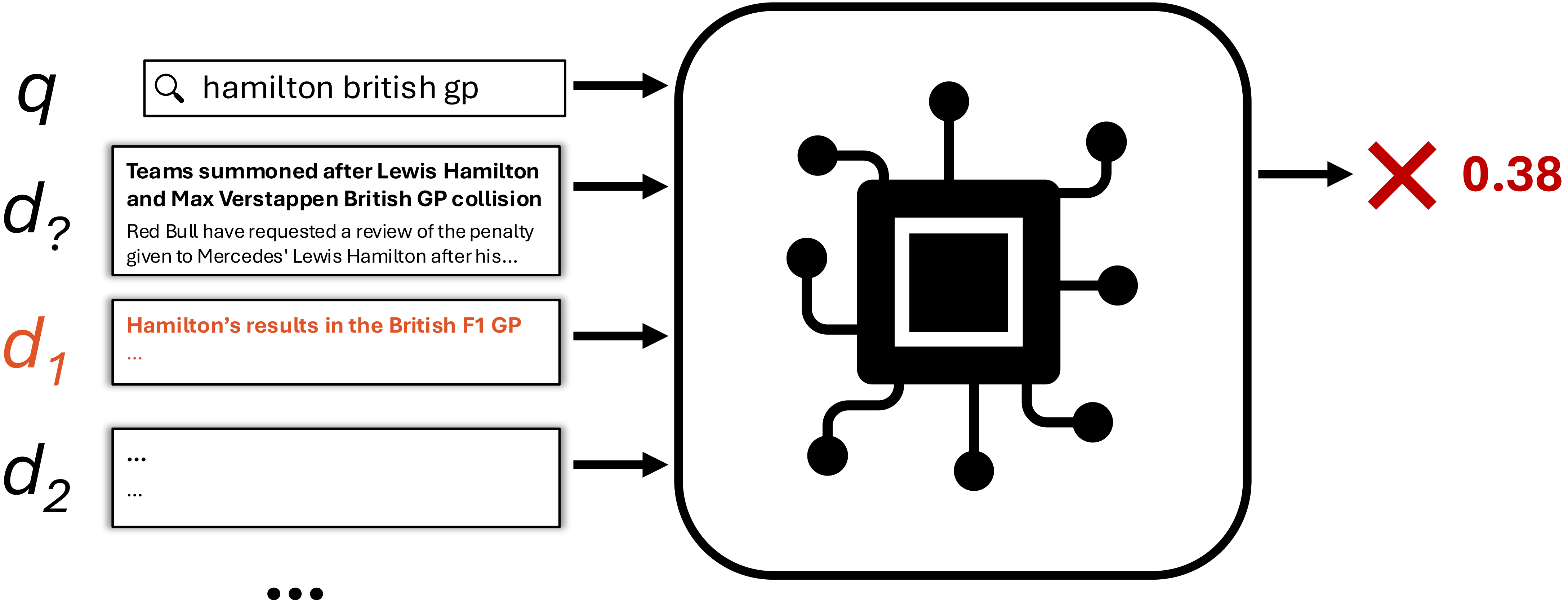
$$\mathrm{Rel}(d_?|q, d_1, d_2, \dots)$$

Is $d_?$ *more* relevant to q *than* d_1, d_2, ...?



$$\mathrm{Rel}(d_?|q, d_1, d_2, \ldots)$$

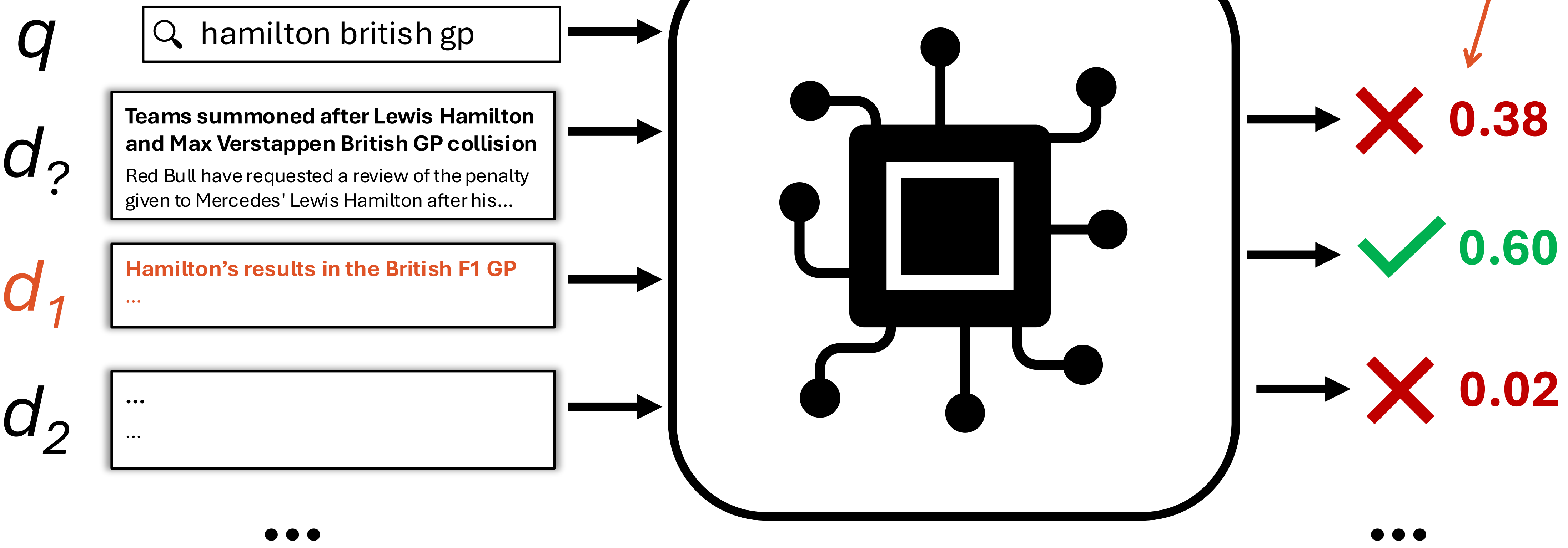
Is $d_?$ *more* relevant to q *than* d_1, d_2, …?



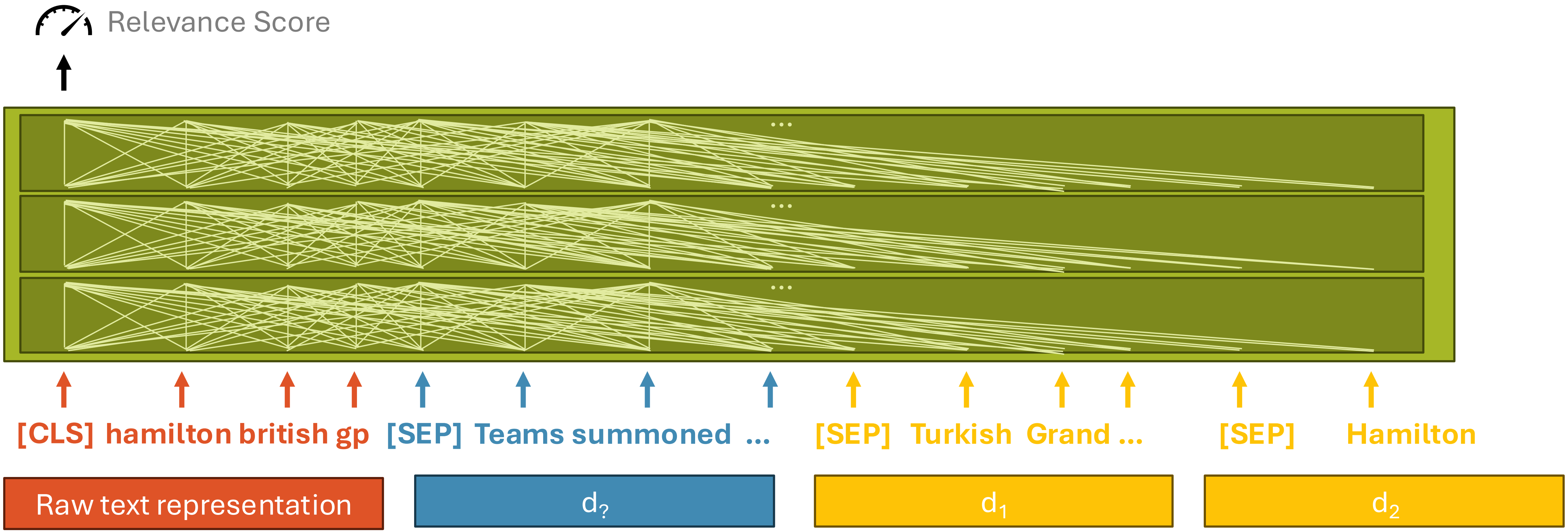
$$\mathrm{Rel}(d_?|q, d_1, d_2, \dots)$$

Is $d_?$ *more* relevant to q *than* d_1, d_2, …?

Score multiple WRT one another at the same time

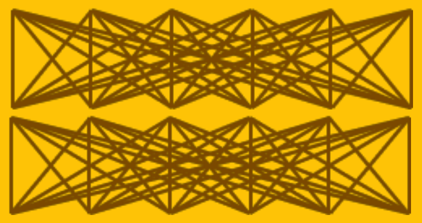


Duo, Listo, etc. Cross Encoder



[DuoBERT], [RankZephyr]

Duo, Listo, etc. Cross Encoder



Duo+

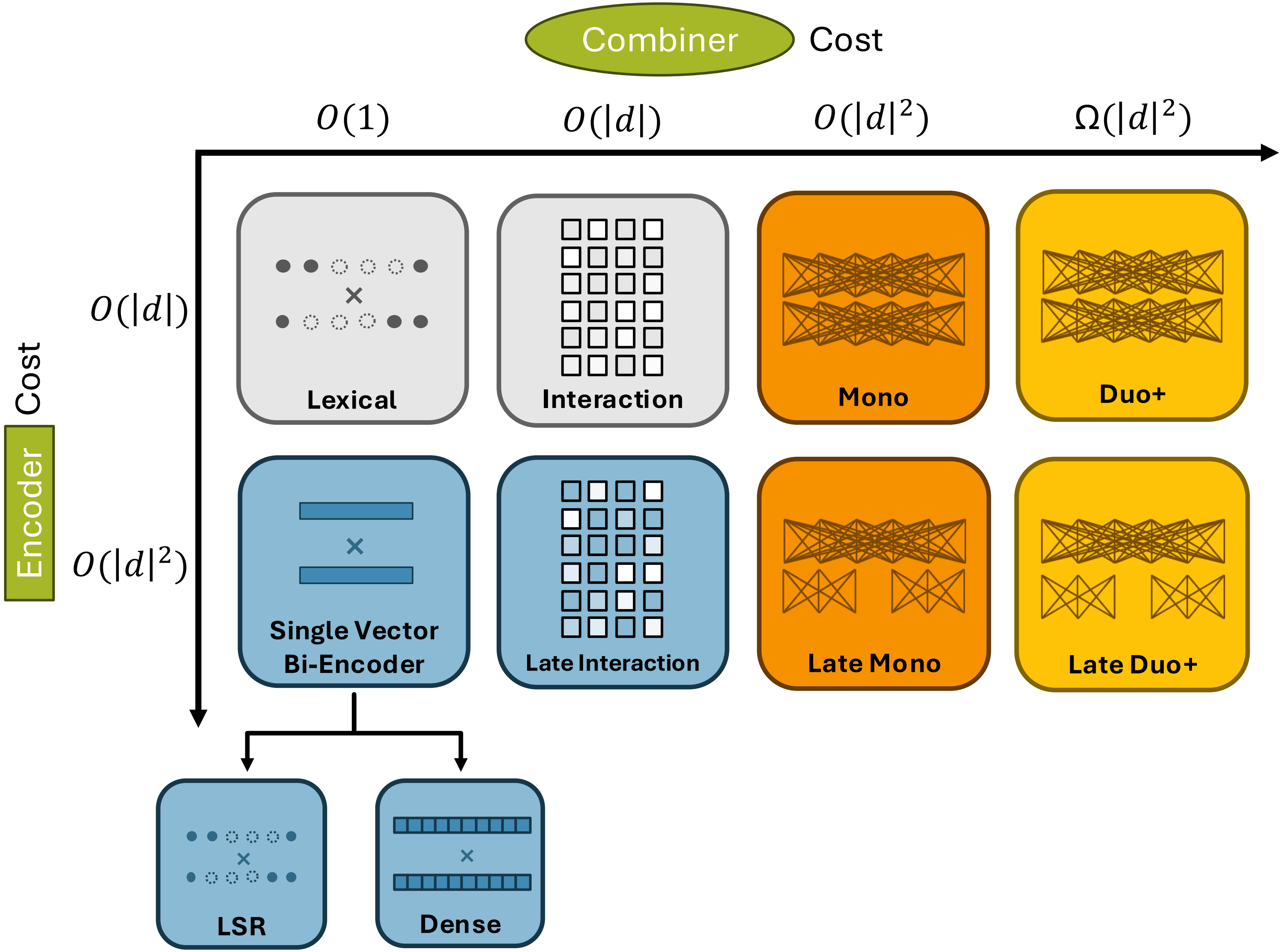
Late Duo+

Uses cross attention between query and multiple documents for combination

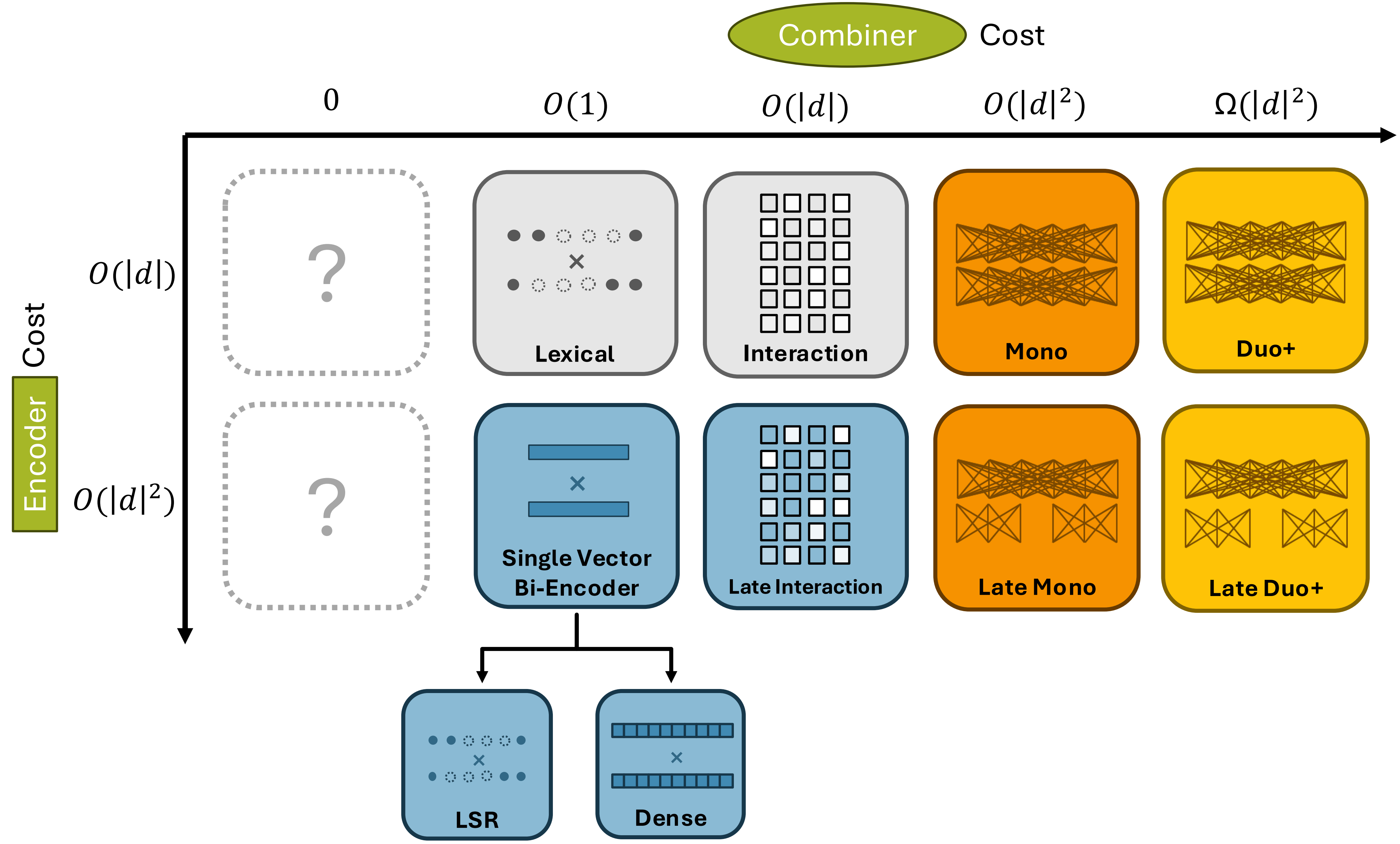
Common models:

DuoBERT [DuoBERT]
ListoBERT [ListoBERT]
RankGPT [RankGPT]
PE-Rank [PERank]
... More!

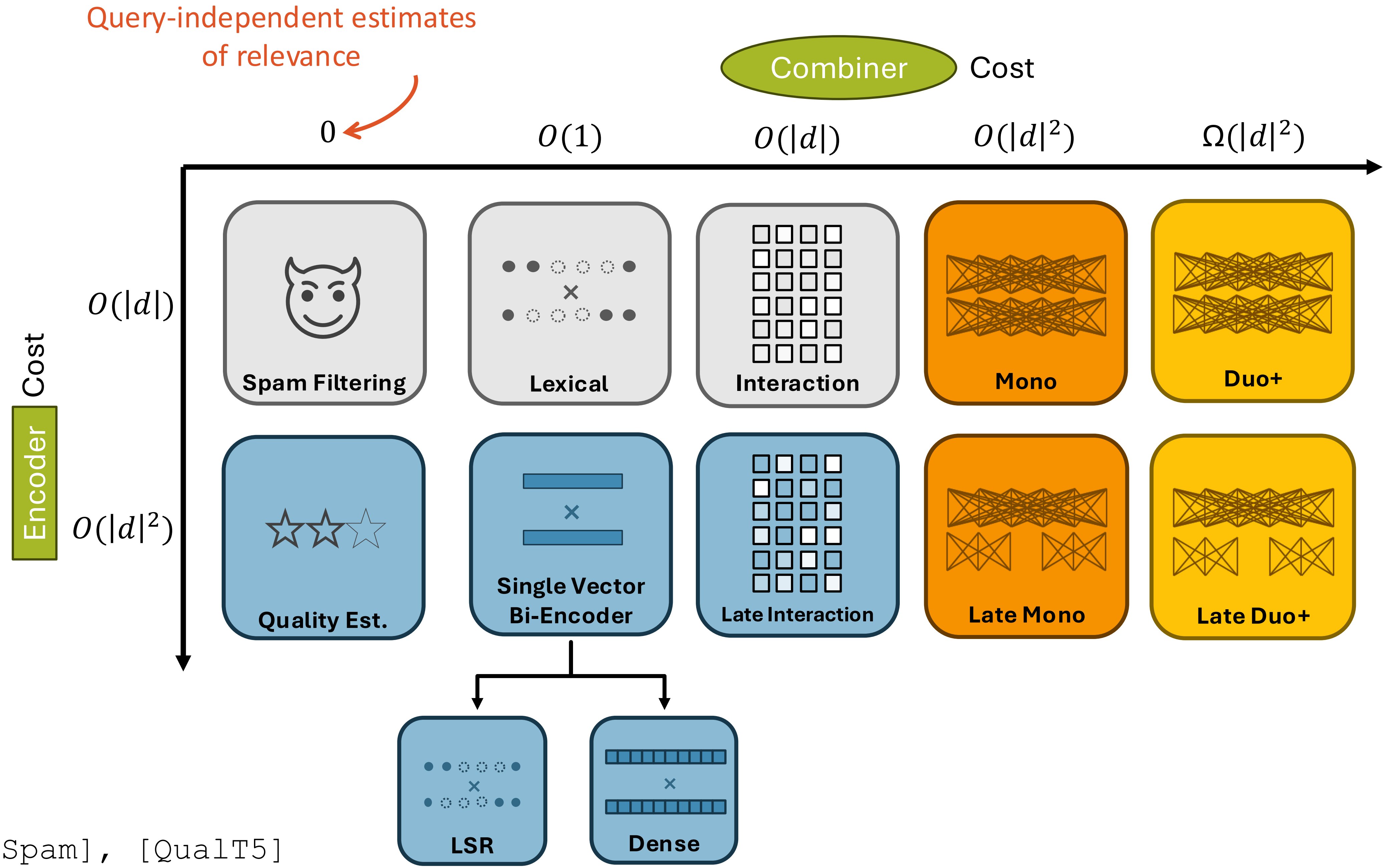
The Table of Relevance Est. Models



The Table of Relevance Est. Models



The Table of Relevance Est. Models



[Spam], [QualT5]

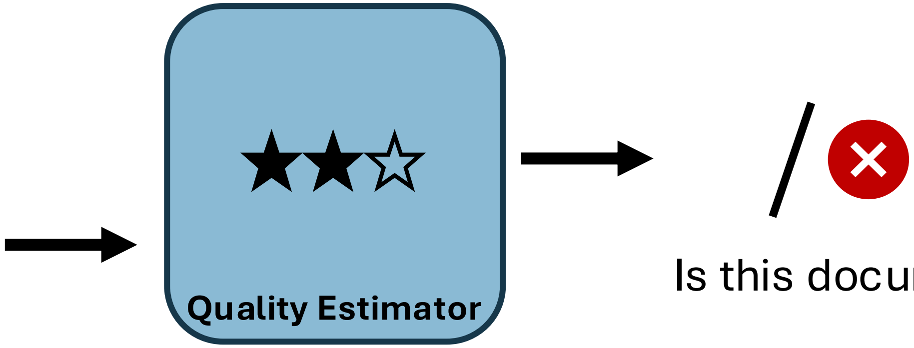
Is this document likely relevant to *any* query?

[QualT5]

Is this document likely relevant to *any* query?

[QualT5]

"Includes Fairway Meadows Apartments Reviews, maps & directions to Fairway Meadows Apartments in Franklin and more from Yahoo US Local Find Fairway Meadows Apartments in Franklin with Address, Phone number from Yahoo US Local. Includes Fairway Meadows Apartments Reviews, maps & directions to Fairway Meadows Apartments in Franklin and more from Yahoo US Local"



Is this document likely relevant to *any* query?

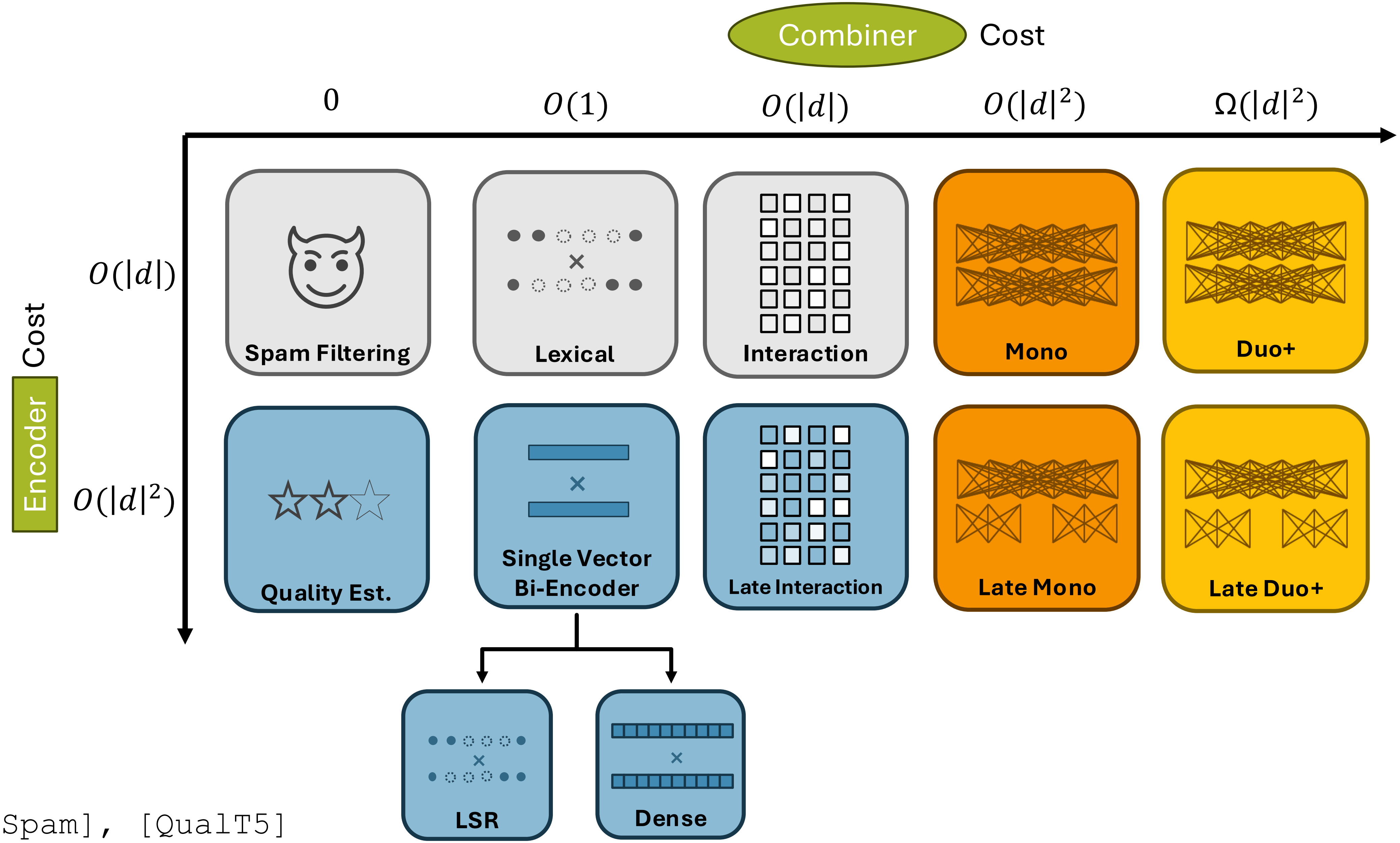
[QualT5]

“Location of Port Arthur. Port Arthur is a small town and former convict settlement on the Tasman Peninsula, in Tasmania, Australia. Port Arthur is one of Australia's most significant heritage areas and an open-air museum. Port Arthur was named after George Arthur, the Lieutenant Governor of Van Diemen's Land. The settlement started as a timber station in 1830, but it is best known for being a penal colony.”

Is this document likely relevant to *any* query?

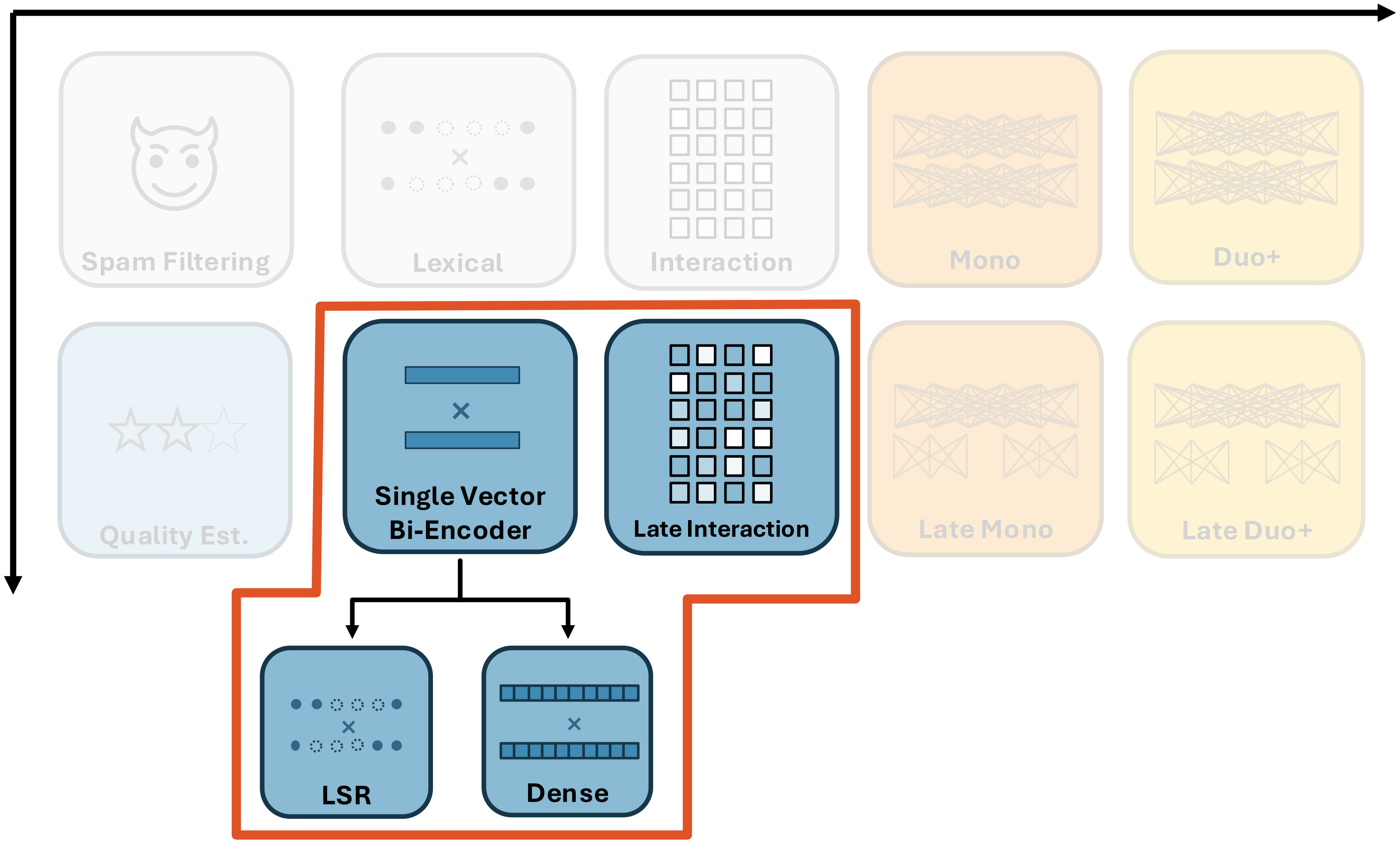
[QualT5]

The Table of Relevance Est. Models

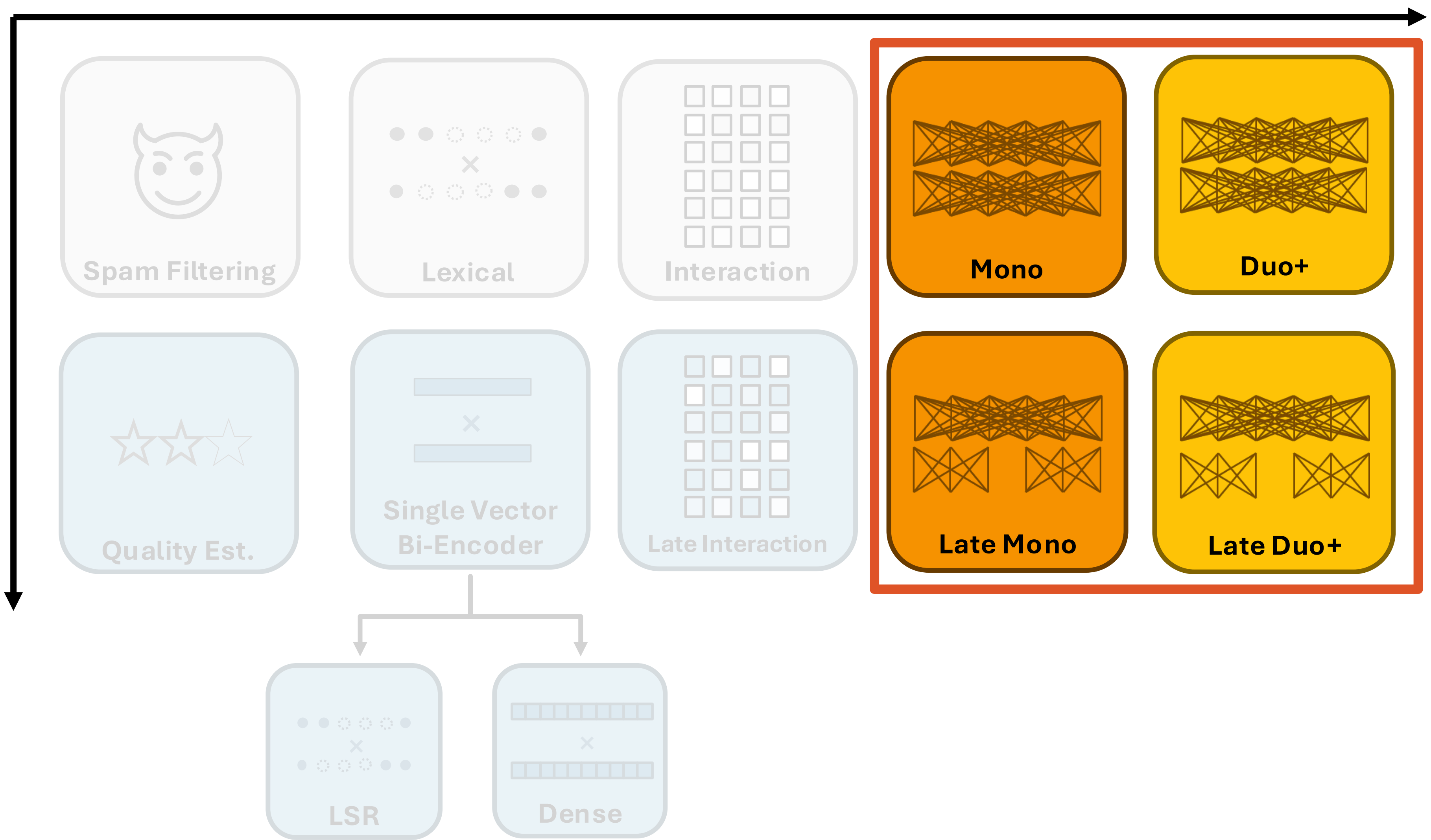


[Spam], [QualT5]

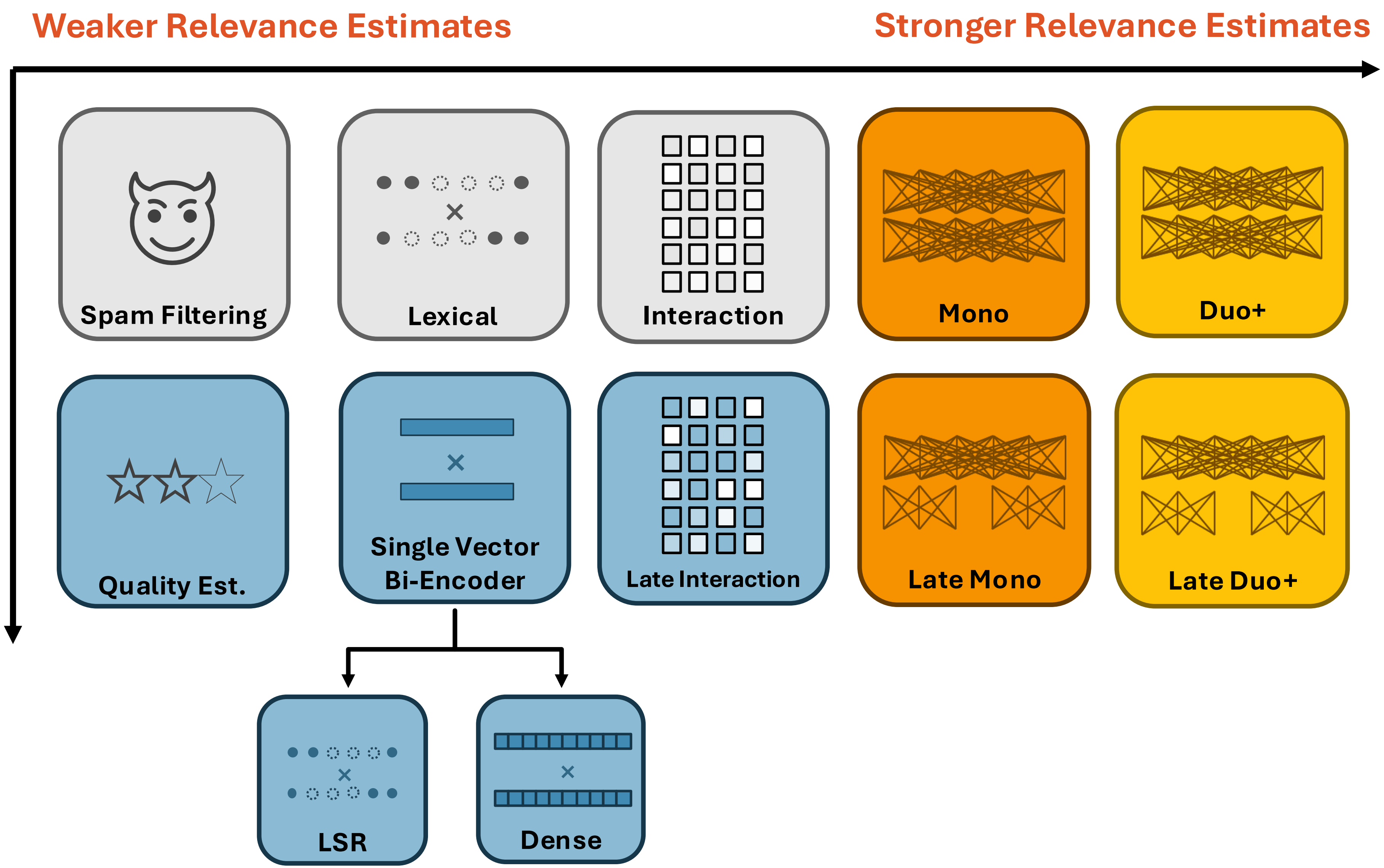
“Bi-Encoders”


Spam Filtering
Lexical
Interaction
Mono
Duo+
Quality Est.
Single Vector Bi-Encoder
Late Interaction
Late Mono
Late Duo+
LSR
Dense

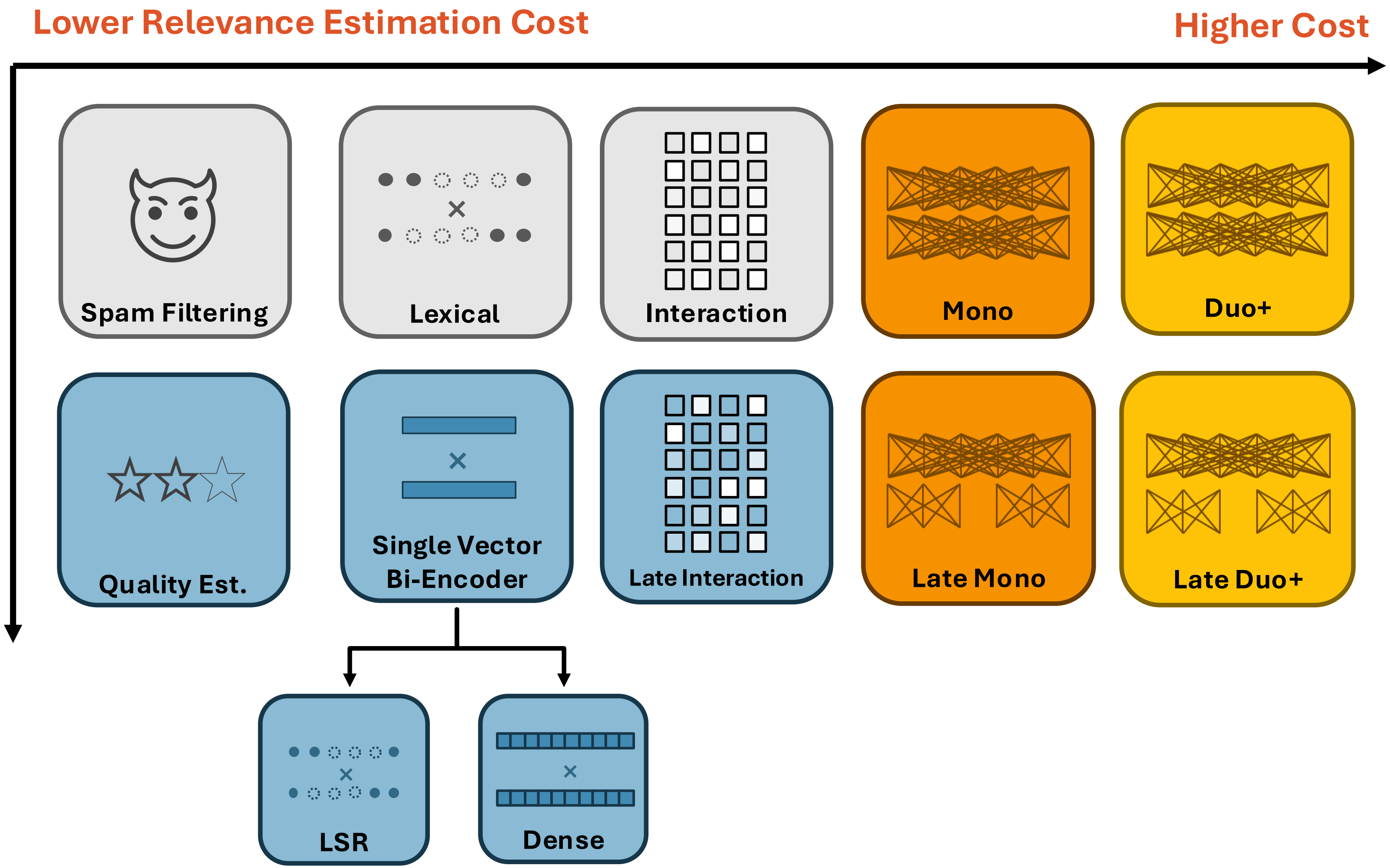
"Cross-Encoders"



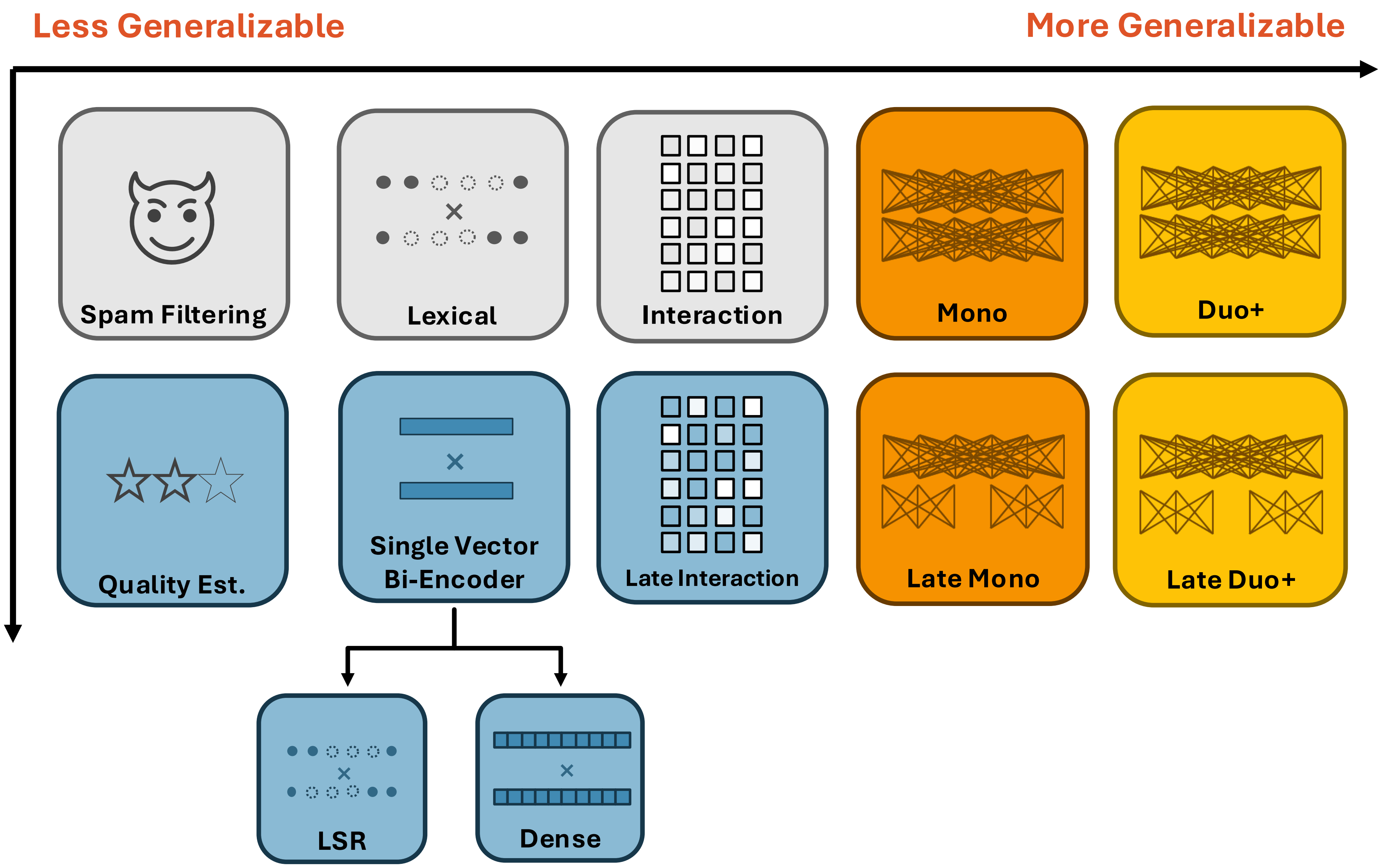
In general...


Weaker Relevance Estimates
Stronger Relevance Estimates
Spam Filtering
Lexical
Interaction
Mono
Duo+
Quality Est.
Single Vector Bi-Encoder
Late Interaction
Late Mono
Late Duo+
LSR
Dense

In general...

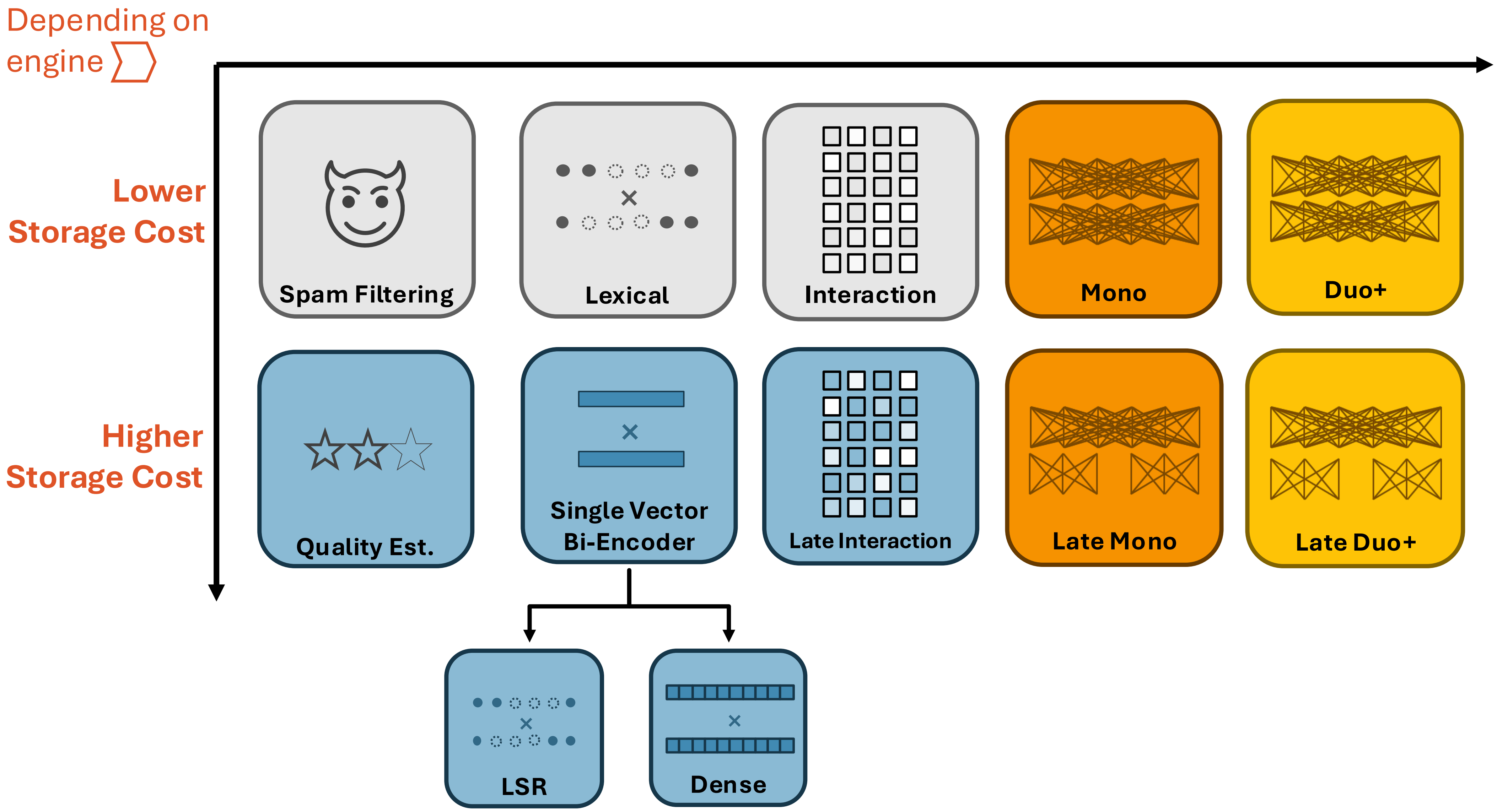

Lower Relevance Estimation Cost
Higher Cost
Spam Filtering
Lexical
Interaction
Mono
Duo+
Quality Est.
Single Vector
Bi-Encoder
Late Interaction
Late Mono
Late Duo+
LSR
Dense

In general...

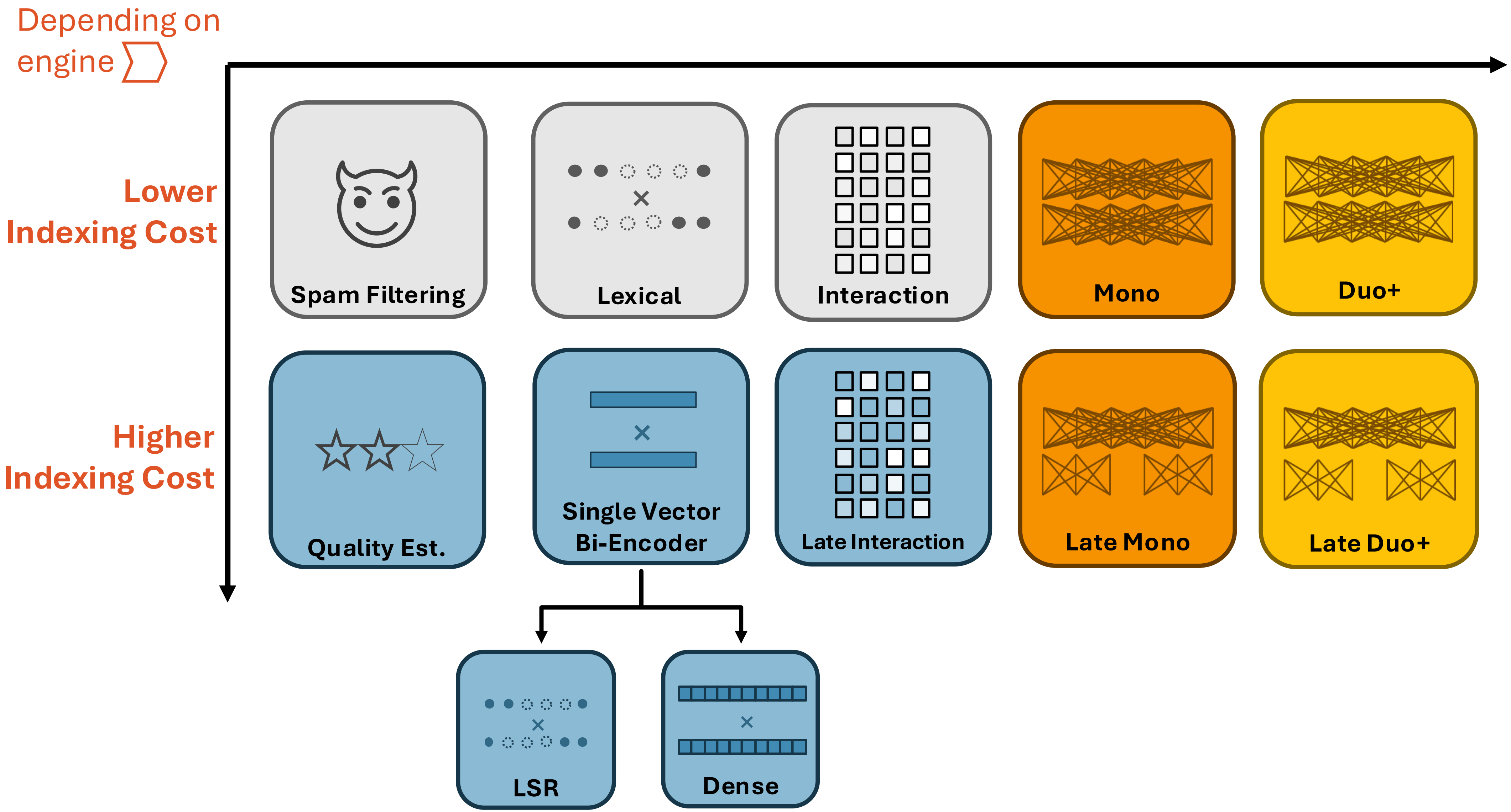


Less Generalizable
More Generalizable
Spam Filtering
Lexical
Interaction
Mono
Duo+
Quality Est.
Single Vector
Bi-Encoder
Late Interaction
Late Mono
Late Duo+
LSR
Dense

In general...


Depending on engine
Lower Storage Cost
Higher Storage Cost
Spam Filtering
Lexical
Interaction
Mono
Duo+
Quality Est.
Single Vector Bi-Encoder
Late Interaction
Late Mono
Late Duo+
LSR
Dense

In general...


Depending on engine
Lower Indexing Cost
Higher Indexing Cost
Spam Filtering
Lexical
Interaction
Mono
Duo+
Quality Est.
Single Vector Bi-Encoder
Late Interaction
Late Mono
Late Duo+
LSR
Dense

Neural Retrieval and Re-ranking

Part 1: Relevance Est. Models

Part 2: Training

Part 3: Engines

Wrap-up

Neural Retrieval and Re-ranking

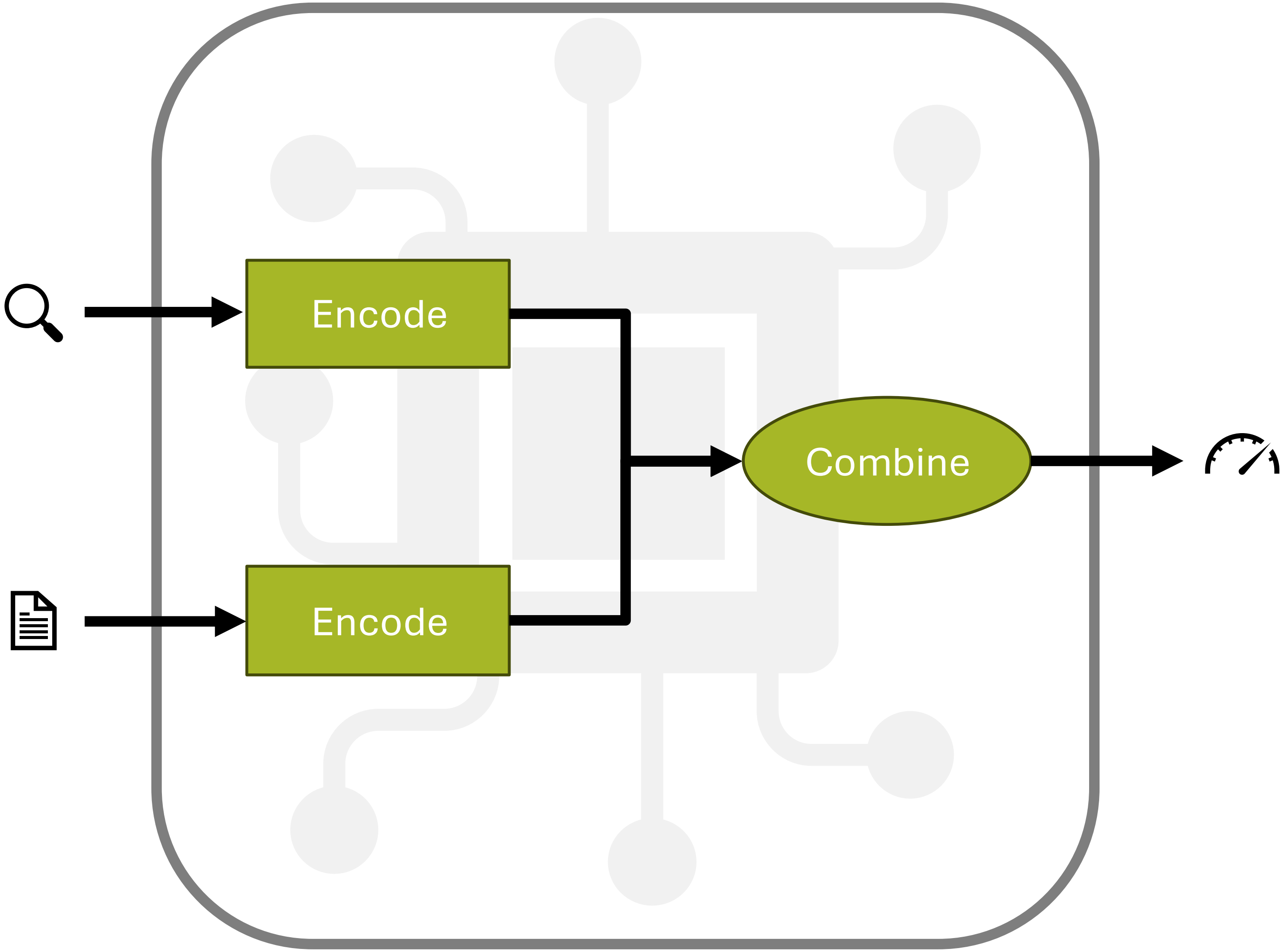
Part 1: Relevance Est. Models

Part 2: Training

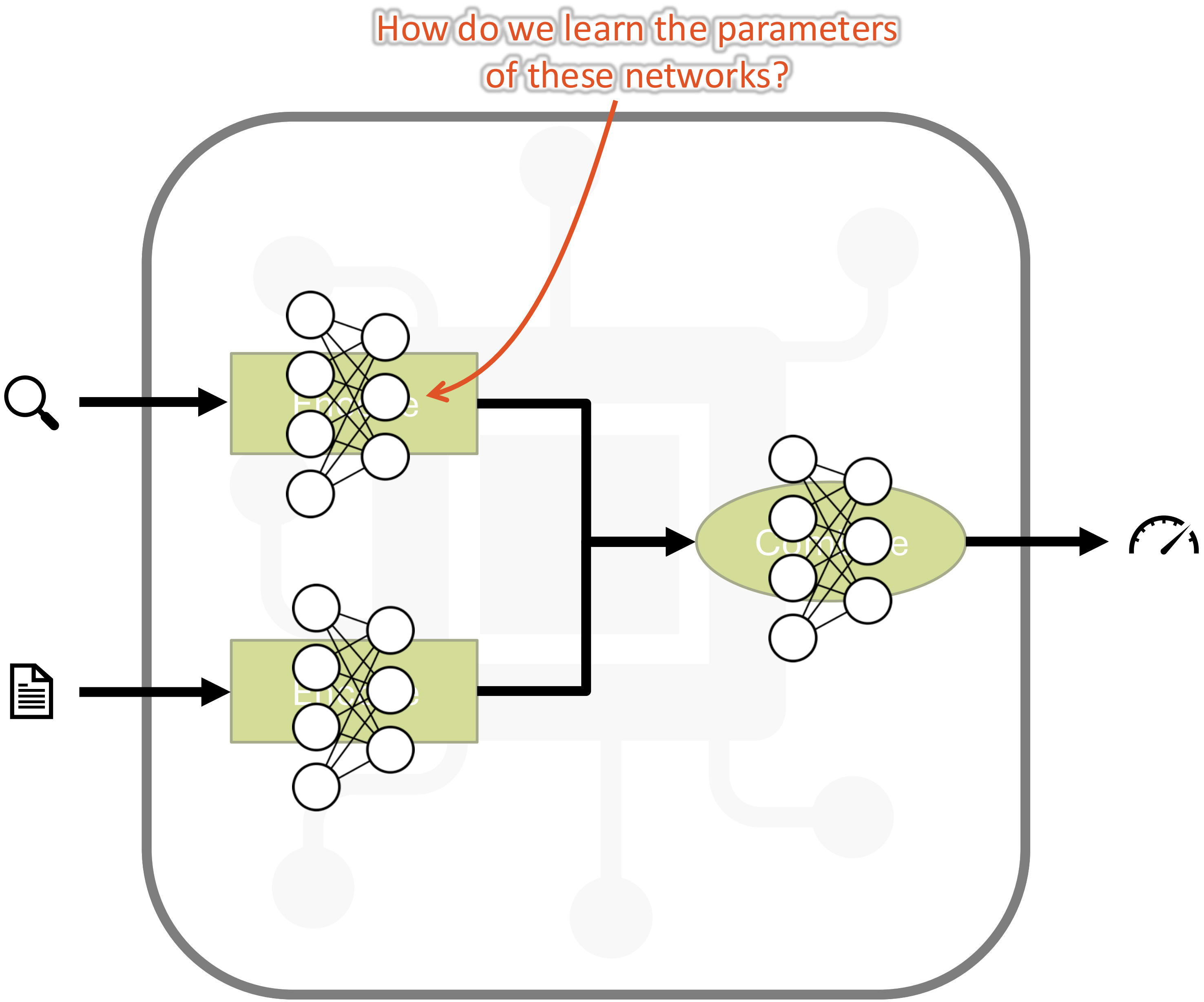
Part 3: Engines

Wrap-up

Generic Relevance Est. Model Structure



Generic Relevance Est. Model Structure



I want to know Lewis Hamilton's performance in the British Formula One Grand Prix

Human-labeled Relevance Est. Data

hamilton british gp

Teams summoned after Lewis Hamilton and Max Verstappen British GP collision

Red Bull have requested a review of the penalty given to Mercedes' Lewis Hamilton after his first-lap collision with Max Verstappen at the British Grand Prix.

Hamilton was ruled to have been "predominantly to blame" for the crash, which resulted in Red Bull driver Verstappen being taken to hospital.

The Briton was given a 10-second penalty but recovered to win the race.

Both teams have been summoned to appear via video link on Thursday.

...

Source: https://www.bbc.co.uk/sport/formula1/57988419

Turkish Grand Prix to replace Canadian Grand Prix on 2021 F1 calendar amid rise in coronavirus cases

The Canadian Grand Prix has been cancelled and will be replaced on its 11-13 June date by a race at Turkey's Istanbul Park track.

It is the second successive year the Montreal event has been cancelled because of the coronavirus pandemic.

Travel restrictions made it impossible to enter Canada without a mandatory 14-day quarantine, said an F1 statement.

Meanwhile the contract for a race on Montreal's Circuit Gilles Villeneuve has been extended by two years.

...

https://www.bbc.co.uk/sport/formula1/56915817

Training Data:

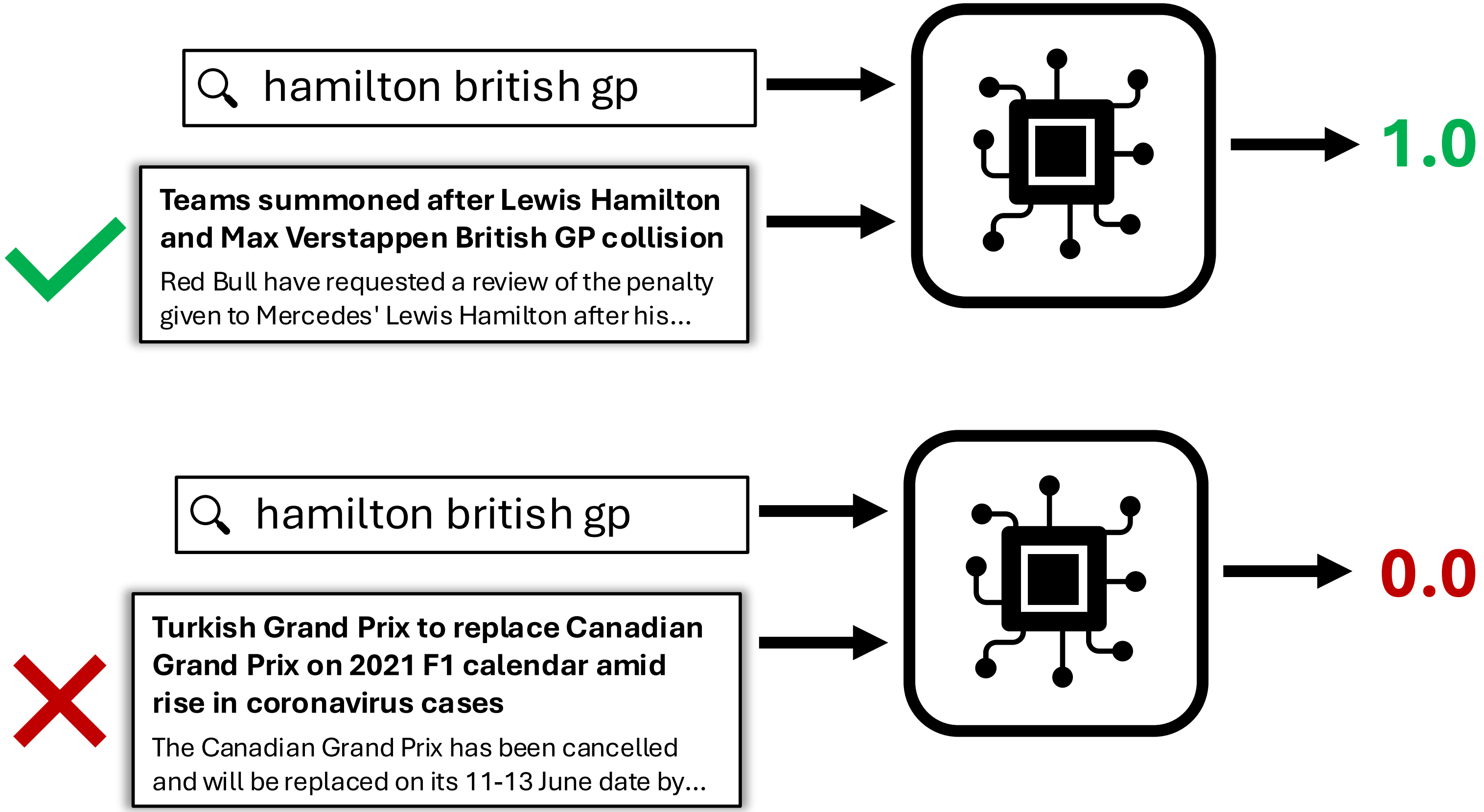
Sample 1:
- Query: hamilton british gp
- Document: Teams summoned ...
- Label: 1 (relevant)

Sample 2:
- Query: hamilton british gp
- Document: Turkish Grand ...
- Label: 0 (not relevant)

...

Pointwise Training

Optimize for label values directly



Example: Mean Squared Error (MSE)

$$L(q, d, l) = (Rel(d|q) - l)^2$$

Alternative: Contrastive “Triples”

hamilton british gp

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...

https://www.bbc.co.uk/sport/formula1/56915817

Training Data:

Sample 1:

- Query: hamilton british gp
- Relevant Doc: Teams summoned ...
- Non-Relevant Doc: Turkish Grand ...

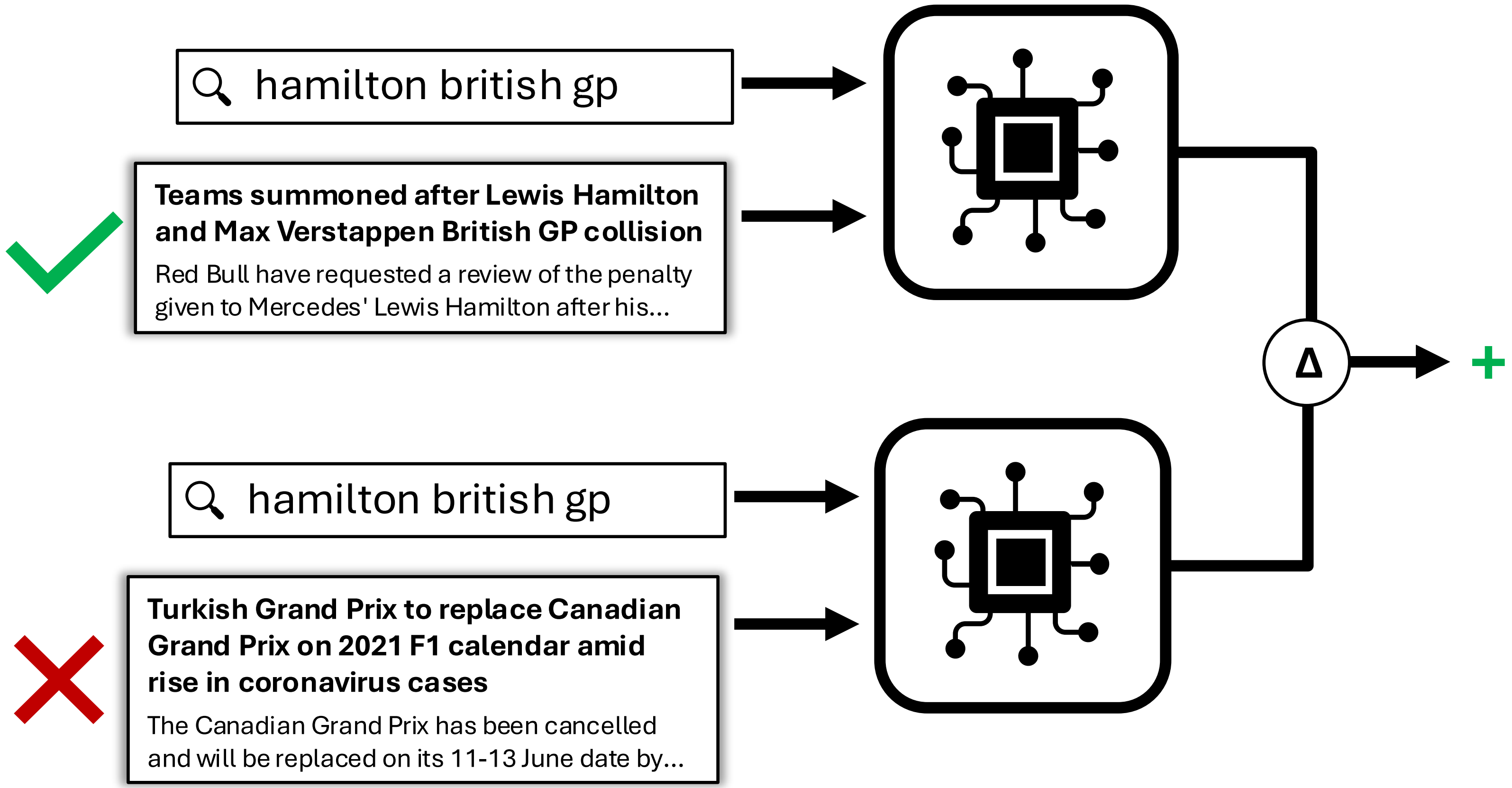
Sample 2:

- ...

...

Pairwise Training

Optimize for relative values of a training “triple”



Example: Cross Entropy Loss (CE)

$$L(q, d_+, d_-) = -\log \frac{e^{Rel(d_+|q)}}{e^{Rel(d_+|q)} + e^{Rel(d_-|q)}}$$

Common Trick: In-batch Negatives

Typical Training Batch:

Query1, $Doc1_+$, $Doc2_-$

Query2, $Doc3_+$, $Doc4_-$

Query3, $Doc5_+$, $Doc6_-$

Query4, $Doc7_+$, $Doc8_-$

…

Common Trick: In-batch Negatives

Typical Training Batch:

Query1, **Doc1$_+$**, **Doc2$_-$**

Query2, Doc3$_+$, Doc4$_-$

Query3, Doc5$_+$, Doc6$_-$

Query4, Doc7$_+$, Doc8$_-$

…

These (essentially random) docs in the batch are probably not relevant to the query

Common Trick: In-batch Negatives

Typical Training Batch:

Query1, Doc1$_+$, Doc2$_-$

Query2, Doc3$_+$, Doc4$_-$

Query3, Doc5$_+$, Doc6$_-$

Query4, Doc7$_+$, Doc8$_-$

…

Re-use other documents in this batch as more negative samples

Common Trick: In-batch Negatives

Typical Training Batch:

Query1, **$Doc1_+$, $Doc2_-$**

Query2, $Doc3_+$, $Doc4_-$

Query3, **$Doc5_+$, $Doc6_-$**

Query4, **$Doc7_+$, $Doc8_-$**

…

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Typical Training Batch:

Query1, **$Doc1_+$, $Doc2_-$**

Query2, **$Doc3_+$, $Doc4_-$**

Query3, **$Doc5_+$, $Doc6_-$**

Query4, **$Doc7_+$, $Doc8_-$**

...

Re-use other documents in this batch as more negative samples

Common Trick: In-batch Negatives

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Query1, **Doc1$_+$, Doc2$_-$**

Query2, **Doc3$_+$, Doc4$_-$**

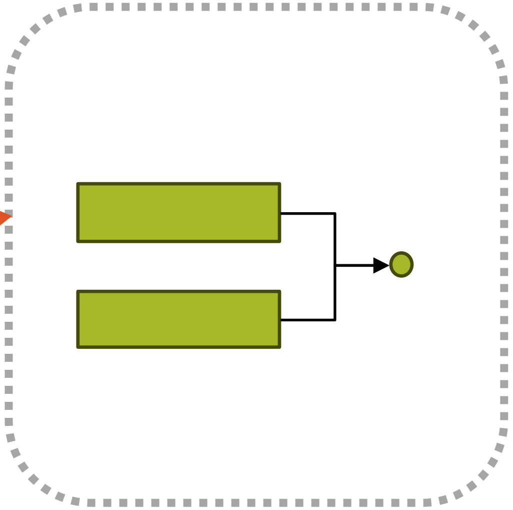
Query3, **Doc5$_+$, Doc6$_-$**

Query4, **Doc7$_+$, Doc8$_-$**

…

Re-use other documents in this batch as more negative samples

Useful when encoder is large, since the doc representations can be reused



Common Trick: In-batch Negatives

Expanded version:

Query1, Doc1$_{+}$, Doc2_, Doc3_, Doc4_, Doc5_, Doc6_, Doc7_, Doc8_

Query2, Doc3$_{+}$, Doc4_, Doc1_, Doc2_, Doc5_, Doc6_, Doc7_, Doc8_

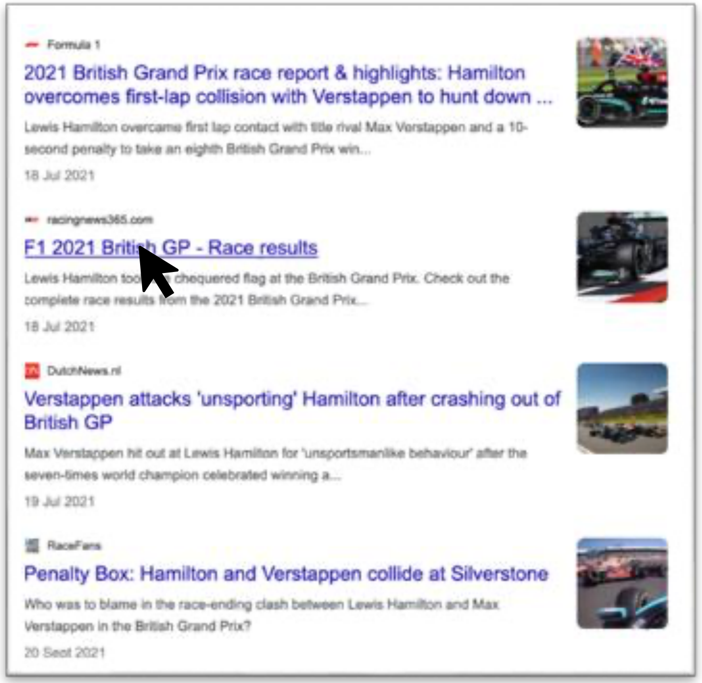
Query3, Doc5$_{+}$, Doc6_, Doc1_, Doc2_, Doc3_, Doc4_, Doc7_, Doc8_

Query4, Doc7$_{+}$, Doc8_, Doc1_, Doc2_, Doc3_, Doc4_, Doc5_, Doc6_

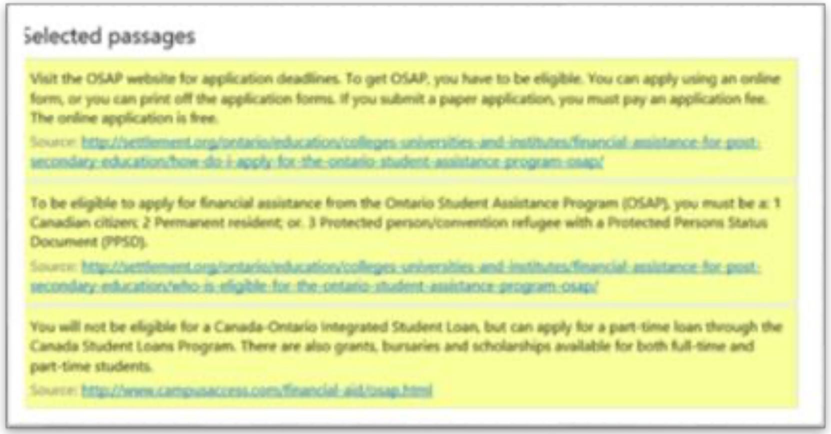
Hard Negative Mining

In practice, you'll often have pairs of queries and **relevant docs**, without **negative docs**.

Clicks:



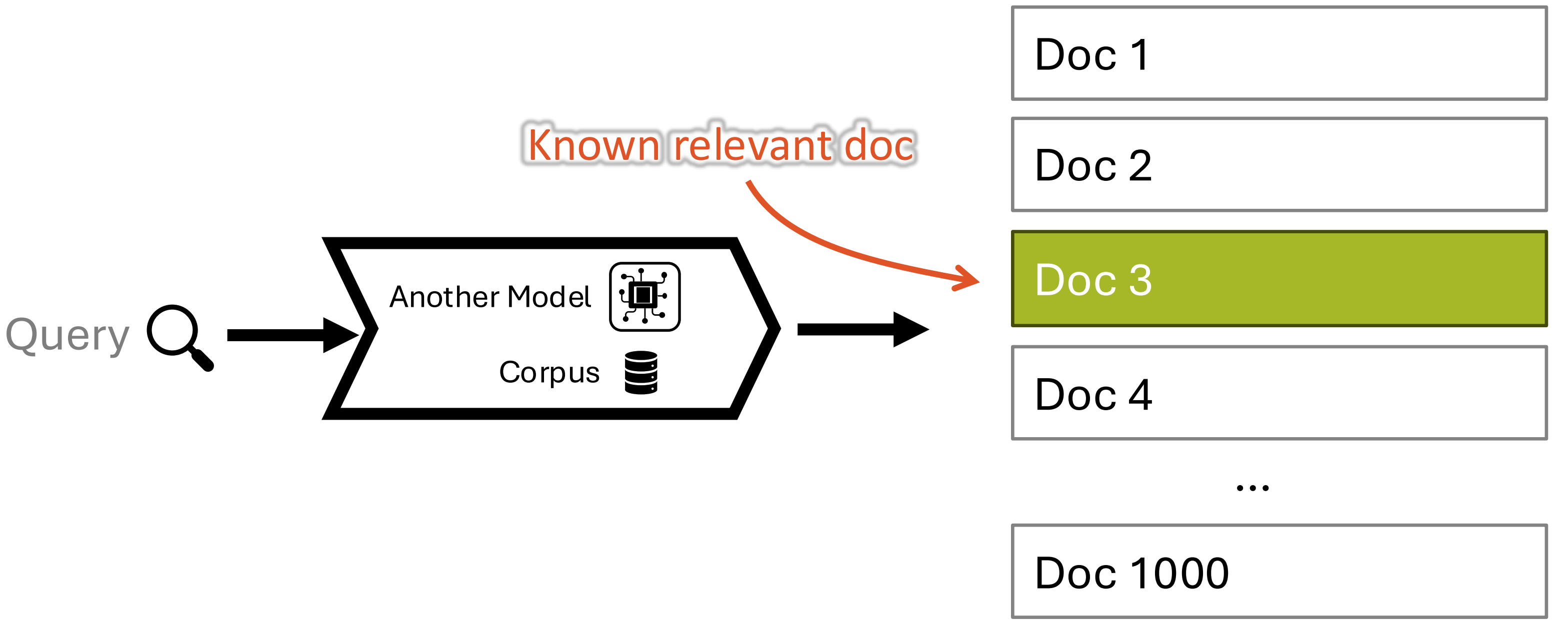
MS-MARCO:



[MSMARCO]

Hard Negative Mining

Select additional negatives from the top results of another engine.



Hard Negative Mining

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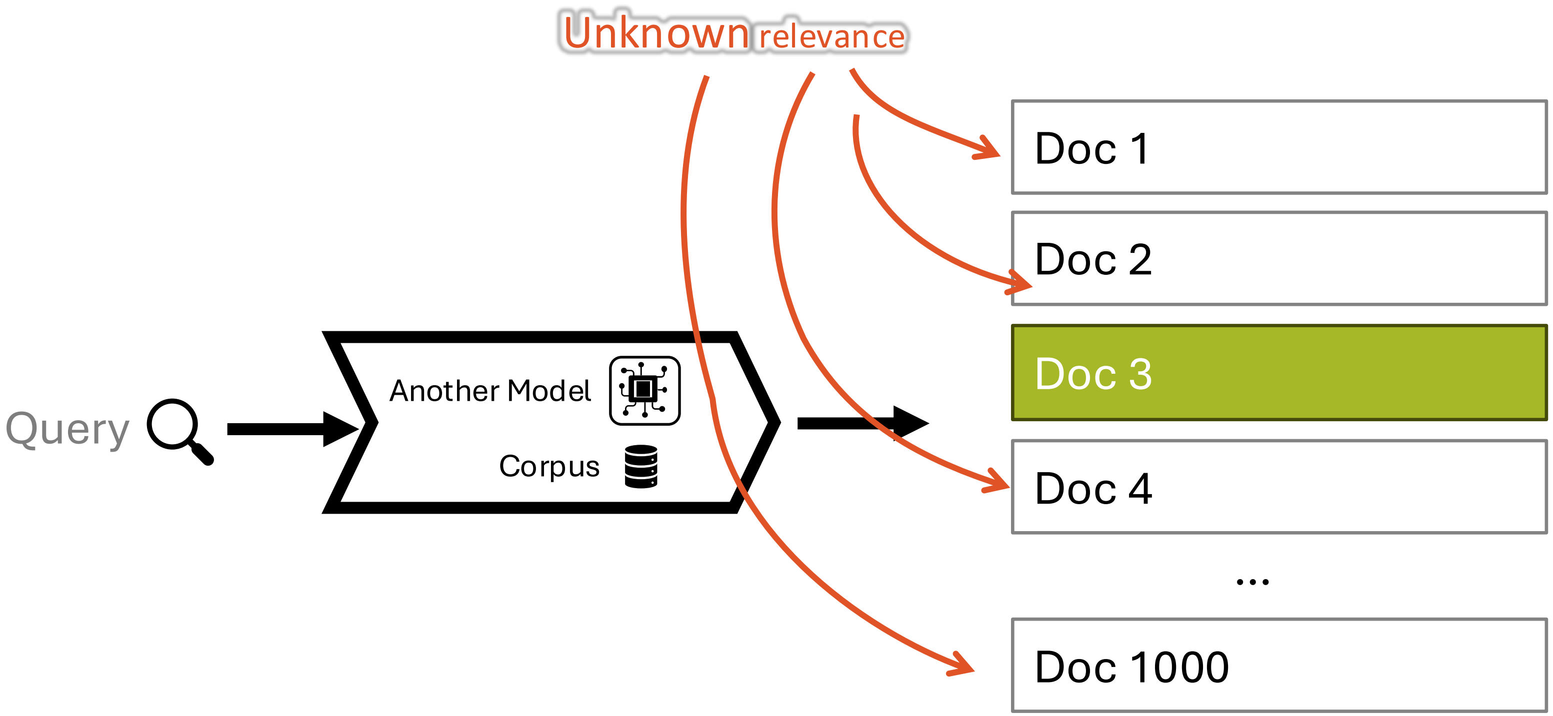

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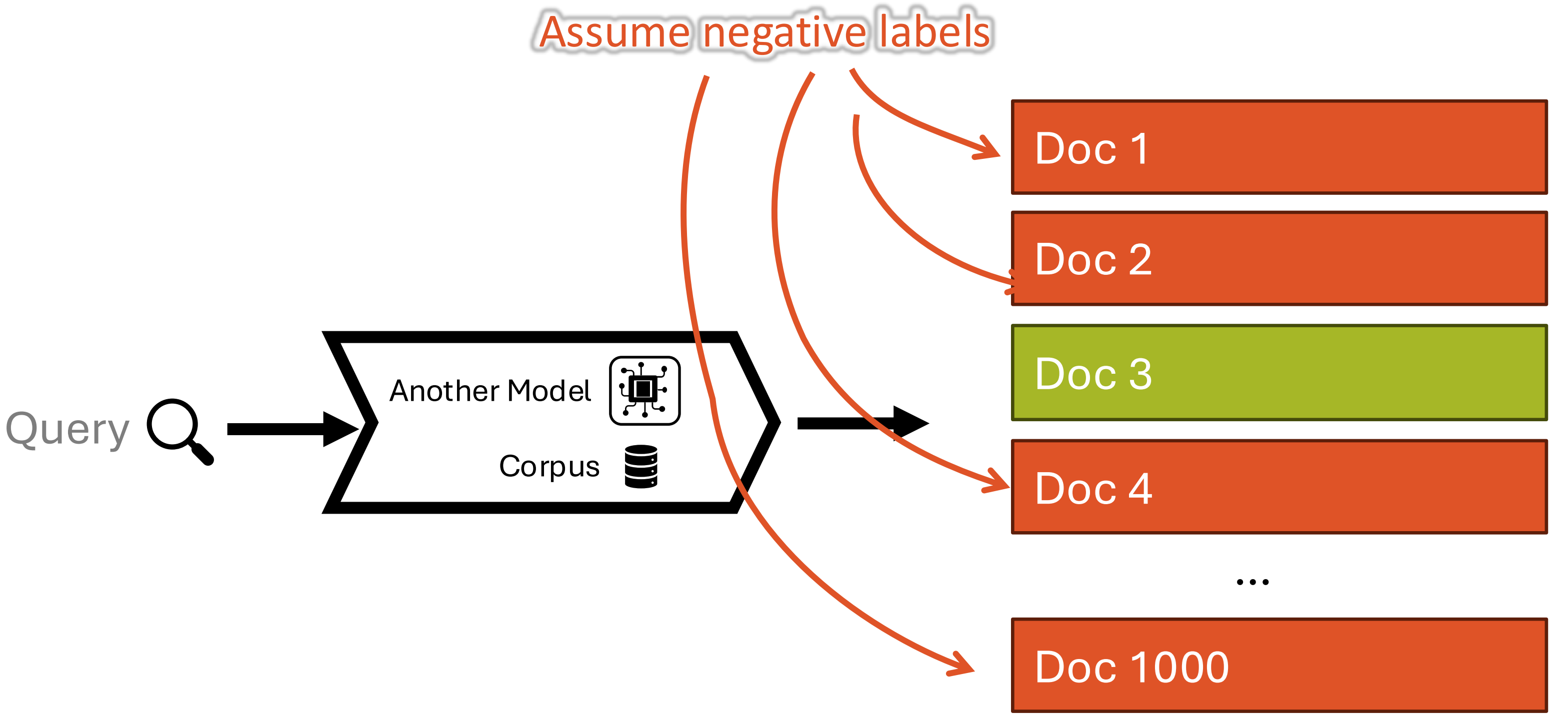
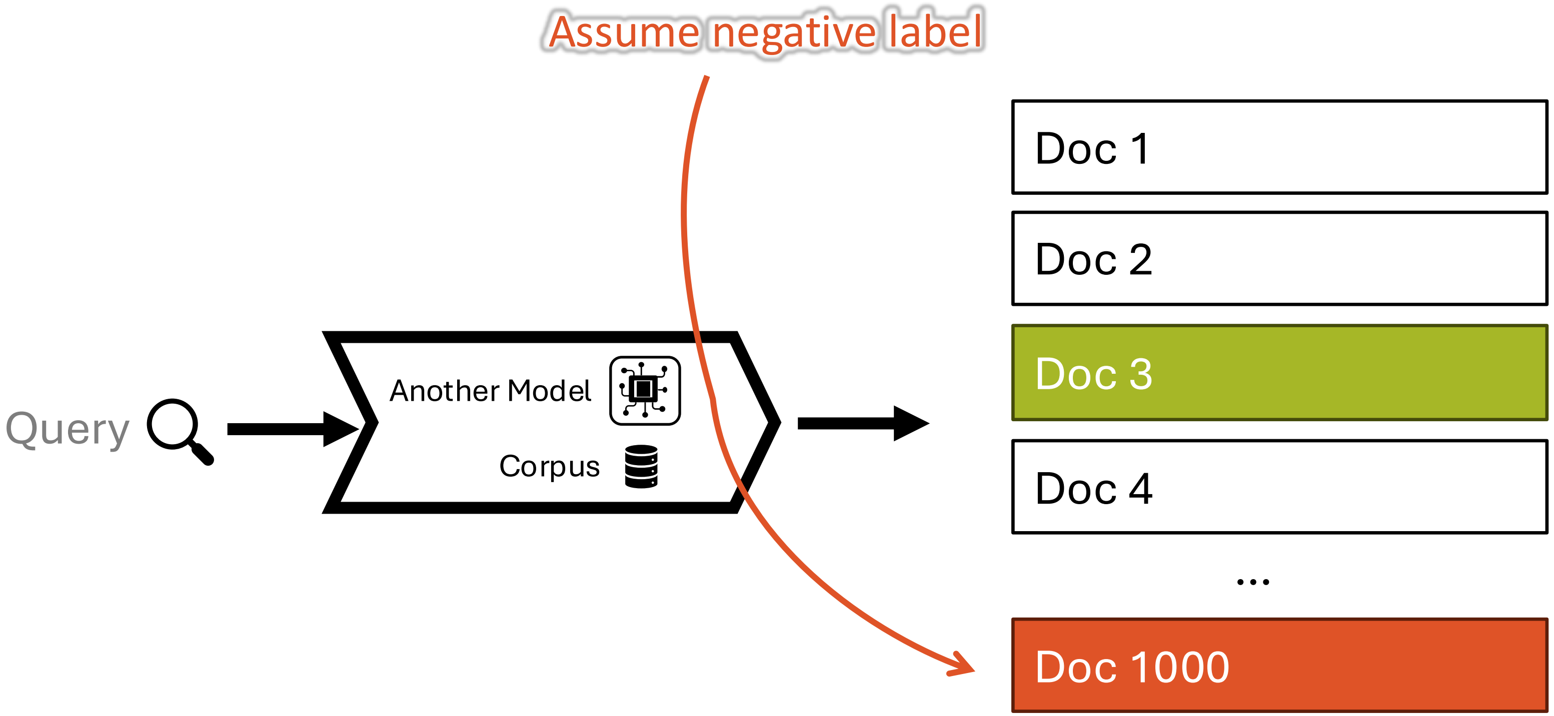


+ Stronger negatives than random selection from corpus

– Possibility of *false* negatives (i.e., relevant docs mislabelled as negative)

Hard Negative Mining

Select additional negatives from the top results of another engine.

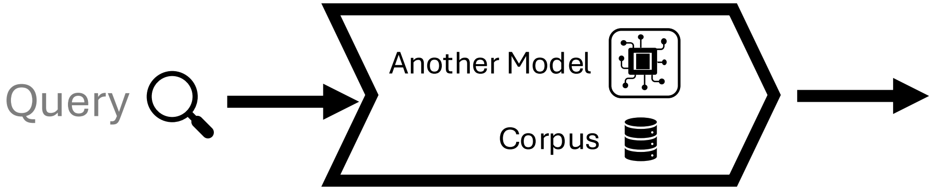


+ Stronger negatives than random selection from corpus

– Possibility of *false* negatives (i.e., relevant docs mislabelled as negative)

Distillation

Train a model to mimic the results of another engine



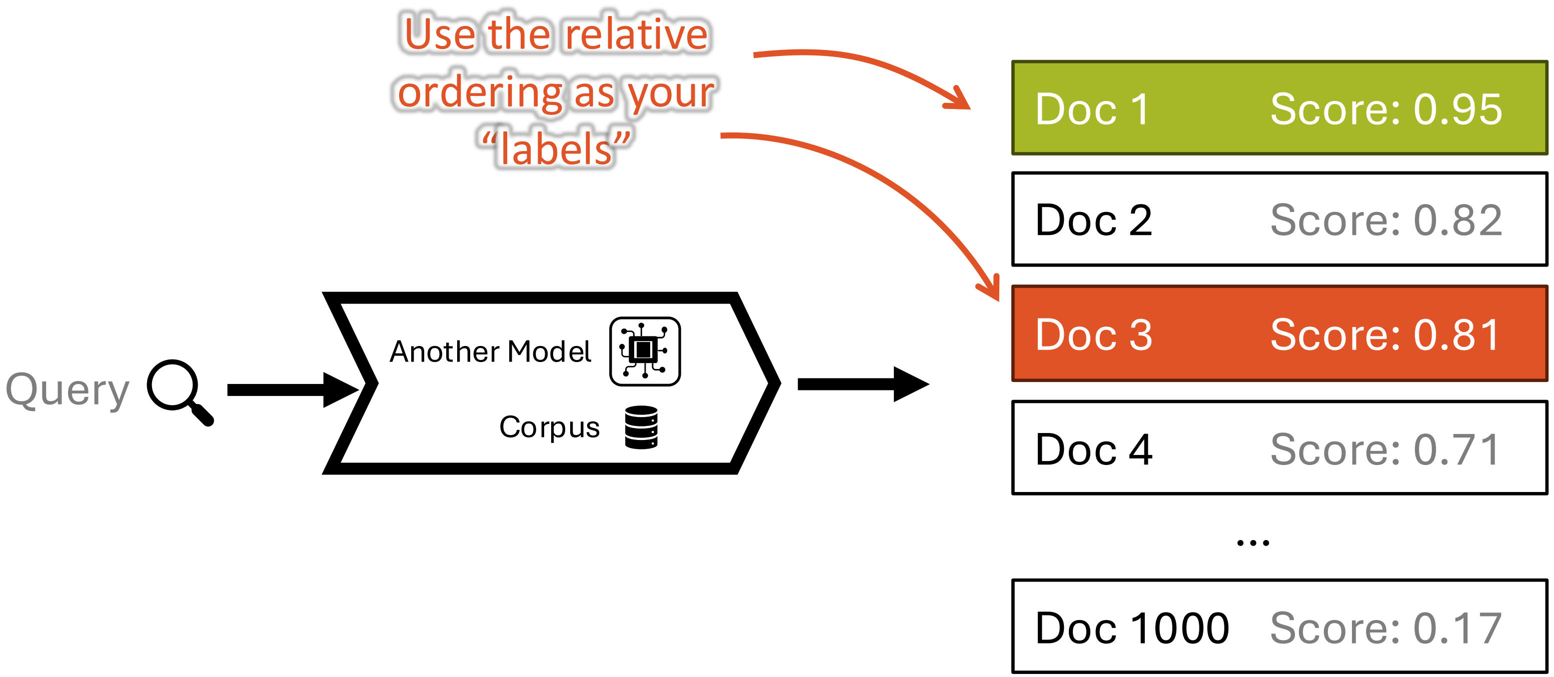
Use the scores as your “labels”

Doc 1	Score: 0.95
Doc 2	Score: 0.82
Doc 3	Score: 0.81
Doc 4	Score: 0.71
…	
Doc 1000	Score: 0.17

Example: KL-Divergence Loss

Distillation

Train a model to mimic the results of another engine

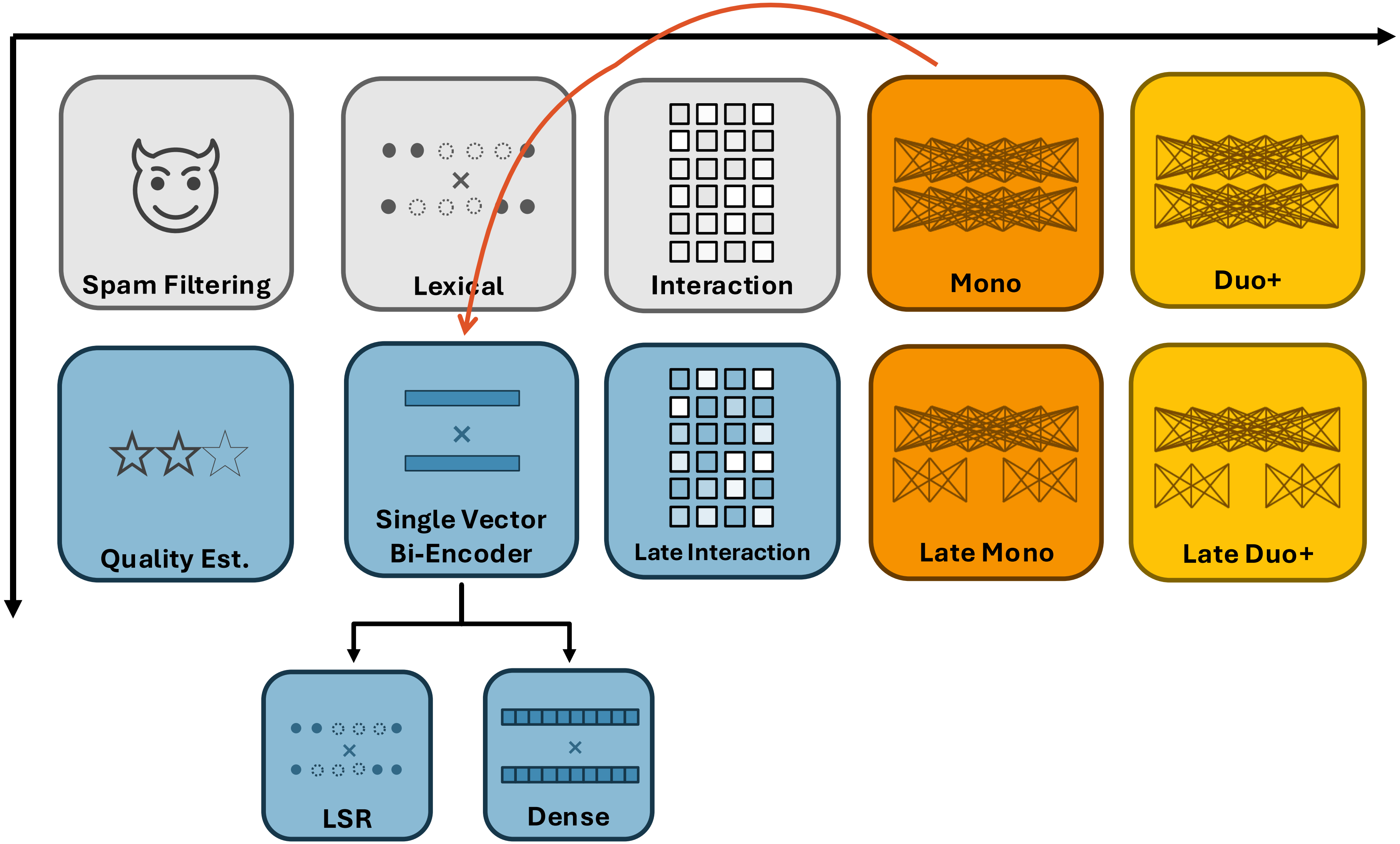


Example: RankNet Loss

[RankNet]

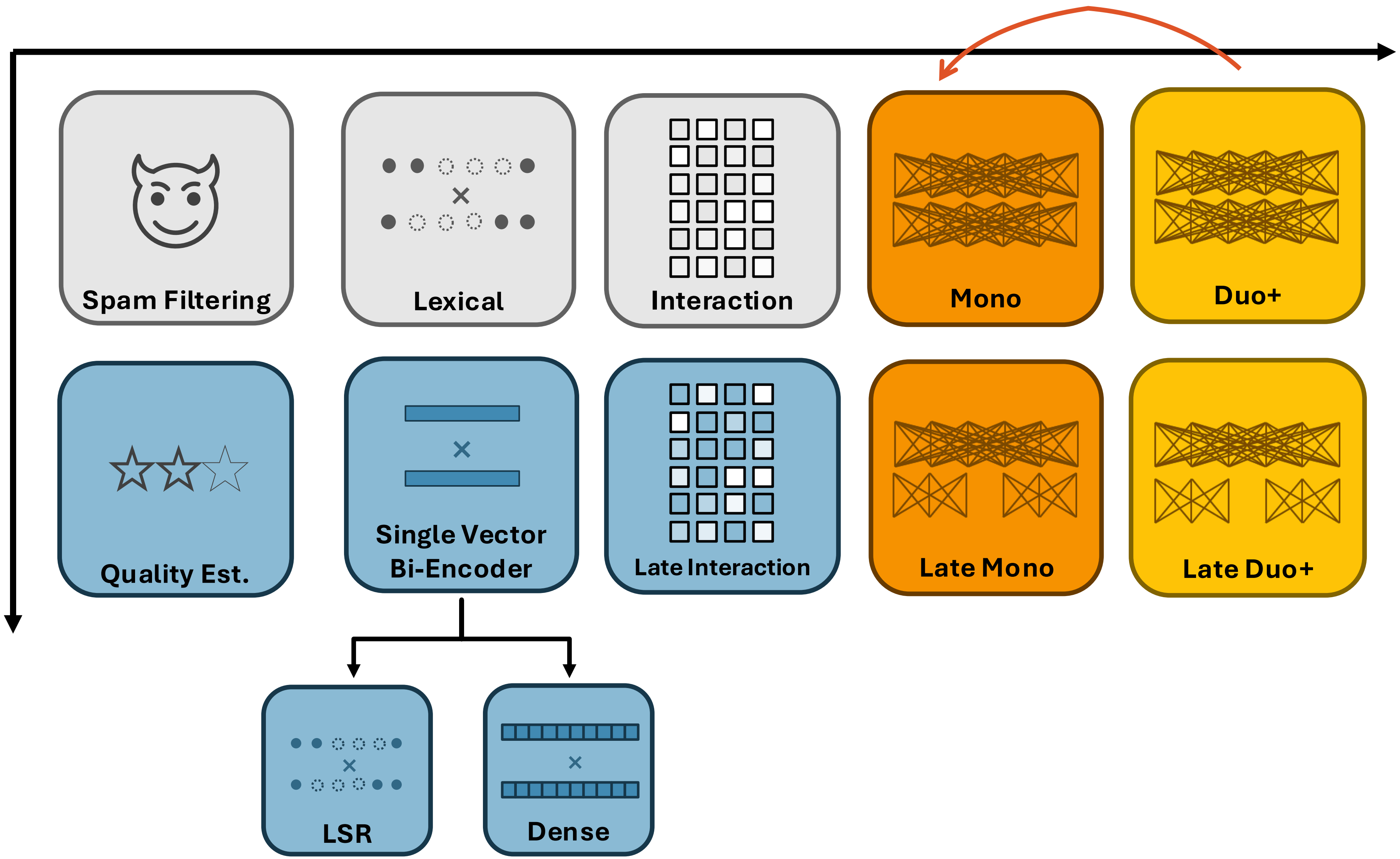
Distillation

Generally, distil stronger -> weaker architecture



Distillation

Generally, distil stronger -> weaker architecture



[DistilCE]

Synthetic Data Generation

Previous approaches work if you're missing positive or negative relevance labels, but what if you're missing **queries**?

[
What incident led to Lewis Hamilton and Max Verstappen's collision at the British GP?,
What penalty was given to Lewis Hamilton after the British GP collision with Max Verstappen?,
Why has Red Bull requested a review of the penalty given to Lewis Hamilton?,
...
]

[InPars], [Promptagator]

Training

In Summary:

You can use human annotations of relevance.

Usually trained by contrasting a relevant doc with a non-relevant one.

Usually helpful to augment human labels with in-batch negatives and hard negatives.

If you already have a strong relevance est. model, you can use distillation.

If you don’t have queries to train on, you can generate them for your documents.

Neural Retrieval and Re-ranking

Part 1: Relevance Est. Models

Part 2: Training

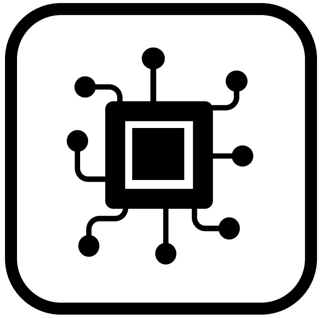
Part 3: Engines

Wrap-up

What we have:

hamilton british gp

Teams summoned after Lewis Hamilton and Max Verstappen British GP collision
Red Bull have requested a review of the penalty given to Mercedes' Lewis Hamilton after his...



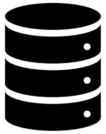
Model

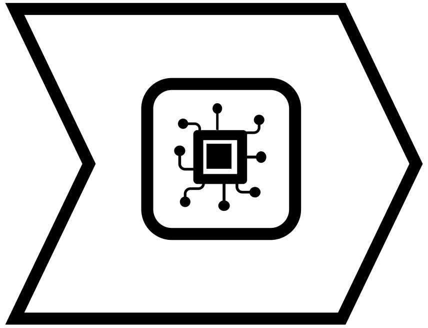
✓ 0.95

What we want:

hamilton british gp

Corpus
Many docs





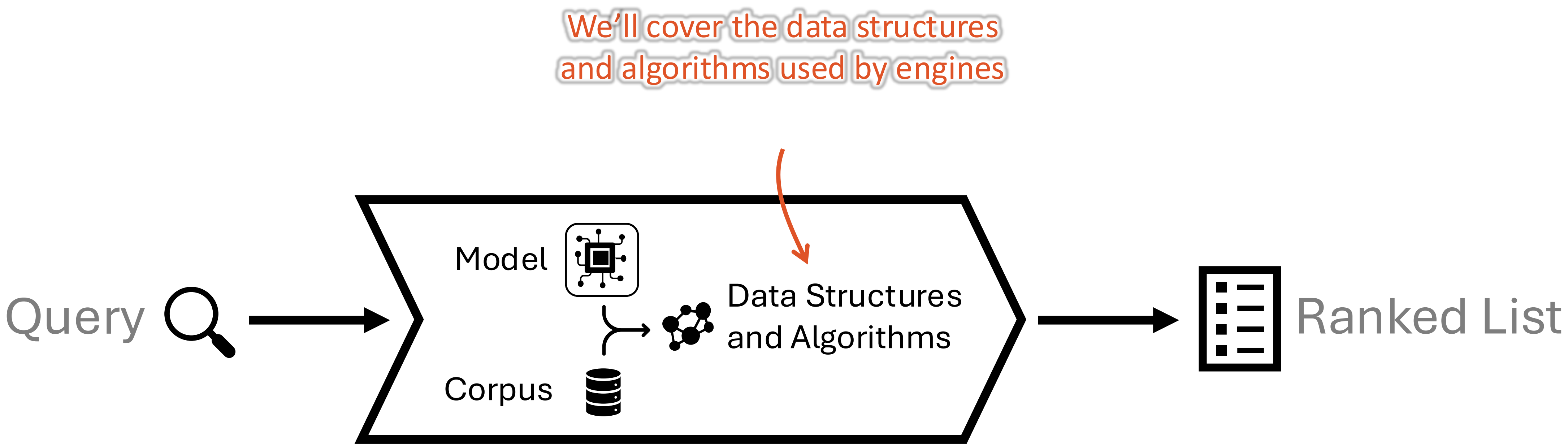
Engine

Ranked List

Docs with highest relevance estimate

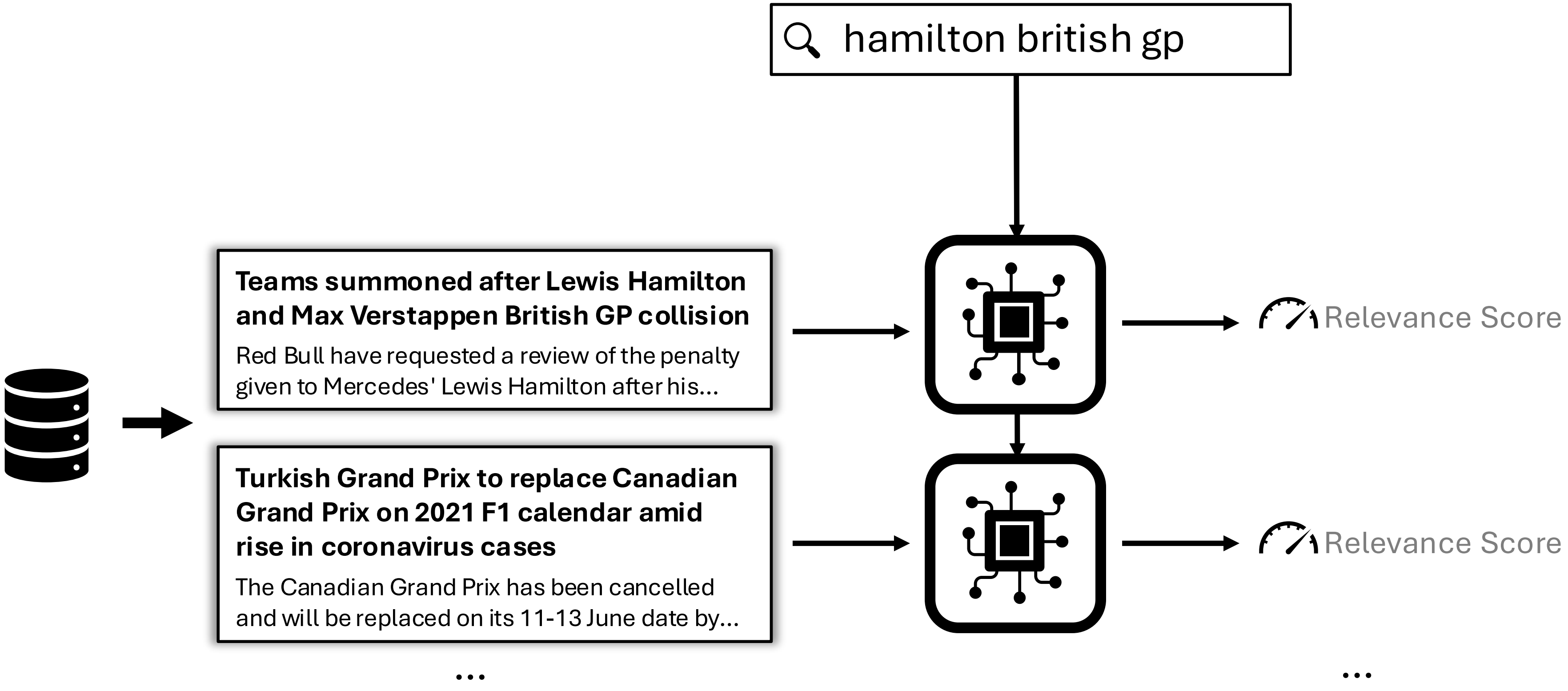
1. **Teams summoned after Lewis Hamilton and Max Verstappen British GP collision**
   Red Bull have requested a review of the penalty given to Mercedes' Lewis Hamilton after his...
2. **Turkish Grand Prix to replace Canadian Grand Prix on 2021 F1 calendar amid rise in coronavirus cases**
   The Canadian Grand Prix has been cancelled and will be replaced on its 11-13 June date by...

...


We'll cover the data structures
and algorithms used by engines
Query
Model
Corpus
Data Structures
and Algorithms
Ranked List

Brute Force Search

Score everything in the corpus!

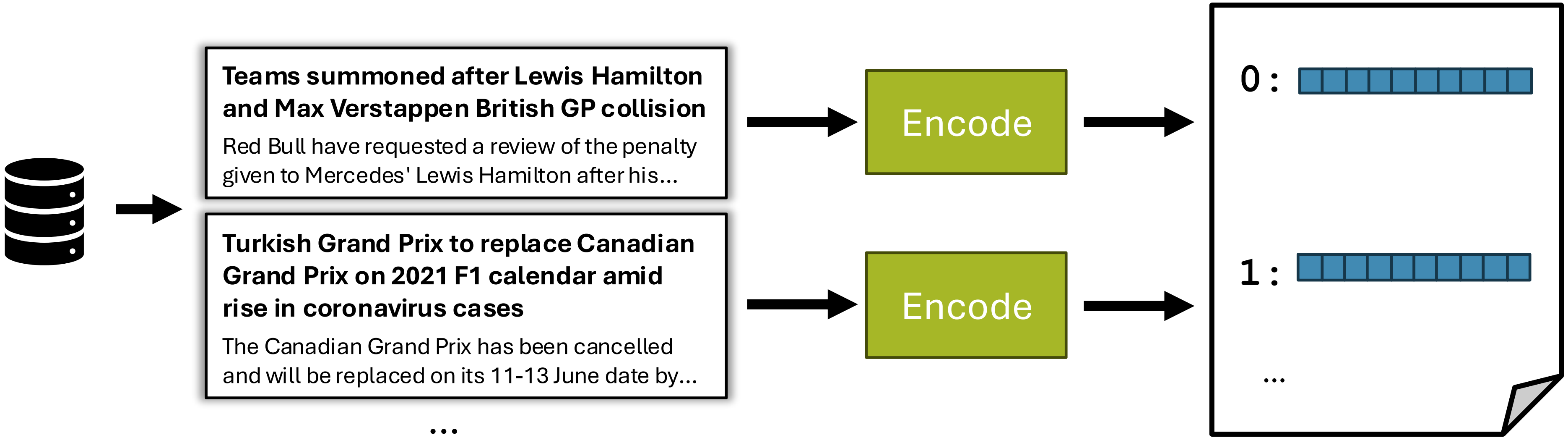


You can speed up brute force search using a forward index

Forward Index

Data Structure: Forward Index (aka Direct Index)

During indexing: Store the representation for each document



Forward Index

Data Structure: Forward Index (aka Direct Index)

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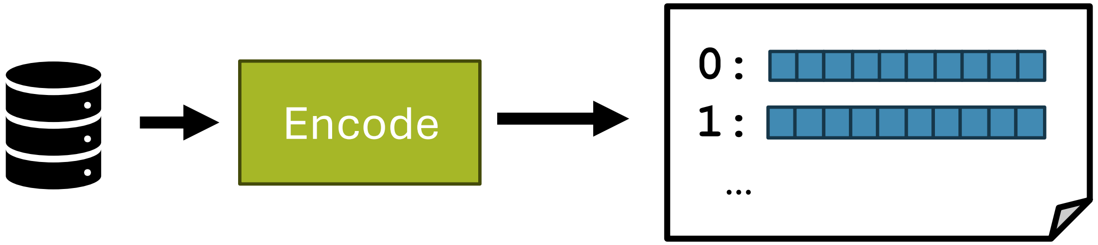
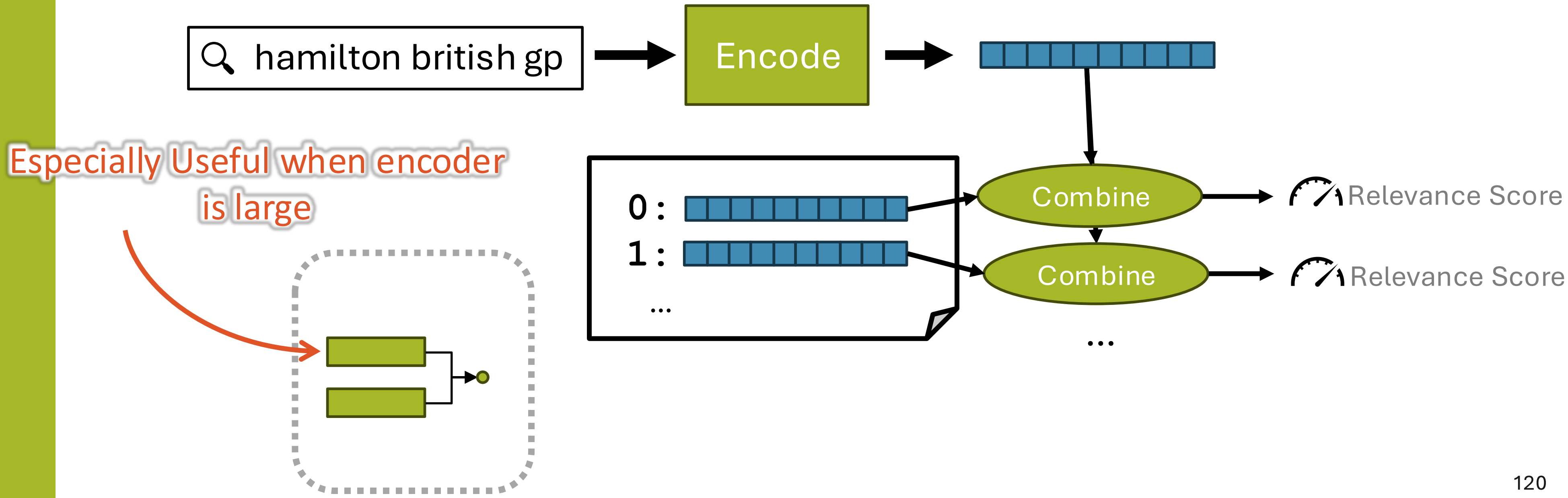
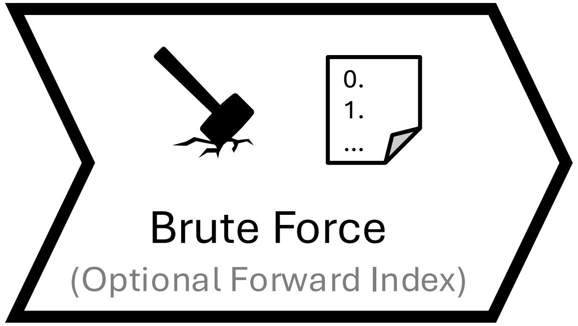


During retrieval: Save compute by using these representations





+ Works with any relevance model

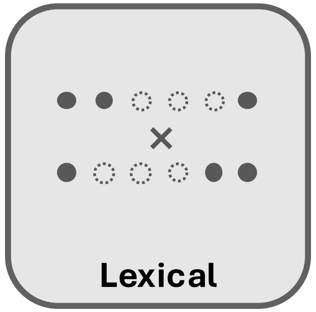
+ Exact top results

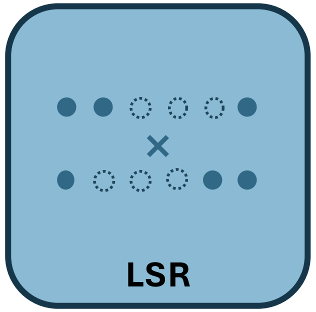
– Not scalable

Let’s be smarter about how to retrieve!

Sparse Retrieval
(with inverted index)

Recall: Sparse Representations





hamilton british gp → Encode → {**hamilton**: **1**, **british**: **1**, **gp**: **1**}

Query terms match in document

Score: >0

Teams summoned after Lewis Hamilton and Max Verstappen British GP collision

Red Bull have requested a review of the penalty given to Mercedes' Lewis Hamilton after his first-lap collision with Max Verstappen at the British Grand Prix.

Hamilton was ruled to have been "predominantly to blame" for the crash, which resulted in Red Bull driver Verstappen being taken to hospital.

The Briton was given a 10-second penalty but recovered to win the race.

Both teams have been summoned to appear via video link on Thursday.

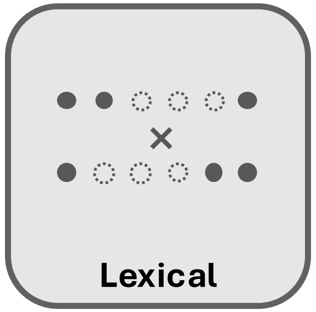
...

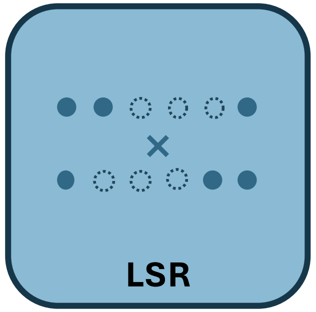
Source: https://www.bbc.co.uk/sport/formula1/57988419

→ Encode → {to: 6, the: 5, **hamilton**: **3**, verstappen: 3, have: 3, teams: 2, summoned: 2, after: 2, lewis: 2, max: 2, **british**: **2**, collision: 2, red: 2, bull: 2, a: 2, penalty: 2, given: 2, was: 2, been: 2, and: 1, **gp**: **1**, …}

Sparse Retrieval
(with inverted index)

Recall: Sparse Representations





amstel river → Encode → `{amstel: 1, river: 1}`

Query terms **do not** match in document

Score: 0

Teams summoned after Lewis Hamilton and Max Verstappen British GP collision

Red Bull have requested a review of the penalty given to Mercedes' Lewis Hamilton after his first-lap collision with Max Verstappen at the British Grand Prix.

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→ Encode →

```
{to: 6, the: 5, hamilton:
3, verstappen: 3, have: 3,
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after: 2, lewis: 2, max:
2, british: 2, collision:
2, red: 2, bull: 2, a: 2,
penalty: 2, given: 2, was:
2, been: 2, and: 1, gp: 1,
...}
```

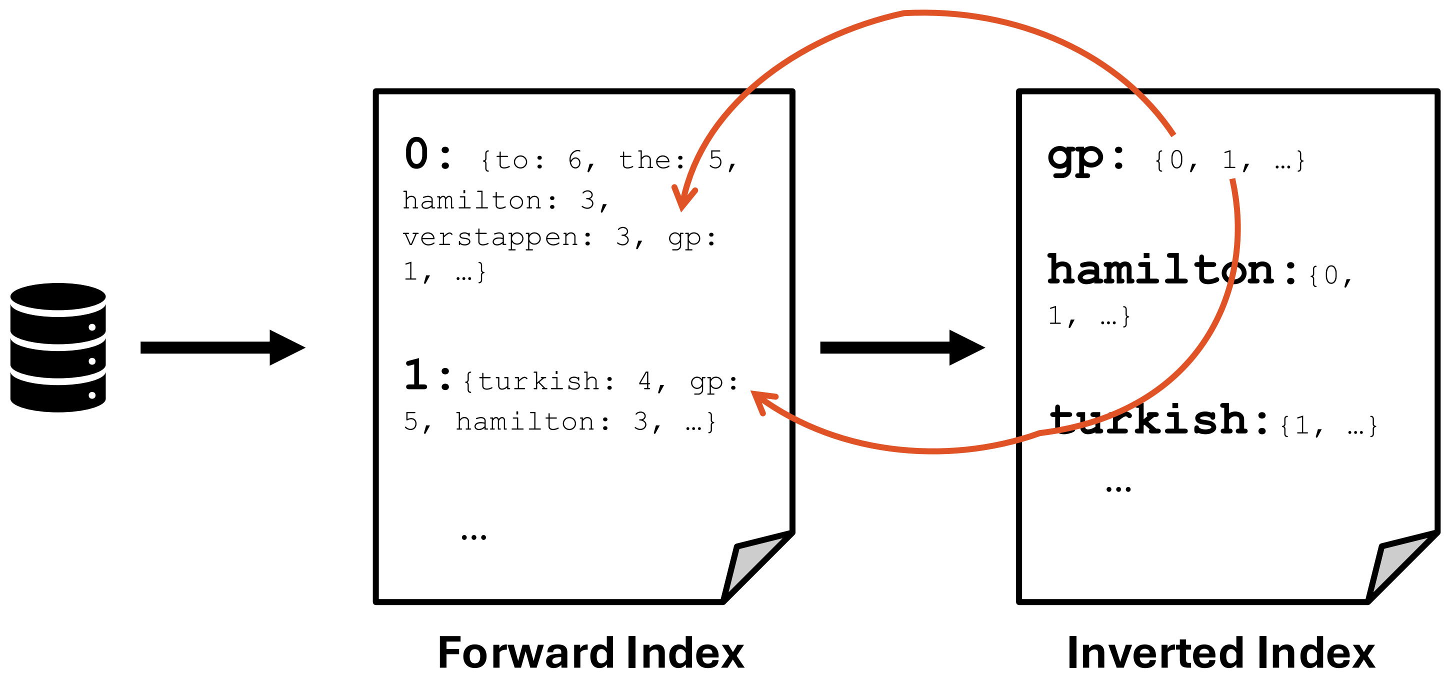
Sparse Retrieval
(with inverted index)

Main idea: We can save a lot of compute by skipping documents that do not have any dimensions that match the query.

Sparse Retrieval
(with inverted index)

Data Structure: Inverted Index (aka Posting Lists)

During indexing: Map terms to documents
(rather than documents to terms)



Sparse Retrieval
(with inverted index)

Data Structure: Inverted Index (aka Posting Lists)

During indexing: Map terms to documents
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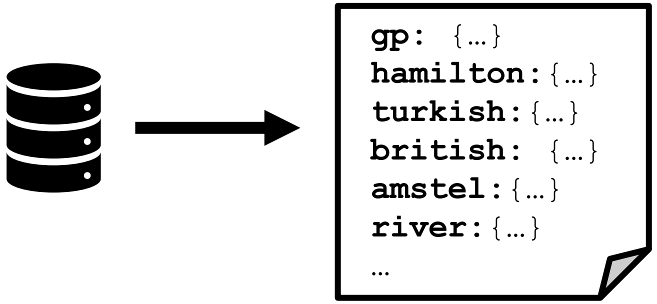
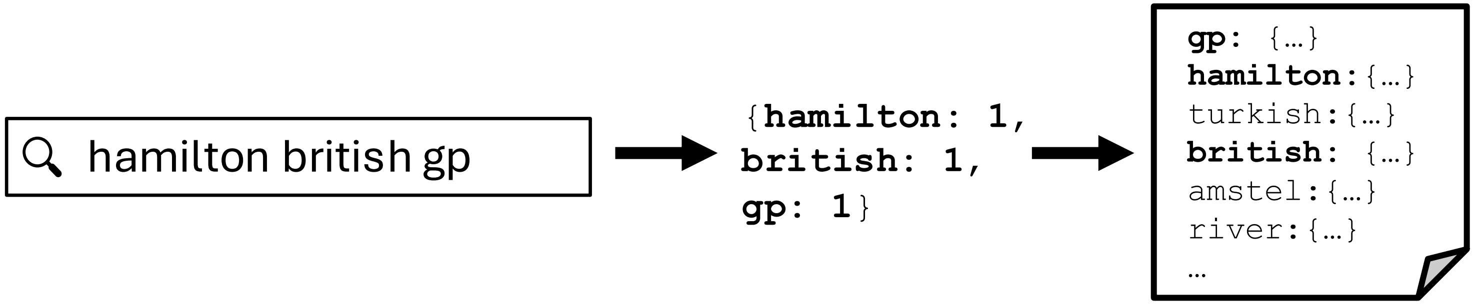


Inverted Index

During retrieval: Only consider docs with matching terms
(there's a LOT of more sophisticated approaches here to speed this up further, see next lecture.)



Inverted Index

Sparse Retrieval
(with inverted index)

Data Structure: Inverted Index (aka Posting Lists)

During indexing: Map terms to documents
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Inverted Index

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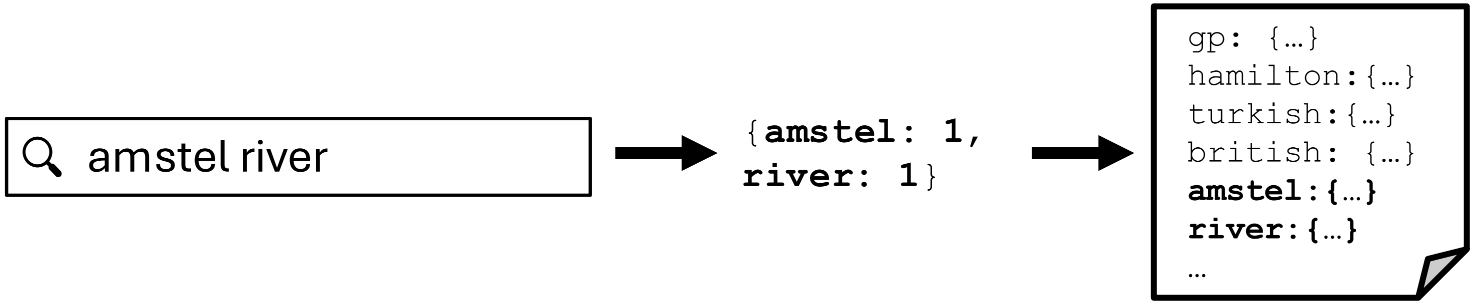
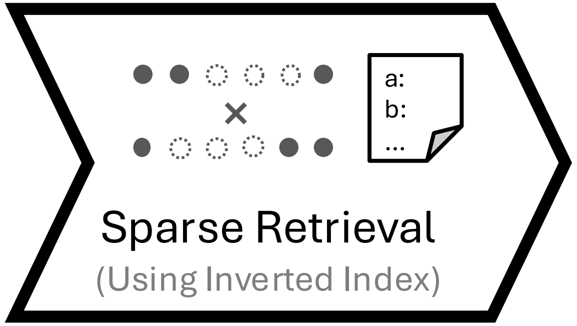


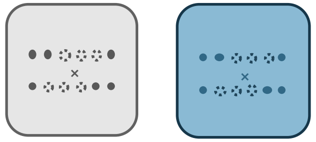
Inverted Index



+ Scalable
(given sufficient sparsity)

+ Exact top results

– Only works with sparse models
(e.g., lexical, LSR)



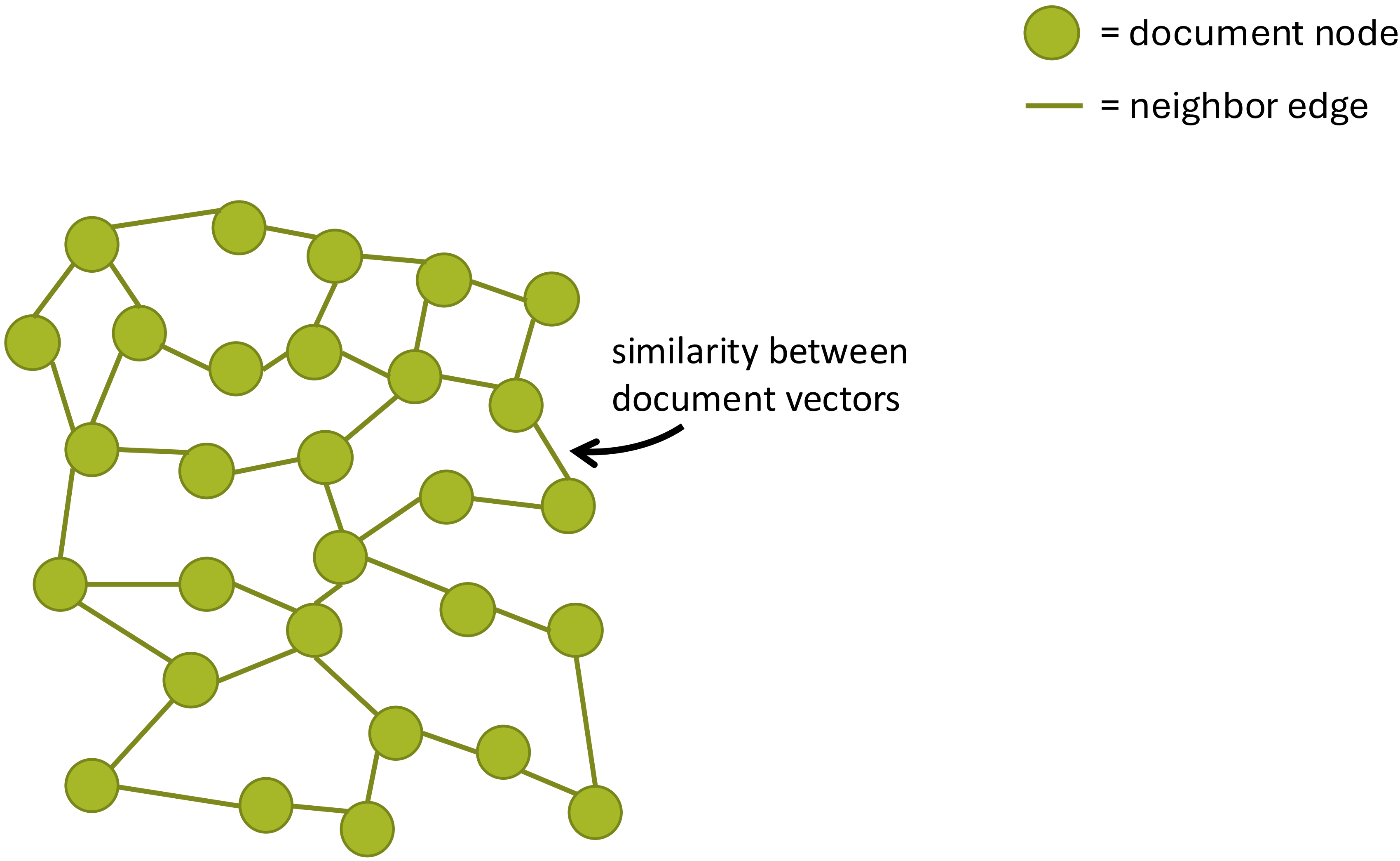
What if we relax the “exact top results” constraint?

If we loosen the constraint for *exact* top-k results, we have an **Approximate Nearest Neighbor** (ANN) Search

A popular ANN approach is HNSW, which is based on the **Cluster Hypothesis:** Documents that are “close” to one another are likely to be relevant to the same queries

Idea: Build a document similarity graph at index time that we can “explore” to find the most similar documents for each query.

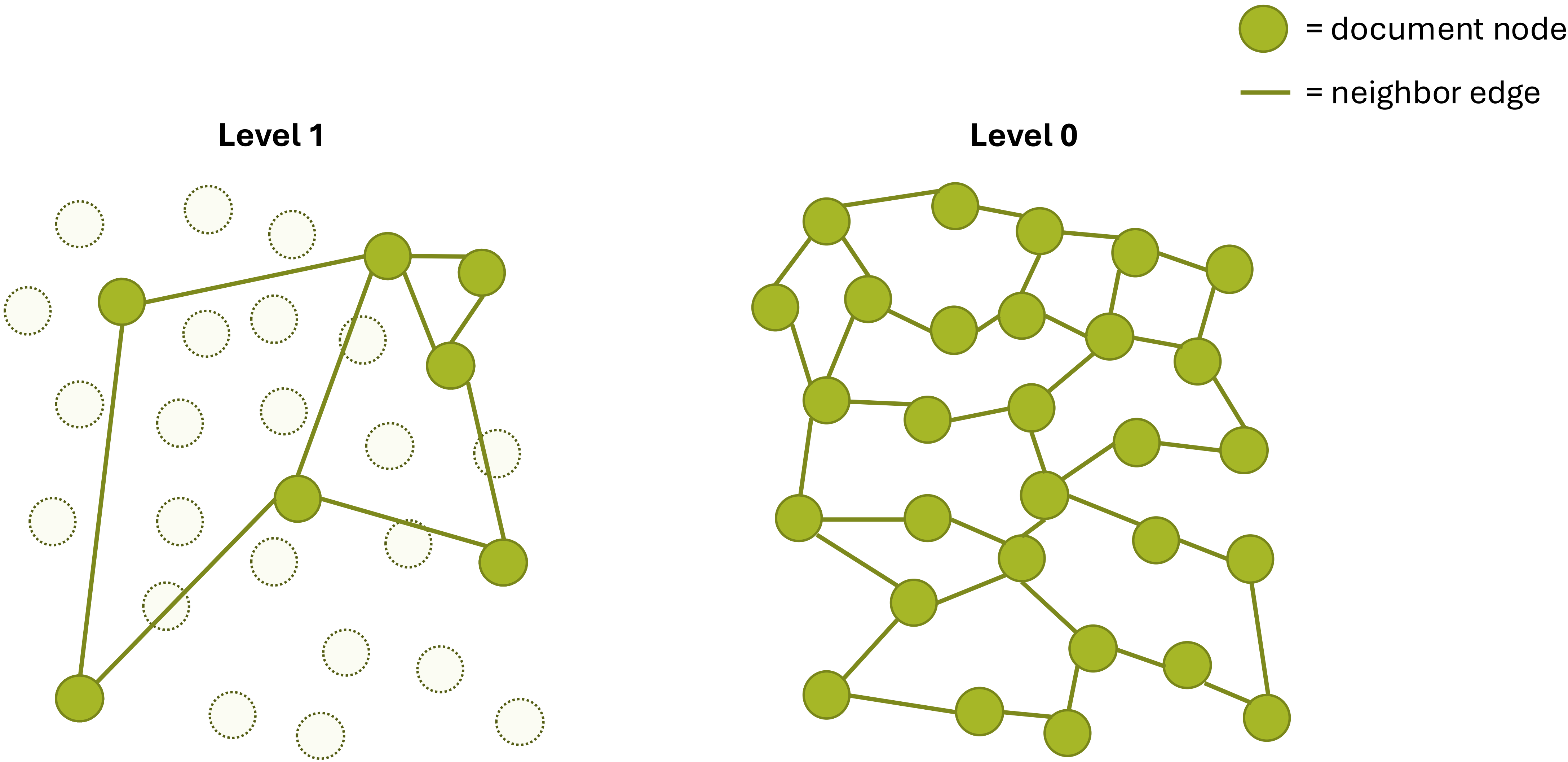
[HNSW]



Where to start the search?

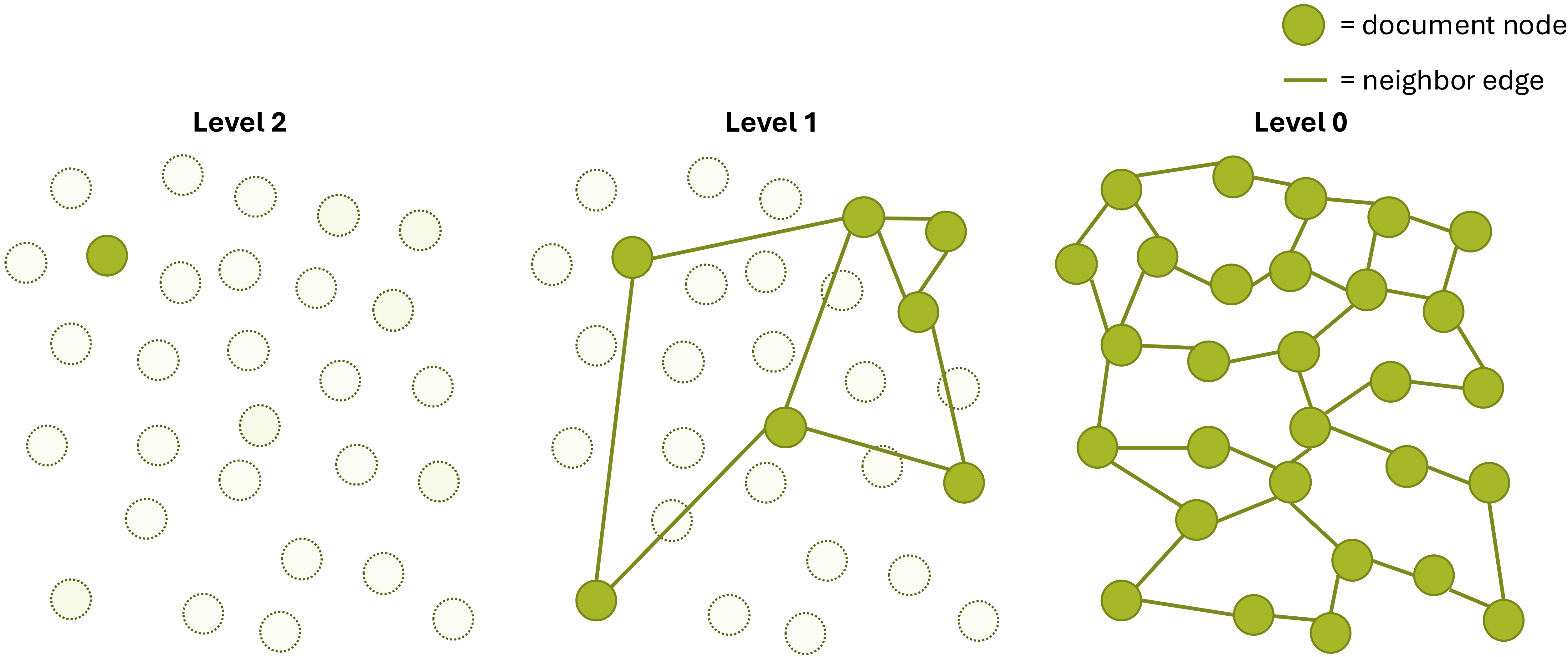
If we pick a random node, we may need to explore the entire graph

[HNSW]

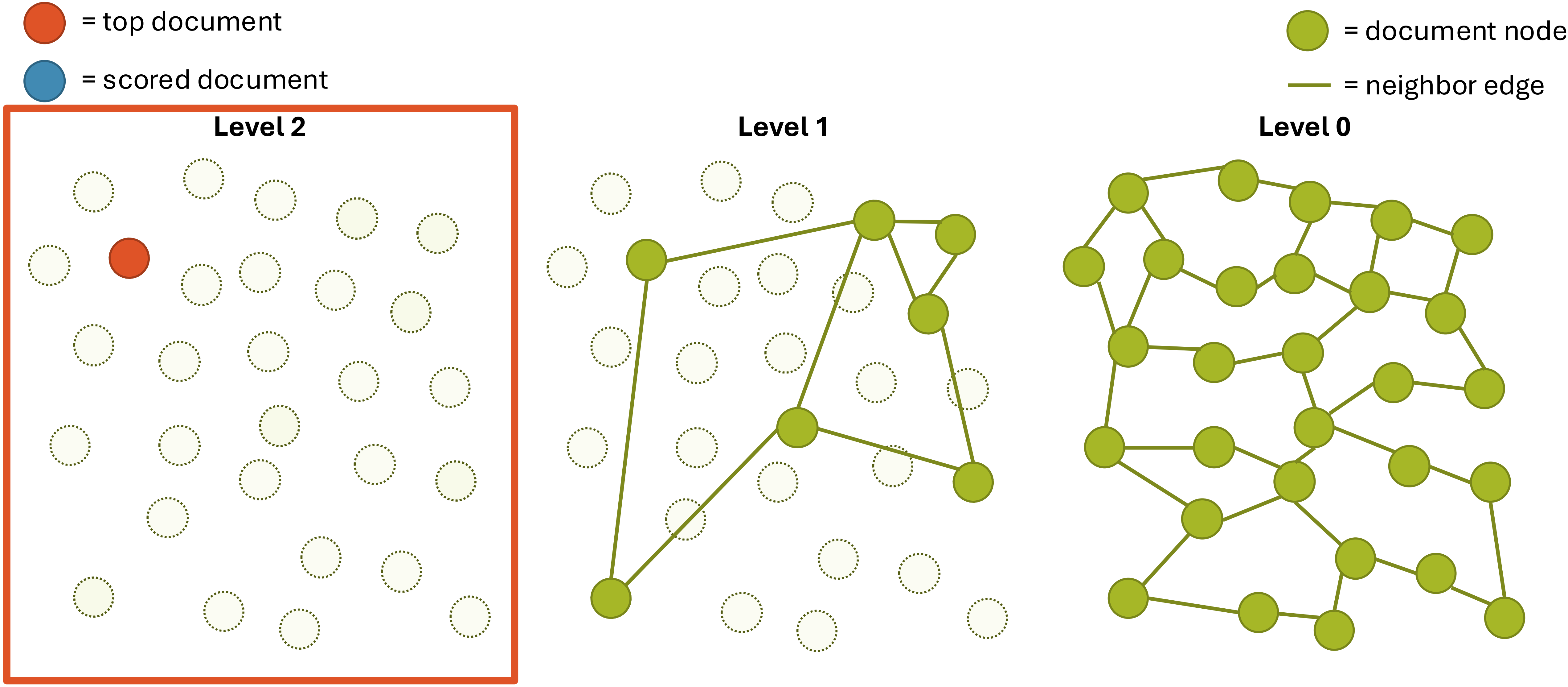


HNSW: Build a hieratical set of NN graphs with subsets of documents

[HNSW]

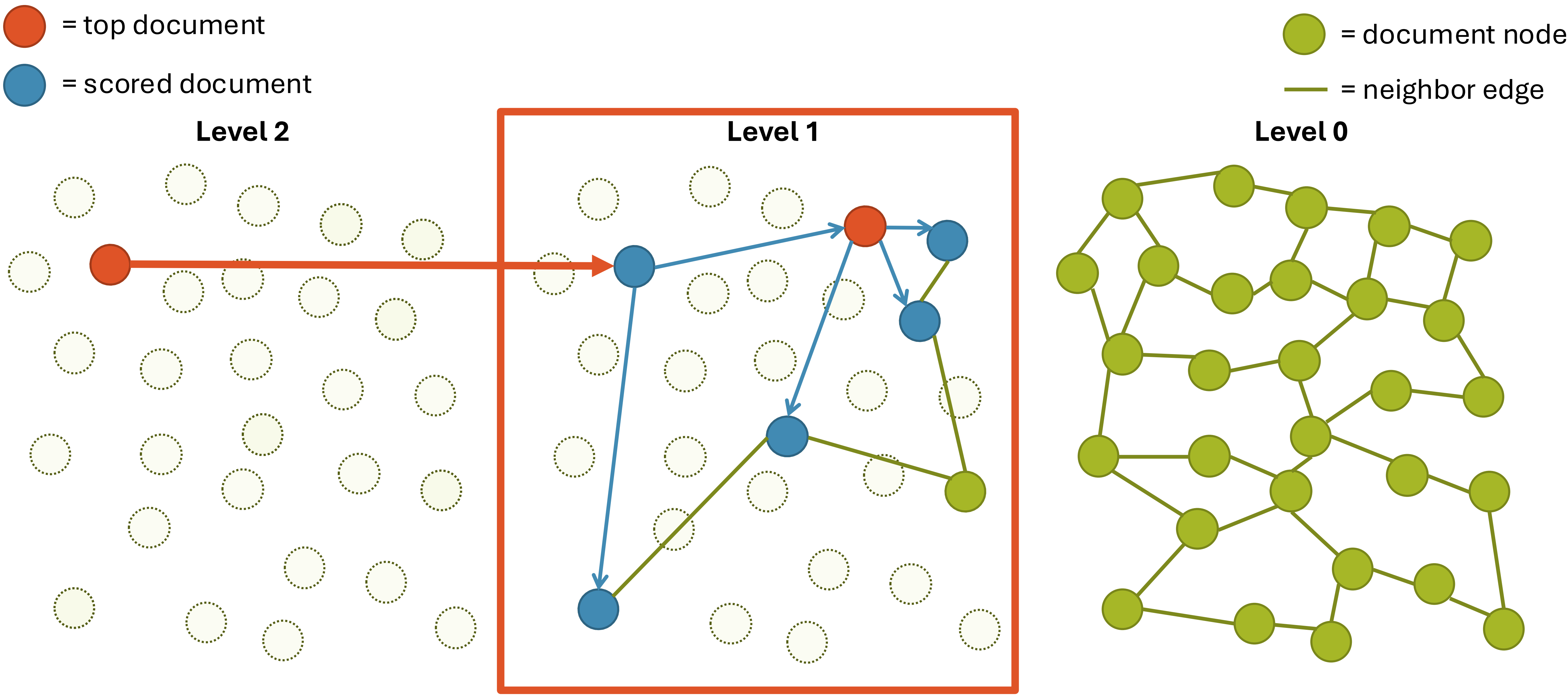


HNSW: Build a hieratical set of NN graphs with subsets of documents



To search, iteratively find closest document to the query by exploring the neighbors from the previous level.

[HNSW]



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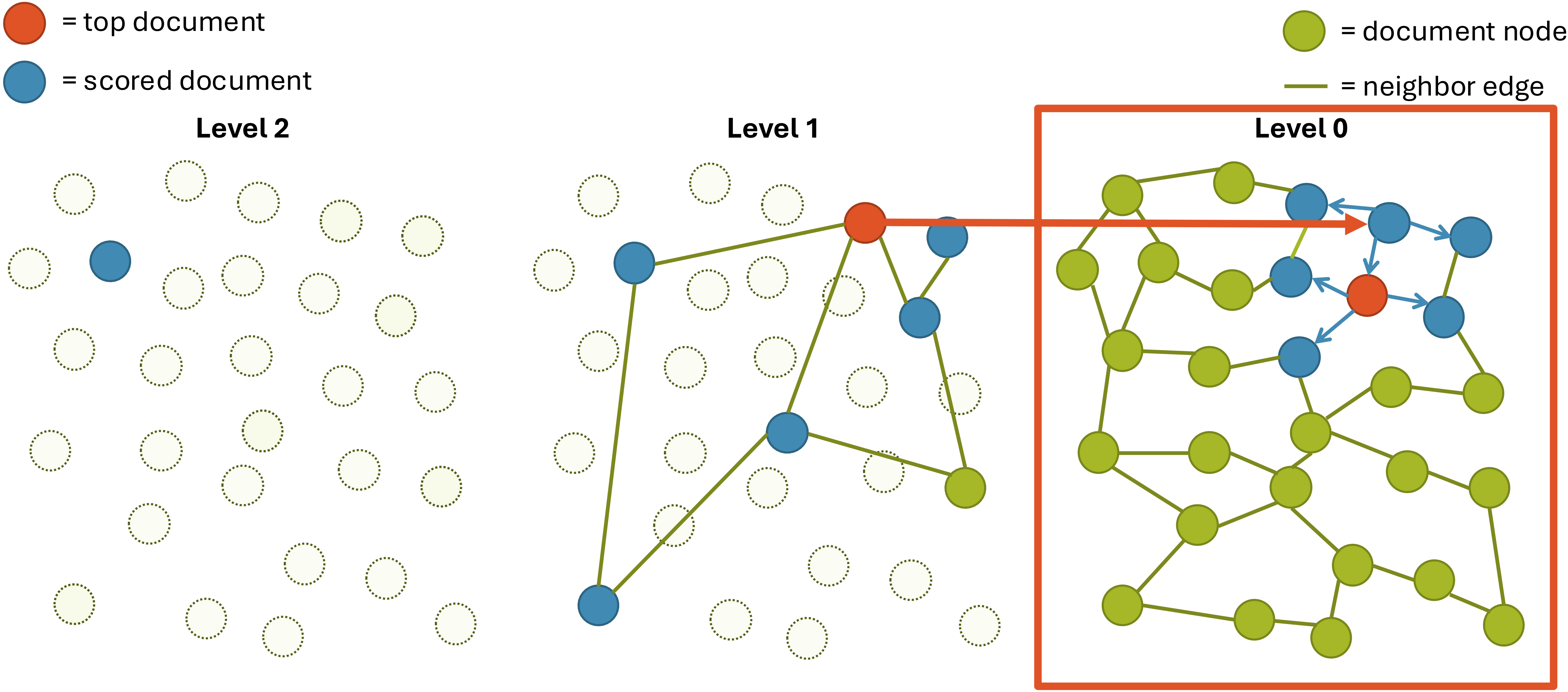
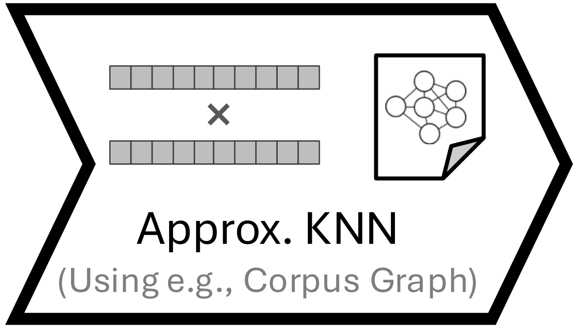
[HNSW]



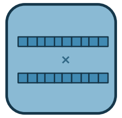
To search, iteratively find closest document to the query by exploring the neighbors from the previous level.

[HNSW]



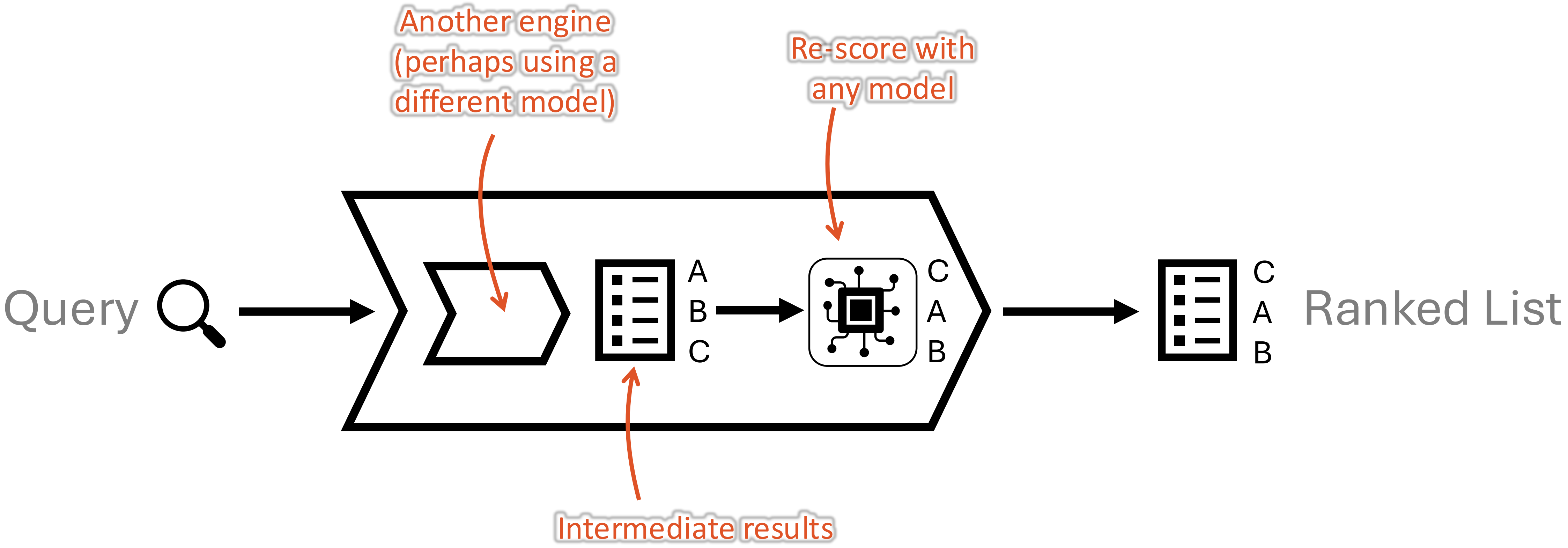
+ Scalable

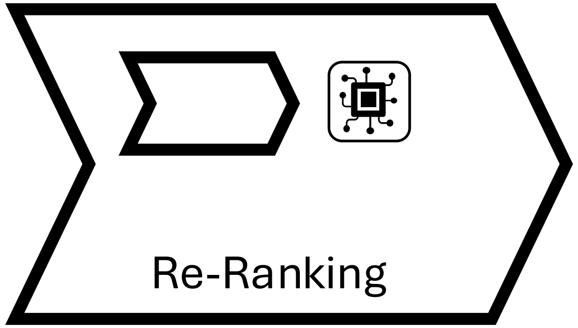
– Approximation of top results (effectiveness degrades)

– Designed for single-rep dense models 

Is there a scalable approach that works with any relevance est. model?

Re-Ranking (aka “cascading”)

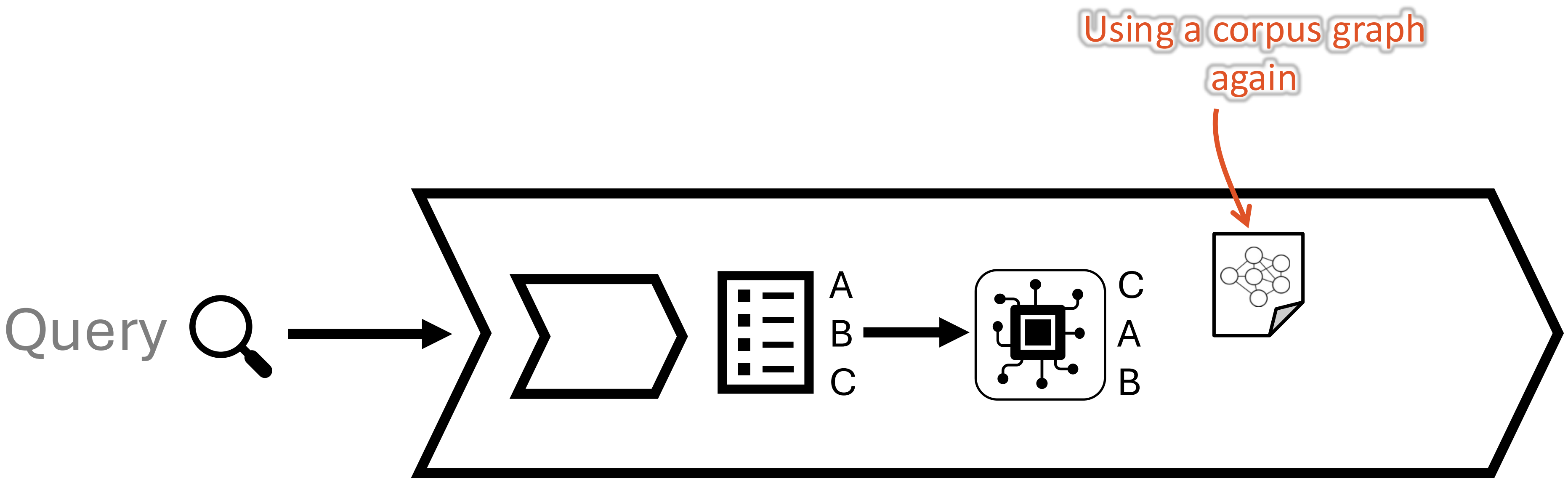




+ Works with any relevance est. model
+ Scalable

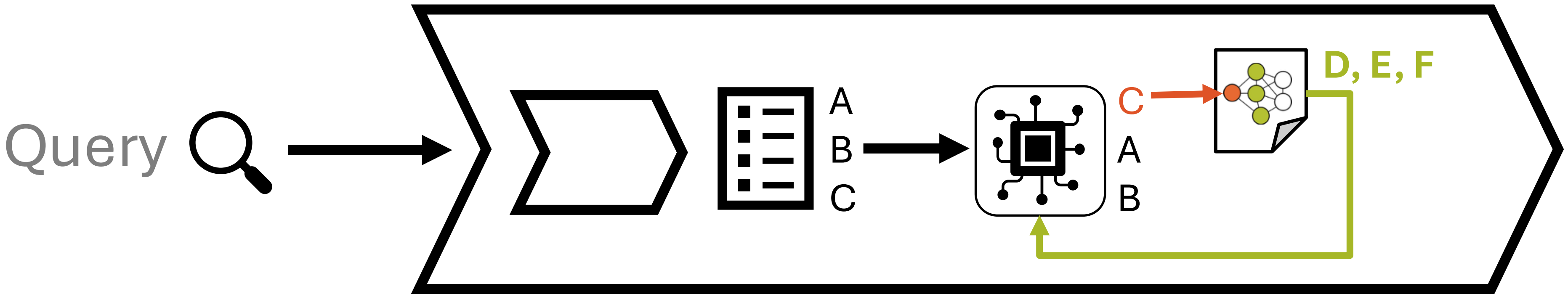
– Approximation of top results
– Limited by performance of first-stage engine
– Requires a first-stage engine

Adaptive Re-Ranking



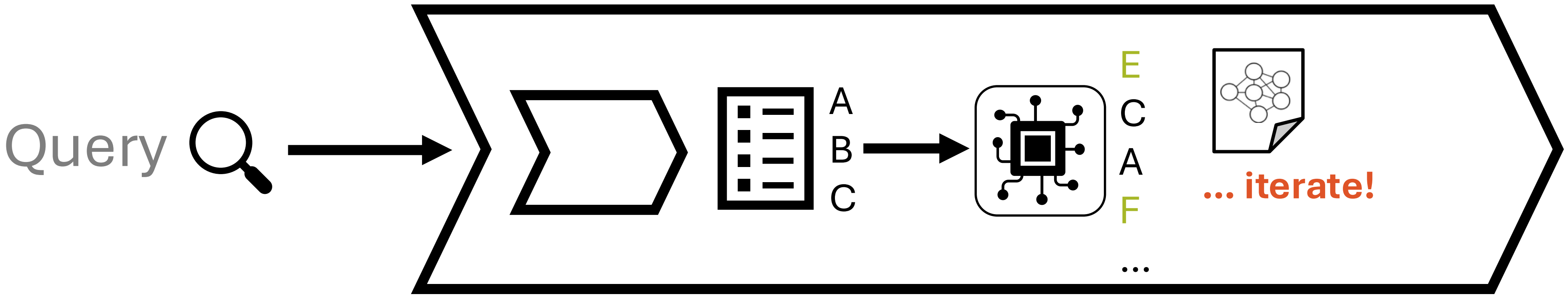
[GAR], [LADR]

Adaptive Re-Ranking

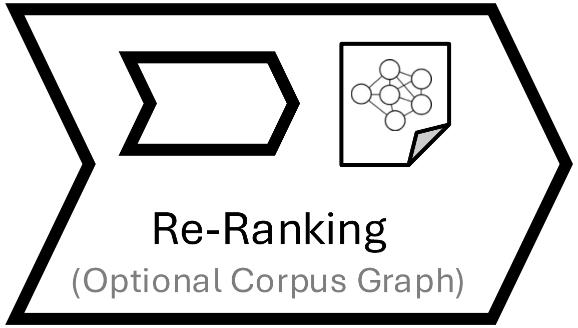


[GAR], [LADR]

Adaptive Re-Ranking



[GAR], [LADR]



+ Works with any relevance est .model

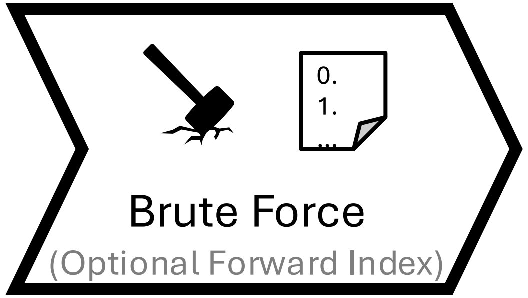
+ Scalable

– Approximation of top results

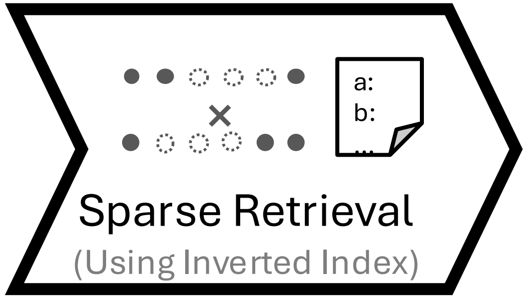
~~– Limited by performance of first-stage engine~~

– Requires a first-stage engine

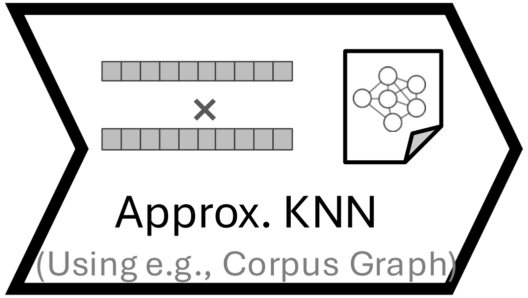
Summary of Engines



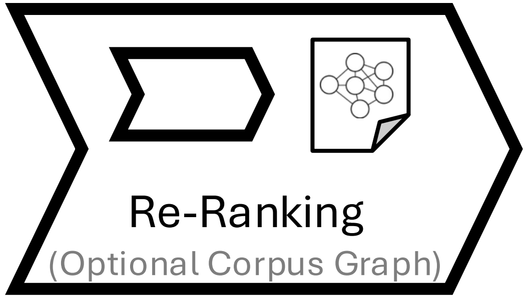
+ Works with any relevance est. .model
+ Exact top results
– Not scalable



+ Scalable
  (given sufficient sparsity)
+ Exact top results
– Only works with sparse models



+ Scalable
– Approximation of top results
– Designed for single-rep dense models



+ Works with any relevance est. model
+ Scalable
– Approximation of top results
– Requires a first-stage engine

Any model can be a retriever,
any model can be a re-ranker.

(It just depends what trade-offs you want.)

Neural Retrieval and Re-ranking

Part 1: Relevance Est. Models

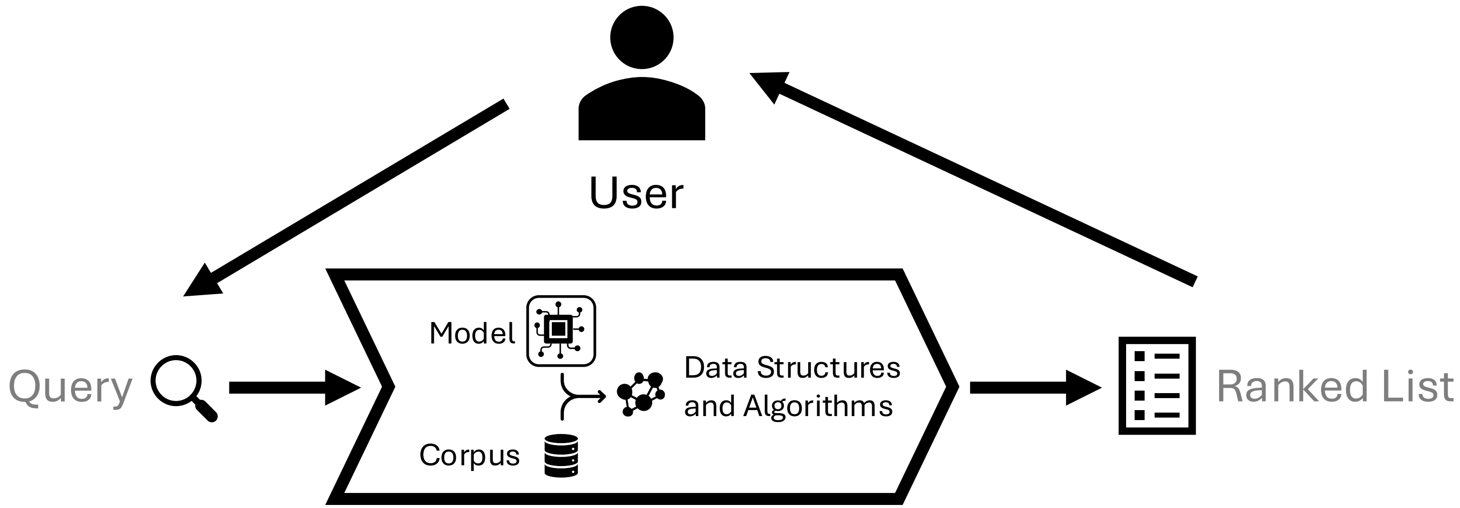
Part 2: Training

Part 3: Engines

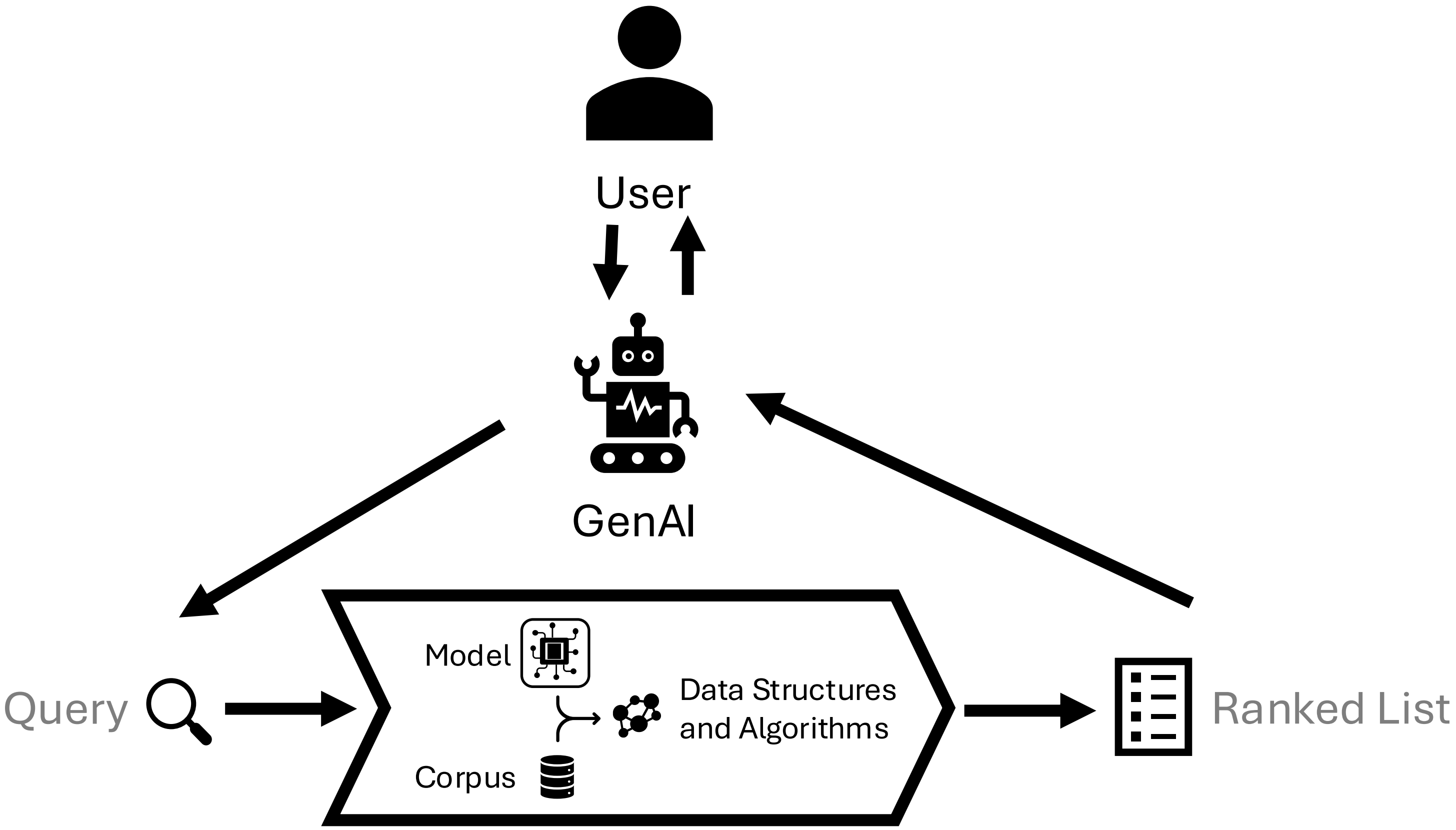
Part 4: Bonus: RAG

Wrap-up

We usually consider users as both the source of queries and the consume of result lists.



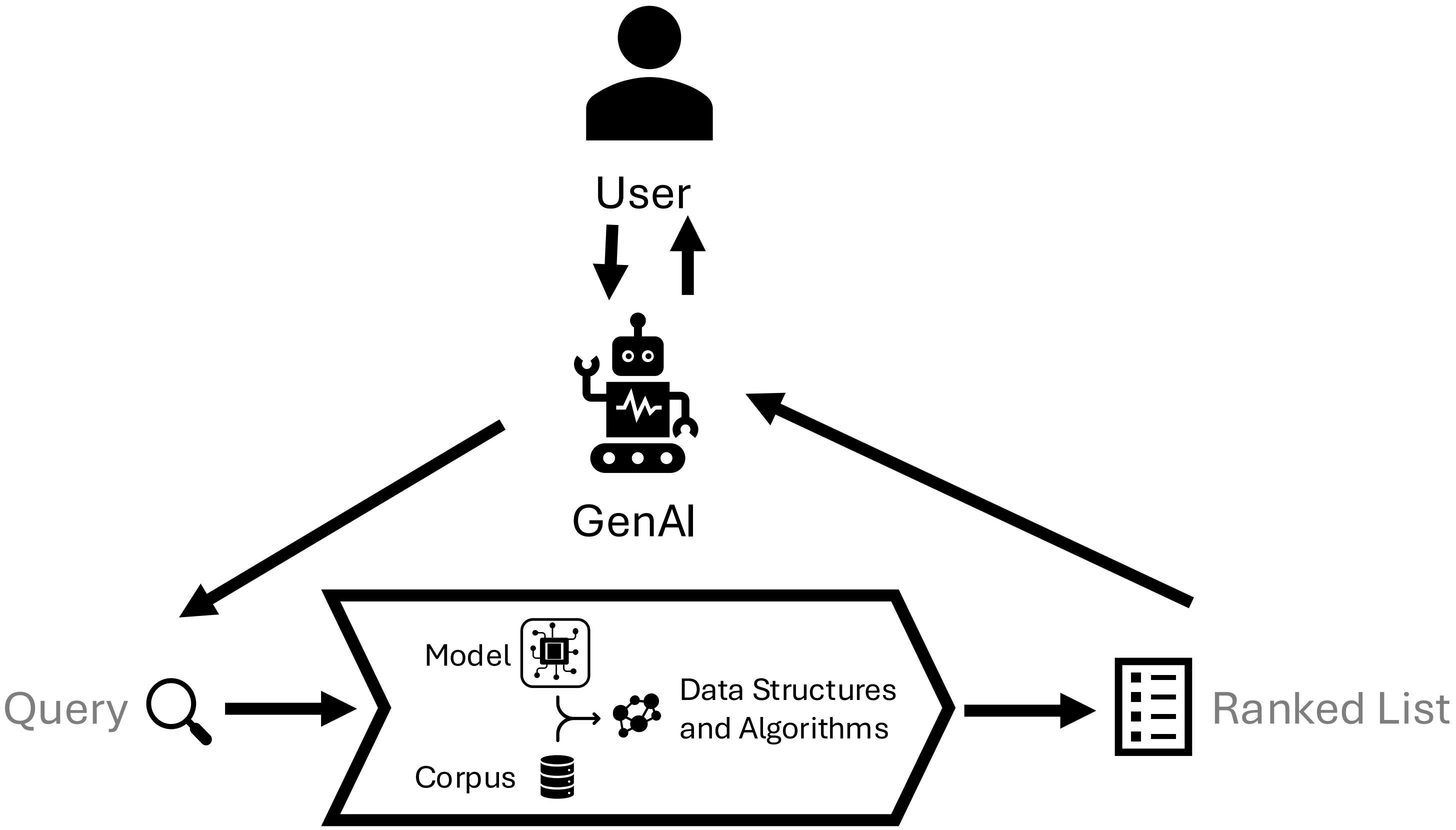
Increasingly, Generative AI acts as an intermediary – issuing queries and consuming results before presenting them to the user. (Retrieval Augmented Generation (RAG))



Why?

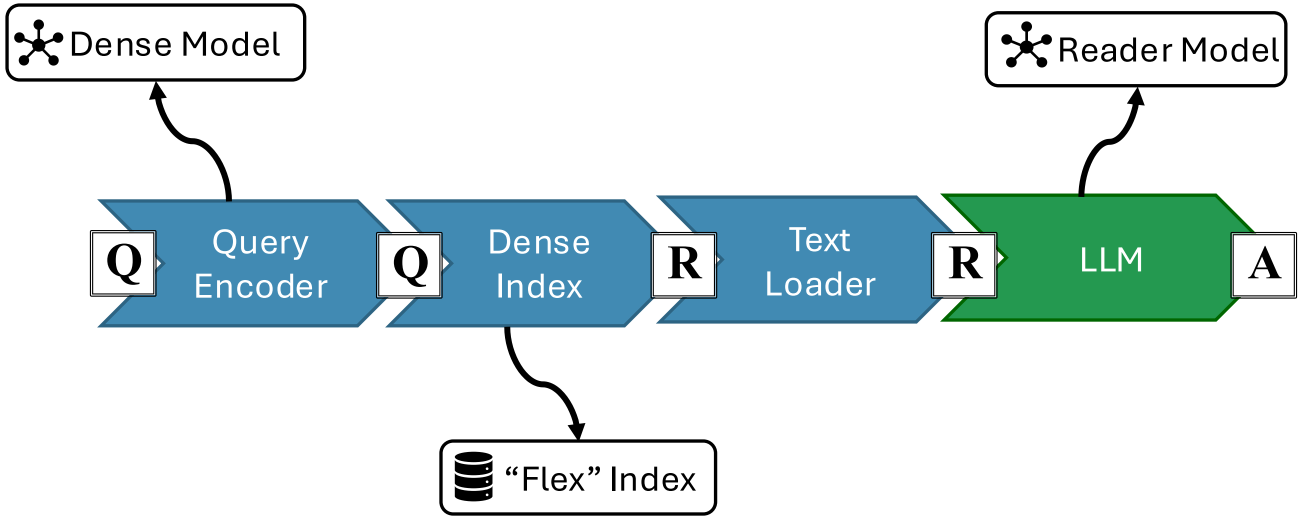
- Can help reduce *hallucinations* from LLMs by *grounding* the generated text

- Allows text to be *attributed* to the retrieved documents.



Can also be expressed as PyTerrier transformers!

Example: Retrieve-then-read pipeline:




```
e5_index = pt.Artifact.from_hf('pyterrier/ragwiki-e5.flex')
e5_rag = (E5() >>
        >> e5_index().retriever()
        >> pt.text.get_text(terrier_index, ['title', "text"])
        >> pyterrier_rag.readers.T5FiD("terrierteam/t5fid_base_nq")
)
```

https://github.com/terrierteam/pyterrier_rag

Neural Retrieval and Re-ranking

Part 1: Relevance Est. Models

Part 2: Training

Part 3: Engines

Wrap-up

Retrieval and Re-ranking

Are you / can you?

familiar with the main types of **relevance est. models**.

describe the main strategies for **training** them.

understand how they are used within an **engine**.

recognize the **trade-offs** of various systems.

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